TAMIL NADU AGRICULTURAL UNIVERSITY

B.Sc. (Hons) Agriculture — First Year, First Semester

Principles of Agricultural Economics

AEC 101

Lecture-wise Theory Study Material

(As per the 2025 Syllabus)

Fifteen Lectures · Definitions, Tables, Worked Numericals and Diagrams
With Exam-Focus Sections, Glossary, Formula Sheet and References

Prepared for examination-oriented self-study
Indian and Tamil Nadu agricultural examples throughout

How to Use This Book

This book follows the **AEC 101 — Principles of Agricultural Economics** syllabus lecture by lecture. Every one of the fifteen lectures in the syllabus is presented as a numbered chapter, in the same order, so that the reader can move from the classroom lecture straight into the matching chapter.

Each chapter is built from the same parts, so the pattern is easy to follow:

- **Headed sections** that expand every sub-topic named in the syllabus line, in simple exam-oriented English.
- **Definition boxes** (shaded) that give the exact wording expected in an answer.
- **Formula boxes** that set out each relationship clearly, one step at a time.
- **Tables** that compare concepts side by side — the fastest way to revise.
- **Worked numericals** with full step-by-step solutions, because marks in this paper are often lost in calculation.
- **Real diagrams** drawn to scale and placed next to the theory they explain.
- **Exam focus** at the end of every chapter: key points, short-answer questions with model answers, and long-answer questions with answer outlines.

Indian and Tamil Nadu examples are used throughout — the minimum support price (MSP), regulated markets (uzhavar sandhai, APMC mandis), the eNAM electronic market, Kisan Credit Card (KCC) farm finance, and GST on farm produce. These examples are the ones most likely to appear in the question paper.

> **How to revise.** First read the headed sections once for understanding. Then, for each chapter, work through the definition boxes and tables until they can be reproduced from memory. Finally, attempt the worked numericals with the solution covered. The formula sheet at the end of the book collects every formula in one place for the last-day revision.

Syllabus Map

The table below shows how the fifteen syllabus lines are covered by the fifteen chapters of this book.

Table 0.1 — Syllabus coverage map

Chapter	Syllabus topic covered
1	Economics: meaning, scope and subject matter; definitions; activities; approaches to economic analysis; micro and macroeconomics; positive and normative analysis
2	Nature of economic theory; rationality assumption; equilibrium; economic laws; basic concepts — goods and services, desire and want, demand, utility, cost and price, wealth, capital, income and welfare
3	Agricultural economics: meaning, definition, characteristics of agriculture, importance and role in economic development; agricultural planning and development; role of economics in agricultural development policies
4	Utility theory — cardinal and ordinal utility; law of diminishing marginal utility; equi-marginal utility; indifference curve analysis and properties; budget line
5	Consumer's equilibrium and derivation of the demand curve; demand — meaning, kinds, law of demand, schedule and curve, determinants; consumer surplus
6	Elasticity of demand — price, income and cross elasticity; factors influencing elasticity; importance; standard of living; Engel's law of family expenditure
7	Production: process, creation of utility; factors of production — definition and characteristics; input-output relationship
8	Laws of returns — law of variable proportions and returns to scale; cost concepts; short-run and long-run cost curves; revenue concepts
9	Supply: stock versus supply, law of supply, supply schedule, supply curve, determinants, elasticity of supply
10	Market structure: meaning and types; perfect and imperfect competition; digital marketing — meaning, scope and role in agricultural markets
11	Distribution theory: factor market and pricing of factors; rent and quasi-rent; wages; population — theories, determinants, current policies and programmes
12	Interest — pure and gross interest; profit — economic profit; forms of business organisations
13	National income — meaning and importance, circular flow, concepts, accounting and approaches to measurement, difficulties in measurement
14	Money — barter and its problems, evolution, meaning and functions, classification, money supply, price index, inflation and deflation; agricultural and public finance
15	Public revenue and expenditure; taxes — direct and indirect, agricultural taxation; economic systems; economic planning; international trade and balance of payments; GST and its implications

Contents

1. Economics: Meaning, Scope and Subject Matter 5
2. Nature of Economic Theory and Basic Concepts 11
3. Agricultural Economics: Meaning, Importance and Role 18
4. Utility Theory: Cardinal and Ordinal Approaches 25
5. Consumer's Equilibrium, Demand and Consumer Surplus 33
6. Elasticity of Demand, Standard of Living and Engel's Law 40
7. Production: Process, Factors and Input–Output Relations 47
8. Laws of Returns, Cost and Revenue 53
9. Supply: Law, Determinants and Elasticity 61
10. Market Structure and Digital Marketing 67
11. Distribution Theory: Rent, Wages and Population 73
12. Interest, Profit and Business Organisations 79
13. National Income: Concepts and Measurement 84
14. Money, Prices and Agricultural Finance 90
15. Public Finance, Economic Systems and International Trade 96
Appendix A — Model Question Papers 103
Appendix B — Practice Numericals with Answers 106
Appendix C — One-Line Revision Points 109
Glossary of Key Terms 111
Formula Sheet 115
References and Suggested Reading 118

LECTURE 1

Economics: Meaning, Scope and Subject Matter

1.1 Why Economics Matters to an Agriculture Student

Every farmer, every village trader, every agricultural officer takes decisions that are essentially economic: what to grow, how much to spend on seed and fertiliser, when to sell, and at what price. Economics is the subject that studies these decisions in a systematic way. For a student of agriculture it is not a distant theory — it is the language in which the business of farming is discussed.

Economics grew out of the practical problem of scarcity. Human wants are unlimited, but the resources used to satisfy them — land, labour, capital and time — are limited. Economics studies how society chooses to use scarce resources to produce goods and services and how those goods and services are distributed among people.

1.2 Meaning of Economics

DEFINITION — ECONOMICS

Economics is the social science that studies the **production, distribution and consumption of goods and services**, and the behaviour of individuals, firms and governments in allocating scarce resources among competing wants.

The word itself comes from the Greek *oikonomia*, meaning household management — managing a household with limited means to satisfy its many needs. The same idea, scaled up, describes the economy of a village, a state, or a country.

Wealth, Welfare and Scarcity Definitions

- **Wealth definition (classical).** Adam Smith (1776) called economics "the science of wealth". This view studied how wealth is produced, accumulated and distributed. It was criticised for concentrating on wealth rather than human well-being.
- **Welfare definition (neoclassical).** Alfred Marshall placed human welfare at the centre: economics is the study of how people earn and use their income to obtain material well-being.
- **Scarcity definition (modern).** Lionel Robbins gave the most widely accepted definition: economics studies human behaviour as a relationship between **unlimited wants and scarce means** that have alternative uses.

ROBBINS' DEFINITION (1932)

"Economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses."

Robbins' definition contains three conditions that create the economic problem: (i) wants are unlimited, (ii) the means to satisfy them are scarce, and (iii) those means have alternative uses. Where all three hold, choice is unavoidable, and the study of choice is economics.

1.3 Scope and Subject Matter of Economics

The scope of economics answers the question: what does economics study, and what does it not study? Its subject matter is organised around two central questions — the **micro** question of individual choice and the **macro** question of the economy as a whole.

Economics is regarded as a science (it uses systematic observation and reasoning) and also as an art (it guides practical policy). It is a **social science** because its subject is human beings, whose behaviour cannot be controlled in a laboratory.

Microeconomics

> **DEFINITION — MICROECONOMICS**
>
> Microeconomics studies the economic behaviour of **individual units** — a single consumer, a single firm, a single market, a single commodity — and how prices are determined in individual markets.

Its main tools are demand and supply, elasticity, utility, production and cost, and the theory of the firm. The word "micro" comes from the Greek for "small". When a Tamil Nadu farmer decides how many acres of paddy to sow, that is a microeconomic decision.

Macroeconomics

> **DEFINITION — MACROECONOMICS**
>
> Macroeconomics studies the economy **as a whole** — aggregate output, national income, total employment, the general price level and the balance of payments.

Its main tools are national income accounting, money and banking, inflation, and fiscal policy. When the Government of India fixes the MSP for paddy for the whole country, or when the Reserve Bank of India changes the repo rate, those are macroeconomic actions.

Table 1.1 — Microeconomics versus Macroeconomics

Basis	Microeconomics	Macroeconomics
Unit of study	Individual consumer, firm, market, commodity	Economy as a whole; aggregates
Central problem	Price determination and resource allocation in a single market	National income, employment and general price level
Key tools	Demand and supply, elasticity, utility, cost and revenue	National income, money supply, inflation, fiscal and monetary policy
Known as	Price theory	Income theory
Approach	Bottom-up ("slicing the economy")	Top-down ("looking at the whole pie")
Agricultural example	A farmer's choice of crop and the price of paddy in a local mandi	MSP for the whole country; agricultural GDP; food inflation

> **Link between the two.** Micro and macro are two views of the same economy. What is true for one firm is not always true for the economy. For example, a single farmer who sells more paddy earns more, but if all farmers sell more, the price may fall and total income may not rise. This is the "fallacy of composition".

1.4 Economic Activities

Economic activity means any human activity undertaken to earn a living or to satisfy wants using scarce resources. Economic activities are traditionally classified into four groups.

- **Production** — creating goods and services by combining factors of production (land, labour, capital, organisation). Example: a farmer producing paddy from land, seed, water and labour.
- **Exchange** — transferring goods and services from one person to another, usually for money. Example: selling paddy in the regulated market.
- **Distribution** — sharing the income generated among the factors of production as rent, wages, interest and profit.
- **Consumption** — using goods and services to satisfy wants. Example: a family cooking and eating the rice it has bought.

Activities that do not aim at earning income — a mother cooking for her family, a student studying for love of the subject — are called **non-economic activities**. The distinction is about the purpose, not the physical act.

1.5 Approaches to Economic Analysis

Economists analyse problems using two broad approaches, which are complementary.

Positive (what is) and Normative (what ought to be)

> **DEFINITIONS**
>
> **Positive economics** deals with facts — it describes what is, and its statements can be tested against evidence. **Normative economics** deals with value judgements — it states what ought to be, and cannot be proved or disproved by facts alone.

Table 1.2 — Positive versus Normative Economics

Basis	Positive economics	Normative economics
Concern	"What is"	"What ought to be"
Nature	Factual, objective, verifiable	Value-based, subjective
Example	"An increase in the MSP of paddy raises the income of paddy growers."	"The Government should raise the MSP of paddy."
Testing	Can be tested with data	Cannot be tested by facts alone
Also called	Descriptive economics	Policy economics / welfare economics

Deductive and Inductive Methods

- **Deductive method** (also called the analytical or abstract method) starts from general principles or assumptions and derives particular conclusions. Example: assuming consumers are rational, we deduce the law of demand. It moves from the general to the particular.
- **Inductive method** starts from observed particular facts and builds up a general rule. Example: observing that as the price of tomatoes falls in many markets, more is bought, we infer the law of demand. It moves from the particular to the general.

Modern economics uses both: theory is built deductively, and tested inductively with data (statistics and econometrics).

1.6 The Central Problems of an Economy

Because resources are scarce, every economy — whether it is a village or a nation — must answer three basic questions:

1. **What to produce** and in what quantities? (More rice or more cotton? More food or more factories?)
2. **How to produce**? (Which combination of labour and machinery? Labour-intensive or capital-intensive?)
3. **For whom to produce**? (How is the output distributed among the people?)

A fourth question, **how efficiently** resources are used and whether the economy grows over time, is also important. These questions are answered differently by different economic systems — by the market in a capitalistic economy, by the state in a socialistic economy, and by a mix of both in India's mixed economy (see Lecture 15).

1.7 Importance of the Study of Economics

- It helps in the **wise use of scarce resources** at the level of the farm, the firm and the nation.
- It provides the tools to **understand prices** — why the price of onion rises, why the price of paddy falls at harvest.
- It guides **government policy** on taxation, subsidies, MSP, credit and public investment.
- It helps in **planning** — deciding how much to produce and invest for future growth.
- It develops **analytical thinking**, useful in every profession, including agricultural extension and agribusiness.

Worked Example 1.1 — Classifying a statement as positive or normative

Problem. Classify each statement as positive or normative: (a) "The MSP of paddy for 2025-26 is ₹2,369 per quintal." (b) "The Government should give a higher MSP to millets." (c) "An increase in fertiliser subsidy increases fertiliser use."

Solution.

- (a) **Positive.** It is a statement of fact that can be checked against the official announcement.
- (b) **Normative.** The word "should" signals a value judgement about what is desirable.
- (c) **Positive.** It states a cause-and-effect relationship that can be tested with data on subsidy and fertiliser consumption.

Rule of thumb: if the sentence can be settled by looking at facts, it is positive; if it depends on opinion or values (often signalled by "should", "ought", "fair", "better"), it is normative.

1.8 Economics and Other Sciences

Economics borrows from and contributes to several other subjects, which shows the breadth of its scope.

- **With mathematics and statistics:** tools for measuring relationships and testing theories.
- **With sociology and psychology:** to understand how customs, habits and expectations shape economic behaviour.
- **With history and political science:** to study how institutions and government shape the economy.
- **With agriculture and technology:** to apply economic principles to farming and innovation.

Because it deals with human welfare, economics is described as the "science of choice" and, by Marshall, as "the study of mankind in the ordinary business of life".

Exam Focus — Lecture 1

Key points to remember

- Economics studies the allocation of scarce resources among unlimited wants.
- The three great definitions: wealth (Adam Smith), welfare (Marshall), scarcity (Robbins).
- Robbins' scarcity definition rests on unlimited wants, scarce means, and alternative uses.
- Microeconomics = individual units and price theory; macroeconomics = aggregates and income theory.
- Positive economics describes "what is"; normative economics prescribes "what ought to be".
- Deductive method moves general to particular; inductive method moves particular to general.
- Every economy answers three questions: what, how and for whom to produce.

Short-answer questions with model answers

Q1. Define economics. Give Robbins' definition.

Model answer. Economics is the social science that studies the production, distribution and consumption of goods and services and how scarce resources are allocated among competing wants. Robbins (1932) defined it as "the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses."

Q2. Distinguish between microeconomics and macroeconomics.

Model answer. Microeconomics studies individual units such as a consumer, firm or single market and is called price theory. Macroeconomics studies the economy as a whole — national income, employment and the general price level — and is called income theory. They are complementary views of the same economy.

Q3. What are the central problems of an economy?

Model answer. (i) What to produce and in what quantities, (ii) how to produce, and (iii) for whom to produce. These arise because resources are scarce and wants are unlimited.

Long-answer questions with answer outlines

Q1. Discuss the meaning, scope and subject matter of economics.

Outline. (i) Introduction — scarcity and choice. (ii) Meaning — three definitions (wealth, welfare, scarcity) with authors. (iii) Scope — micro and macro; economics as a science and an art; a social science. (iv) Subject matter — production, exchange, distribution, consumption; the central problems. (v) Conclusion — importance for agriculture and policy.

Q2. Explain positive and normative economics with examples from Indian agriculture. Distinguish them clearly.

Outline. (i) Define positive economics with the MSP example. (ii) Define normative economics with a "should" example on MSP or subsidies. (iii) Table of differences — concern, nature, testing. (iv) Role of each in policy making. (v) Conclusion — both are needed; positive analysis informs, normative analysis guides.

LECTURE 2

Nature of Economic Theory and Basic Concepts

2.1 What is Economic Theory?

Economic theory is a body of general statements that explain how people behave in economic matters. It is built by observing human behaviour, simplifying it into assumptions, and then reasoning from those assumptions to conclusions. A theory is useful if it explains observed behaviour and predicts behaviour in new situations.

> **DEFINITION — ECONOMIC THEORY**
>
> Economic theory is a set of generalisations about economic behaviour, expressed as laws, principles or models, that help to explain and predict how individuals, firms and governments use scarce resources.

Assumptions and Abstractions

Because the real world is complicated, a theory must simplify. These simplifications are called assumptions. Common assumptions in economics include:

- **Ceteris paribus** — "other things being equal". When we say "as price rises, quantity demanded falls", we assume that income, tastes and the prices of other goods do not change.
- **Rationality** — consumers try to get the greatest satisfaction and firms try to get the greatest profit.
- **Perfect knowledge** — buyers and sellers are assumed to know the relevant prices.

Assumptions are not claims that the world is exactly like this; they are deliberate simplifications that make analysis possible. A good theory uses few assumptions and explains much.

2.2 The Rationality Assumption

> **DEFINITION — RATIONALITY**
>
> A person is said to be **rational** when, given a set of wants and a limited income, they choose the combination of goods and services that gives them the maximum satisfaction, and when a firm chooses the combination of inputs and output that gives it the maximum profit.

Rationality means that economic agents weigh benefits against costs before deciding. The rational consumer compares the satisfaction (utility) from a rupee spent on one good with that from a rupee spent on another. The rational farmer compares the extra output (marginal product) from an extra bag of fertiliser with its extra cost. Rationality does not mean selfishness; a person who gives to charity is still rational if that giving is what they most want to do with their money.

Limitations of the rationality assumption

- People do not always have full information; they may buy on habit or on the advice of others.
- Emotion, tradition and social custom influence decisions — a farmer may grow a traditional crop out of habit.

- Bounded rationality: people "satisfice" (accept a good-enough choice) rather than always optimise.

Even with these limitations, the assumption is a powerful starting point because it gives clear, testable predictions.

2.3 The Concept of Equilibrium

> **DEFINITION — EQUILIBRIUM**
>
> Equilibrium is a state of rest or balance in which no economic agent has any incentive to change their behaviour, because all forces acting on them are in balance.

In a market, equilibrium occurs at the price where the quantity demanded equals the quantity supplied. At that price there is neither a shortage nor a surplus, so buyers and sellers are content and the price tends to stay put. If price is above equilibrium, a surplus appears and price falls; if price is below equilibrium, a shortage appears and price rises. This self-correcting movement is the market's tendency toward equilibrium.

Types of equilibrium

- **Stable equilibrium** — if disturbed, forces push the system back toward the original position (a ball in a bowl).
- **Unstable equilibrium** — if disturbed, forces push the system further away (a ball on top of a hill).
- **Consumer's equilibrium** — the combination of goods that gives maximum satisfaction for a given income.
- **Producer's / firm's equilibrium** — the output at which the firm maximises profit (where marginal cost equals marginal revenue).

2.4 Economic Laws as Generalisations of Human Behaviour

> **DEFINITION — ECONOMIC LAW**
>
> An economic law is a statement of a tendency — a generalisation about how people are likely to behave under given conditions. It states what tends to happen, not what must happen in every single case.

Economic laws differ from the laws of physics in an important way. The law of gravitation operates with certainty; an economic law such as the law of demand only states a tendency, because human beings are free to act differently. Economic laws are therefore:

- **Statements of tendency**, not certainties.
- **Conditional** — they hold only when "other things remain equal".
- **Based on human behaviour**, which is variable across time and place.
- **Not exact** like physical laws, because they cannot be verified in a controlled laboratory.

Examples of economic laws: the law of demand, the law of diminishing marginal utility, the law of variable proportions, and the law of supply. All are tendencies, and all assume ceteris paribus.

2.5 Basic Concepts — Goods and Services

DEFINITIONS

A **good** is a tangible (physical) thing that satisfies a human want and has the power to command a price. A **service** is an intangible act or effort that satisfies a want — it is consumed as it is produced.

Classification of goods

- **Free goods and economic goods.** Free goods (air, sunshine) exist in abundance and command no price. Economic goods (rice, tractors) are scarce and command a price.
- **Consumer goods and producer goods.** Consumer goods satisfy wants directly (food, clothing). Producer goods (capital goods) are used to produce other goods (seeds, fertiliser, tractors).
- **Perishable, durable and non-durable goods.** Perishable goods (milk, vegetables) must be used quickly; durable goods (a tractor) last for years; non-durable goods (soap) are used up quickly.
- **Private and public goods.** Private goods are consumed by one person at a time (a bag of rice). Public goods are available to all (a village road, flood-control embankments); one person's use does not reduce another's.
- **Intermediate and final goods.** Intermediate goods are used up in producing final goods; final goods are ready for consumption.

Characteristics of goods and services

- **Utility** — the power to satisfy a want.
- **Scarcity** — limited in supply relative to demand (for economic goods).
- **Transferability** — can be transferred from one person to another.
- **Services are intangible, inseparable, perishable and variable** — for example, the advice of an agricultural extension officer is intangible, produced and consumed together, cannot be stored, and varies from officer to officer.

2.6 Desire and Want

DEFINITIONS

A **desire** is a mere wish to have something. A **want** is a desire backed by the ability and willingness to pay for it. A want, when it becomes effective and specific, becomes **demand**.

Desire is unlimited in number. Want is desire made effective by purchasing power. Demand is want backed by ability and willingness to pay, at a given price and time. Thus:

> Desire → Want (desire + ability to pay) → Demand (want + willingness to pay at a given price)

Characteristics of wants

- **Unlimited in number** — wants multiply as civilisation advances.
- **Competitive** — because means are limited, wants compete with one another for satisfaction.

- **Recurring** — many wants (food, water) repeat regularly.
- **Complementary** — some wants must be satisfied together (a car and fuel).
- **Varying in intensity** — urgent wants (food) are satisfied before less urgent ones (luxuries).
- **Grow with civilisation** — new products and habits create new wants.

2.7 Demand, Utility, Cost and Price

- **Demand** is the quantity of a good that consumers are willing and able to buy at a given price during a given period. Willingness without ability, or ability without willingness, is not demand.
- **Utility** is the power of a good or service to satisfy a human want. It is subjective — the same good gives different utility to different people, and to the same person at different times.
- **Cost of production** is the total money value of the resources used to produce a good — that is, the sum of payments to the factors of production.
- **Price** is the money value at which a good is exchanged. Price is determined by the interaction of demand and supply, and in the long run it tends to cover the cost of production.

Price ⟷ Demand and Supply | Normal price tends to equal Cost of Production

2.8 Wealth, Capital, Income and Welfare

DEFINITIONS

Wealth is the stock of all goods and assets that have value and can be exchanged for money. **Capital** is that part of wealth which is used to produce further wealth. **Income** is the flow of money or goods received over a period. **Welfare** is the state of well-being of a person or society.

Wealth — characteristics

- It is a **stock** measured at a point of time (as on 31 March).
- It must have **utility**, **scarcity** and **transferability**.
- It may be **material** (land, buildings, machinery) or **personal** (skills, health) or **national** (roads, forests).

Capital — characteristics

- It is a **produced means of production** — man-made, not a free gift of nature.
- It is **productive** — it yields a return (interest) and helps create more output.
- It is **mobile** (can be moved) and **depreciable** (wears out over time).
- It is created by **saving and investment**; it requires time to build up.
- In farming, capital includes tractors, pumpsets, irrigation structures, and livestock used for work.

Income — characteristics

- It is a **flow** measured over a period (per month, per year).
- It arises from work (wages), property (rent), capital (interest) or enterprise (profit).

- Income may be **money income** or **real income** (the goods and services money can buy).

Table 2.1 — Stock versus Flow, and Wealth versus Income

Concept	Nature	Meaning	Agricultural example
Wealth	Stock	Assets with value at a point of time	Farm land, tractor, livestock owned on a date
Capital	Stock	Wealth used to produce more wealth	Pumpset, irrigation channel
Income	Flow	Receipts over a period	Paddy sale proceeds during a year
Welfare	State	Level of well-being	Household's nutrition and health status

Welfare is broader than income. Income is a means to welfare, but welfare also depends on health, education, leisure, the environment and the distribution of income. This is why economists distinguish between growth (rising income) and development (rising welfare).

Worked Example 2.1 — Classifying concepts

Problem. Classify each item as stock or flow: (a) a farmer's herd of 20 cows, (b) monthly milk income, (c) a bank loan of ₹50,000 outstanding, (d) annual interest paid on that loan.

Solution. (a) Stock — measured at a point of time. (b) Flow — received per month. (c) Stock — the outstanding balance on a date. (d) Flow — paid over a period. **Rule:** if it is measured "as on a date", it is a stock; if "per month/per year", it is a flow.

2.9 Economics as a Science and as an Art

Economics is both a science and an art. As a **science**, it is a systematic body of knowledge based on observation and reasoning. As an **art**, it puts that knowledge to practical use in solving real problems of policy and business.

- **As a positive science**, economics describes and explains economic behaviour (what is).
- **As a normative science**, it prescribes what ought to be done to achieve a goal.
- **As an art**, it guides the farmer, the firm and the government in taking decisions.

Economics differs from the natural sciences because it studies human beings who can change their behaviour. Its conclusions are therefore tendencies rather than exact laws. This is why economics is called a **social science**.

2.10 Consumer Goods versus Producer Goods

Goods can be grouped by the purpose they serve. This distinction matters because the demand for producer goods is a **derived demand** — it comes from the demand for the final goods they help produce.

Table 2.2 — Consumer goods versus producer goods

Basis	Consumer goods	Producer (capital) goods
Purpose	Satisfy wants directly	Used to produce other goods
Demand	Direct demand	Derived demand

Basis	Consumer goods	Producer (capital) goods
Durability	May be durable or perishable	Usually durable
Example (farm)	Rice, milk, vegetables	Seed, fertiliser, tractor, pumpset
Effect of income	Rises with income (normal goods)	Depends on the demand for final goods

Exam Focus — Lecture 2

Key points to remember

- Economic theory simplifies reality using assumptions such as ceteris paribus and rationality.
- Rationality means maximising satisfaction (consumer) or profit (firm).
- Equilibrium is a state of rest where no agent wants to change behaviour; market equilibrium occurs where demand equals supply.
- Economic laws are statements of tendency, conditional and not exact.
- Goods are classified as free/economic, consumer/producer, perishable/durable, private/public.
- Desire + ability to pay = want; want + willingness to pay = demand.
- Wealth, capital and income: the first two are stocks, income is a flow; welfare is broader than income.

Short-answer questions with model answers

Q1. What is meant by ceteris paribus?

Model answer. Ceteris paribus is a Latin phrase meaning "other things being equal". It is an assumption used in economic analysis which holds all other relevant factors constant while studying the effect of one factor. For example, the law of demand states that, other things being equal, quantity demanded falls as price rises.

Q2. Distinguish between a good and a service.

Model answer. A good is a tangible thing that satisfies a want and can be stored and transferred (rice, a tractor). A service is an intangible act that satisfies a want, is produced and consumed at the same time, cannot be stored, and varies in quality (the advice of an extension officer).

Q3. Define wealth and capital. How do they differ?

Model answer. Wealth is the stock of all goods and assets that have value. Capital is that part of wealth which is used to produce further wealth. All capital is wealth, but all wealth is not capital: a house used for living is wealth but not capital, whereas a tractor used for ploughing is both wealth and capital.

Long-answer questions with answer outlines

Q1. Explain the nature of economic theory, including the rationality assumption and the concept of equilibrium.

Outline. (i) Economic theory as generalisation and abstraction. (ii) Assumptions — ceteris paribus, rationality, perfect knowledge. (iii) Rationality — meaning, consumer and firm, limitations. (iv) Equilibrium — definition, market equilibrium, types, stable/unstable. (v) Conclusion.

Q2. Discuss the basic concepts of goods, wants, demand, utility, cost, price, wealth, capital, income and welfare.

Outline. Define each concept in turn with an agricultural example. Use the table on stock versus flow. Emphasise the desire → want → demand chain and the distinction between income and welfare.

LECTURE 3

Agricultural Economics: Meaning, Importance and Role

3.1 Meaning of Agricultural Economics

Agricultural economics is the branch of economics that applies economic principles to the special problems of agriculture. It studies how farmers and the agricultural sector use scarce resources — land, labour, capital and management — to produce food and fibre, and how those products are marketed and consumed.

> **DEFINITION — AGRICULTURAL ECONOMICS**
>
> Agricultural economics is that branch of economics which deals with the allocation of scarce resources in agriculture and with the production, distribution and consumption of agricultural products. It is the application of the principles of economics to the production and distribution of food and fibre.

It is both a **science** (it uses economic theory and statistical methods) and an **art** (it guides farm and policy decisions). Because agriculture has features that differ from industry, general economics alone is not enough; the principles must be adapted to farming conditions.

3.2 Characteristics of Agriculture

Agriculture is different from industry in several important ways. These differences explain why agricultural economics is a distinct subject.

- **Dependence on nature.** Output depends on rainfall, temperature, soil and pests — factors the farmer cannot fully control. This makes farm output uncertain and risky.
- **Biological process and time lag.** Crops and livestock take time to grow. The farmer must decide today, but the result comes months later, so supply cannot be adjusted quickly.
- **Seasonality.** Production and income are seasonal; most income comes at harvest, but expenses continue throughout the year. This creates a need for credit.
- **Perishability of produce.** Milk, fruits and vegetables spoil quickly, forcing early sale, often at low prices.
- **Small and scattered holdings.** In India most farms are small and fragmented, which raises the cost of mechanisation and marketing.
- **Land as an essential, fixed factor.** The area of land is fixed; output is increased mainly by using more labour, capital and technology on the same land.
- **Family labour and unorganised nature.** A large part of farm work is done by family members, and the sector is highly unorganised compared with industry.
- **Law of diminishing returns operates strongly.** On a fixed piece of land, adding more and more input eventually yields less extra output.

- **Joint products and by-products.** One crop may give several products (paddy gives rice and straw; cattle give milk, dung and calves).
- **Price and income instability.** Inelastic demand for food and the cobweb effect cause farm prices to swing widely.

Table 3.1 — Agriculture versus Industry (why agriculture is a special subject)

Basis	Agriculture	Industry
Nature of process	Biological; depends on nature	Mechanical; controlled environment
Time lag	Long and fixed (season to harvest)	Short and adjustable
Risk	High (weather, pests, disease)	Lower; more predictable
Supply response	Slow; cannot change output quickly	Fast; output can be raised quickly
Unit size	Small, scattered holdings	Large, concentrated units
Organisation	Largely unorganised	Organised, corporatised
Produce	Perishable and bulky	Often storable

3.3 Importance of Agriculture in the Indian Economy

Agriculture is the backbone of the Indian economy. Its importance can be grouped under economic, social and environmental heads.

Economic importance

- **Share in national income.** Agriculture and allied activities contribute a significant share of India's gross value added, though this share has fallen over the decades as industry and services have grown.
- **Employment.** Agriculture is the largest employer in the country, providing livelihood directly or indirectly to nearly half the workforce.
- **Source of food and raw material.** It feeds the nation and supplies raw material to sugar, textile, jute, edible-oil and food-processing industries.
- **Source of foreign exchange.** Agricultural and processed products are important exports — rice, spices, tea, coffee, cotton and marine products.
- **Market for industrial goods.** Farmers buy fertiliser, tractors, pumpsets and consumer goods, providing a market for industry.
- **Capital formation and savings.** Agriculture provides savings that can be invested in other sectors.

Social and environmental importance

- **Rural livelihood and poverty reduction.** Most of the rural poor depend on agriculture; raising farm incomes is central to reducing poverty.
- **Food security.** Domestic production insulates the country from global food shocks.
- **Environmental services.** Agriculture manages soil, water and biodiversity, and affects the climate.

> **Tamil Nadu context.** Tamil Nadu is a leading producer of rice, sugarcane, cotton, banana, coconut and pulses. Delta districts such as Thanjavur and Tiruvarur are the rice bowl of the state, while the Cauvery delta and the Thamirabarani basin support intensive cultivation. The state's **Uzhavar Sandhai** (farmers' markets) allow growers to sell fruits and vegetables directly to consumers, reducing the middlemen's margin.

3.4 Role of Agriculture in Economic Development

In the early stages of development, agriculture plays several roles that support the growth of the whole economy. Economists describe these as the "contributions" of agriculture.

- **Product contribution.** It supplies food to a growing non-farm population and raw materials to industry.
- **Factor contribution.** It releases labour and capital to other sectors as productivity rises.
- **Market contribution.** Rural households demand industrial goods, creating a domestic market.
- **Foreign-exchange contribution.** Agricultural exports earn the foreign exchange needed to import machinery and technology.

As an economy develops, the share of agriculture in national income and employment falls — a pattern noted by economists and confirmed by India's own experience. But the absolute importance of agriculture does not fall; a growing economy needs a growing food supply. This is why agricultural development remains central even in a modern economy.

3.5 Agricultural Planning and Development in India

Agricultural planning in India began in earnest with the Five Year Plans after independence. The aim shifted over time from raising food-grain output to ensuring food security, raising farm incomes and promoting sustainable agriculture.

Major milestones

- **Green Revolution (mid-1960s onwards).** Introduction of high-yielding varieties of wheat and rice, along with assured irrigation and fertiliser, made India self-sufficient in food grains.
- **White Revolution (Operation Flood).** Made India the largest producer of milk through cooperative dairying.
- **Institutional credit.** Expansion of cooperative banks, regional rural banks and later the **Kisan Credit Card (KCC)**, which gives farmers timely and flexible credit for crop production and inputs.
- **Minimum Support Price (MSP) and procurement.** A guaranteed floor price for major crops, with public procurement for the food-security system.
- **Marketing reforms.** Model APMC Act, direct marketing, contract farming and the electronic National Agriculture Market (**eNAM**) to create a unified national market.
- **Mission-mode programmes.** National Food Security Mission, Rashtriya Krishi Vikas Yojana, Pradhan Mantri Krishi Sinchayee Yojana (irrigation), and soil-health and crop-insurance schemes.
- **Income support.** Pradhan Mantri Kisan Samman Nidhi (PM-KISAN) provides direct income support to small and marginal farmers.

Objectives of agricultural planning

- Attaining self-sufficiency in food grains and ensuring national food security.
- Increasing agricultural productivity per hectare and per worker.
- Reducing rural poverty and raising the standard of living of farmers.
- Providing adequate and timely agricultural credit and crop insurance.
- Promoting sustainable use of land, water and biodiversity.

3.6 Role of Economics in Agricultural Development Policies

Economics provides the analytical foundation for designing and evaluating agricultural policies. Its role can be seen in several areas.

- **Price policy.** Economics helps fix the MSP, set procurement prices and decide on buffer stocks, using the concepts of demand, supply and elasticity.
- **Resource allocation.** It guides the efficient allocation of land, water and credit among crops and regions.
- **Cost–benefit analysis.** It evaluates irrigation projects, subsidies and schemes to see whether benefits exceed costs.
- **Credit and finance.** It designs interest subvention, credit limits and repayment schedules suited to the seasonality of farming.
- **Marketing and trade.** It analyses market structure, price spread and the effects of import and export policy on farmers.
- **Welfare and equity.** It assesses who gains and who loses from a policy, so that support can be targeted to small and marginal farmers.

Worked Example 3.1 — Computing the share of agriculture in the economy

Problem. In a district, the gross value added (GVA) of agriculture is ₹1,200 crore and the total GVA of the district is ₹3,000 crore. Agriculture employs 60% of the workers. Comment on the structure of the district economy.

Solution.

- Share of agriculture in GVA = (1,200 / 3,000) × 100 = **40%**.
- Share of agriculture in employment = **60%**.
- **Comment.** Agriculture produces 40% of output but employs 60% of workers. This means productivity per worker in agriculture is lower than in the non-farm sectors, so there is "disguised unemployment" — more people work on farms than are really needed. Policy should raise farm productivity and create non-farm rural jobs.

3.7 Problems of Indian Agriculture

Despite its importance, Indian agriculture faces several persistent problems that policy must address.

- **Small and fragmented holdings** that raise the cost of mechanisation and irrigation.
- **Low productivity per hectare and per worker** compared with developed countries.
- **Dependence on the monsoon** and inadequate irrigation in many regions.
- **Indebtedness** and dependence on costly informal credit.
- **Poor marketing infrastructure** — storage, transport and processing gaps that cause post-harvest losses.
- **Price and income instability** caused by inelastic demand and supply shocks.
- **Disguised unemployment** — too many workers on too little land.
- **Land degradation** and declining soil health due to overuse of inputs.

Addressing these problems requires investment in irrigation and storage, credit and insurance, extension services, market reform and the promotion of farmer producer organisations.

Exam Focus — Lecture 3

Key points to remember

- Agricultural economics applies economic principles to farming and the food economy.
- Agriculture is special because it depends on nature, has a time lag, is seasonal, risky and small-scale.
- Its importance: national income, employment, food, raw material, exports, market and savings.
- Its contributions to development: product, factor, market and foreign exchange.
- Agricultural planning milestones: Green Revolution, Operation Flood, KCC, MSP, eNAM, PM-KISAN.
- Economics guides price policy, resource allocation, cost–benefit analysis, credit, marketing and equity.
- As the economy develops, the share of agriculture in income and employment falls, but its absolute importance remains.

Short-answer questions with model answers

Q1. Define agricultural economics.

Model answer. Agricultural economics is the branch of economics that deals with the allocation of scarce resources in agriculture and with the production, distribution and consumption of agricultural products. It applies economic principles to the special conditions of farming.

Q2. Why is agriculture a special subject compared with industry?

Model answer. Because agriculture depends on nature, has a long biological time lag, is seasonal and risky, produces perishable goods, is carried on in small scattered holdings, and responds slowly to price changes. These features require principles different from those used for industry.

Q3. What is the role of the Kisan Credit Card?

Model answer. The KCC provides farmers timely and flexible credit for crop production, inputs and household needs through a simple, renewable card, reducing dependence on costly informal lenders and suiting the seasonal pattern of farm expenses.

Long-answer questions with answer outlines

Q1. Explain the importance of agriculture in the Indian economy and its role in economic development.

Outline. (i) Introduction — agriculture as the backbone. (ii) Economic importance — income, employment, food, raw material, exports, market, savings. (iii) Contributions to development — product, factor, market, foreign exchange. (iv) Social and environmental importance. (v) Tamil Nadu examples — Uzhavar Sandhai, delta agriculture. (vi) Conclusion.

Q2. Discuss agricultural planning and development in India, and the role of economics in agricultural development policies.

Outline. (i) Meaning and objectives of agricultural planning. (ii) Milestones — Green Revolution, Operation Flood, credit and KCC, MSP, eNAM, mission programmes, PM-KISAN. (iii) Role of eco-

nomics — price policy, resource allocation, cost–benefit analysis, credit, marketing, welfare. (iv) Conclusion.

LECTURE 4

Utility Theory: Cardinal and Ordinal Approaches

4.1 Meaning of Utility

DEFINITION — UTILITY

Utility is the power or capacity of a good or service to satisfy a human want. It is subjective and depends on time, place and person.

Utility is not the same as usefulness. A cigarette may have utility for a smoker though it is not useful for health. Utility is measured by the satisfaction a consumer feels, and it changes from person to person and from moment to moment.

Types of utility

- **Total utility (TU)** — the sum of satisfaction obtained from all units of a commodity consumed.
- **Marginal utility (MU)** — the additional satisfaction obtained from consuming one more unit of a commodity.
- **Average utility (AU)** — total utility divided by the number of units consumed.

$$MU_n = TU_n - TU_{n-1} \quad | \quad TU = \Sigma\, MU \quad | \quad AU = TU / Q$$

4.2 Cardinal Utility Approach

The cardinal (or marginal utility) approach was developed by Alfred Marshall. It assumes that utility can be measured in definite numbers called **utils**, just as weight is measured in kilograms.

Assumptions of cardinal utility

- Utility is measurable in cardinal numbers (1, 2, 3 ...).
- Utility is additive — total utility is the sum of marginal utilities.
- The consumer is rational and wants to maximise total utility.
- Income is limited and prices are given.
- Marginal utility of money is constant.

4.3 Law of Diminishing Marginal Utility

STATEMENT

The law of diminishing marginal utility states that, other things being equal, as a consumer consumes more and more units of a commodity, the marginal utility obtained from each successive unit goes on falling.

Suppose a thirsty person drinks glasses of water. The first glass gives great satisfaction, the second less, the third still less. After some point, an additional glass gives zero utility (the person is fully satisfied) and beyond that it may give negative utility (discomfort). This is the law in everyday life.

Assumptions of the law

- The units consumed are standard and homogeneous.
- Consumption is continuous without long gaps.
- The consumer's tastes and preferences do not change.
- Utility is measurable and the marginal utility of money is constant.

Exceptions (where the law may not hold)

- Hobbies and collections (a stamp collector may value each new stamp more).
- Drinking and gambling, where the habit grows.
- Misunderstanding of the good — a person may enjoy an item more as they learn to use it.

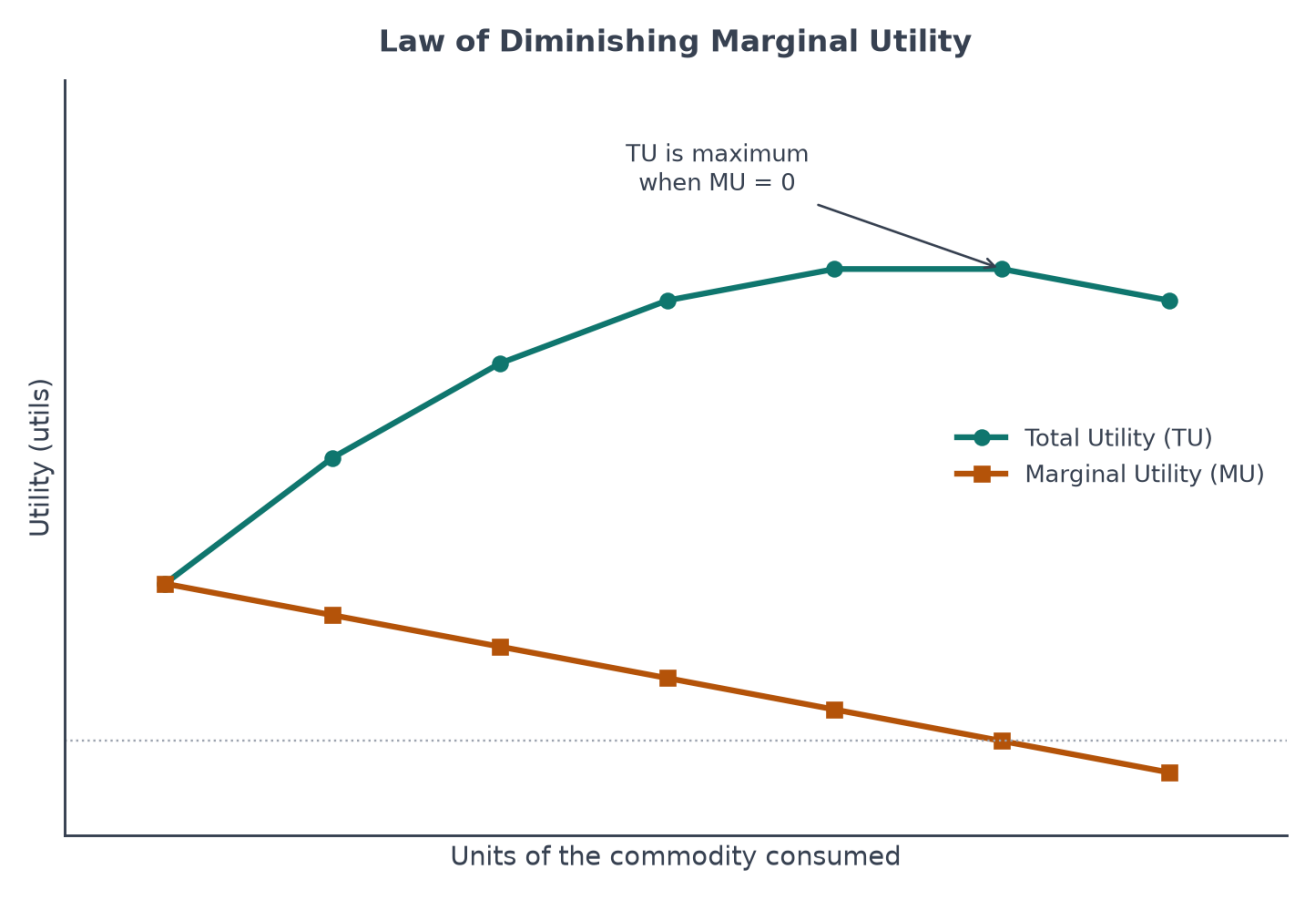


Figure 4.1 — Total utility rises while marginal utility is positive, reaches a maximum when marginal utility is zero, and falls when marginal utility becomes negative.

The table below shows the relationship numerically. Total utility rises as long as marginal utility is positive; it is maximum where marginal utility is zero; and it falls when marginal utility becomes negative.

Table 4.1 — Total utility and marginal utility schedule

Units consumed (Q)	Total utility (TU)	Marginal utility (MU)
1	20	20
2	36	16
3	48	12

Units consumed (Q)	Total utility (TU)	Marginal utility (MU)
4	56	8
5	60	4
6	60	0
7	56	−4

4.4 Law of Equi-Marginal Utility (Principle of Maximum Satisfaction)

> **STATEMENT**
>
> The law of equi-marginal utility states that a consumer maximises satisfaction by spending income in such a way that the marginal utility of the last rupee spent on each commodity is the same.

A consumer with limited income and several goods will shift spending from goods where a rupee yields less utility to goods where a rupee yields more utility, until the marginal utilities per rupee are equal. At that point, no further rearrangement can raise total satisfaction.

$$MU_x / P_x = MU_y / P_y = MU_z / P_z = \ldots = MU \text{ of money}$$

Example. If a rupee spent on bananas gives 10 utils and a rupee spent on milk gives 6 utils, the consumer will buy more bananas and less milk. As more bananas are bought, their marginal utility falls, until both give equal utility per rupee. This is the point of maximum satisfaction.

Worked Example 4.1 — Applying equi-marginal utility

Problem. A consumer has ₹7 to spend on two goods, X (price ₹2) and Y (price ₹1). The marginal utilities are given. Find the combination that maximises utility.

Units	MU of X (₹2 each)	MU of Y (₹1 each)
1	20	9
2	16	8
3	12	6
4	8	5
5	4	4

Solution. We compare MU per rupee: for X, MU÷2; for Y, MU÷1.

- 1st X gives 10 per rupee; 1st Y gives 9 per rupee → buy X first (10 > 9).
- Now compare 2nd X (16÷2 = 8) with 1st Y (9) → buy Y (9 > 8).
- Compare 2nd X (8) with 2nd Y (8) → equal; buy either. Buy 2nd X.
- Compare 3rd X (12÷2 = 6) with 2nd Y (8) → buy 2nd Y.
- Compare 3rd X (6) with 3rd Y (6) → equal; buy 3rd Y.

Spending: 2 X = ₹4 and 3 Y = ₹3, total ₹7. Marginal utility per rupee: X = 6, Y = 6. They are equal, so this is the point of maximum satisfaction. **Answer: 2 units of X and 3 units of Y.**

4.5 Ordinal Utility Approach — Indifference Curve Analysis

The ordinal approach, developed by J. R. Hicks and R. G. D. Allen, does not measure utility in numbers. It only requires that the consumer can **rank** combinations of goods as more preferred, less preferred, or equally preferred.

> **DEFINITION — INDIFFERENCE CURVE**
>
> An indifference curve is a curve that shows all the different combinations of two goods which give the consumer the same level of satisfaction, so that the consumer is indifferent between them.

Assumptions of indifference curve analysis

- The consumer is rational and ranks preferences consistently.
- Utility is ordinal — only ranking is needed, not measurement.
- The two goods are substitutable for each other.
- Consumer's income, tastes and prices of the two goods are given.

Properties of indifference curves

- **Downward sloping.** To keep satisfaction constant, if the consumer has more of one good they must have less of the other.

- **Convex to the origin.** This follows from the diminishing marginal rate of substitution.
- **Higher curves mean higher satisfaction.** An indifference curve to the right represents a larger bundle and hence greater utility.
- **Indifference curves never intersect.** If two curves crossed, the same bundle would give two different levels of satisfaction, which is a contradiction.
- **Indifference curves do not touch the axes** (under the usual assumption that the consumer wants some of both goods).

MARGINAL RATE OF SUBSTITUTION (MRS)

The MRS of X for Y is the number of units of Y the consumer is willing to give up to obtain one more unit of X, while remaining equally satisfied. It is the slope of the indifference curve, and it diminishes as we move down the curve.

4.6 Budget Line

DEFINITION — BUDGET LINE

The budget line (or price line) shows all the combinations of two goods that a consumer can buy with a given income at given prices. It is the boundary of what is affordable.

$$P_x \cdot Q_x + P_y \cdot Q_y = M \quad | \quad \text{Slope of budget line} = -P_x / P_y$$

Assumptions of the budget line

- The consumer's income is fixed.
- The prices of the two goods are given and constant.
- The consumer spends the whole income on the two goods.

Changes in the budget line

- **Change in income** shifts the whole line outward (more income) or inward (less income), without changing its slope.
- **Change in the price of one good** rotates the line — a fall in the price of X makes the line steeper and extends the X-intercept.

Limitations of indifference curve analysis

- It assumes the consumer knows and can rank all combinations perfectly, which may not hold.
- It deals with only two goods at a time; real consumers choose among many.
- Indifference curves are difficult to draw or observe in practice.
- It assumes consistent, rational preferences, which behavioural economics questions.

Applications

- Explaining consumer's equilibrium and deriving the demand curve (Lecture 5).
- Showing the effect of a change in income or price on the quantity bought.

- Analysing the choice between work and leisure, and between present and future consumption.
- Evaluating the effect of subsidies and taxes on a household's buying pattern.

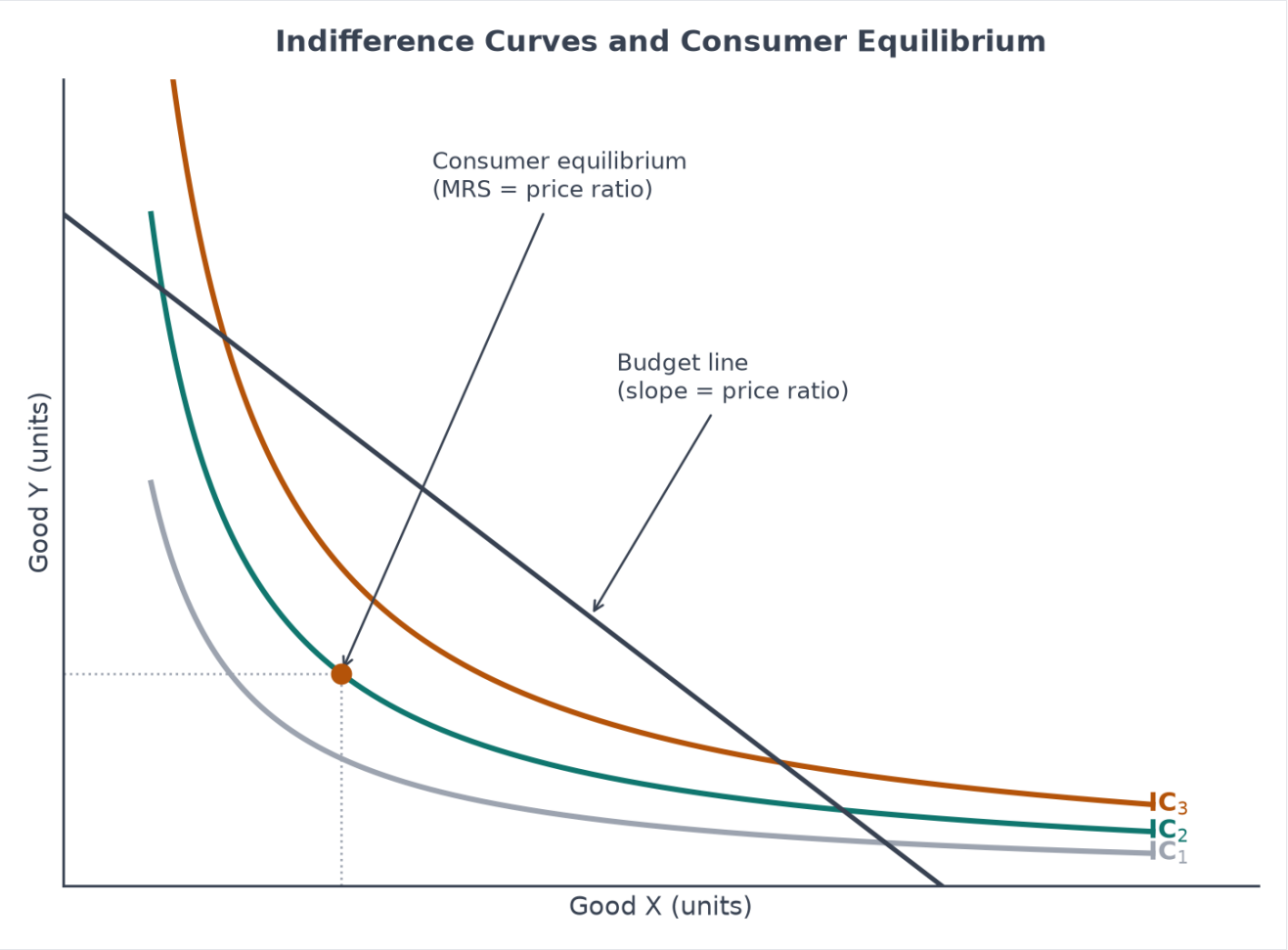


Figure 4.2 — Three indifference curves and a budget line. The consumer is in equilibrium at the point where the budget line is tangent to the highest attainable indifference curve (IC_2), where the MRS equals the price ratio.

4.7 Criticism of the Cardinal Approach and Importance of the Ordinal Approach

The cardinal approach assumes that utility can be measured in exact numbers. Critics point out several difficulties:

- Utility is subjective and psychological; it cannot be measured in numbers like weight or length.
- The unit "util" has no objective standard.
- Utility is not always additive; the satisfaction from goods taken together may differ from the sum of separate satisfactions.
- The marginal utility of money is not constant; it falls as income rises.

Because of these difficulties, the ordinal approach of Hicks and Allen became the accepted method. It needs only that the consumer can **rank** combinations, which is a much weaker and more realistic assumption.

Importance of indifference curve analysis

- It explains consumer's equilibrium without measuring utility.
- It shows the effect of a change in income (income effect) and of a change in price (price effect) on demand.
- It is used to analyse the choice between work and leisure, and between present and future consumption.
- It helps evaluate the welfare effect of subsidies, taxes and price controls.

Exam Focus — Lecture 4

Key points to remember

- Utility is the power to satisfy a want; total utility is the sum of marginal utilities.
- Cardinal approach measures utility in utils (Marshall); ordinal approach only ranks preferences (Hicks).
- Law of diminishing marginal utility: MU falls as more units are consumed; TU is maximum where MU = 0.
- Equi-marginal utility: $MU_x/P_x = MU_y/P_y$ gives maximum satisfaction.
- Indifference curves are downward sloping, convex to the origin, never intersect, and higher curves mean higher satisfaction.
- The budget line shows affordable combinations; its slope is the price ratio.
- Consumer equilibrium: budget line tangent to the highest indifference curve (MRS = price ratio).

Short-answer questions with model answers

Q1. State the law of diminishing marginal utility.

Model answer. Other things being equal, as a consumer consumes more units of a commodity, the marginal utility from each successive unit falls. Total utility rises while marginal utility is positive, is maximum when marginal utility is zero, and falls when marginal utility is negative.

Q2. What is an indifference curve? Give two of its properties.

Model answer. An indifference curve shows all combinations of two goods that give the consumer equal satisfaction. Two properties: it slopes downward (more of one good requires less of the other), and it is convex to the origin because the marginal rate of substitution diminishes.

Q3. State the law of equi-marginal utility.

Model answer. A consumer maximises satisfaction by allocating income so that the marginal utility of the last rupee spent on each commodity is equal, i.e. $MU_x/P_x = MU_y/P_y$.

Long-answer questions with answer outlines

Q1. Explain the law of diminishing marginal utility with a schedule and diagram, and state its assumptions and exceptions.

Outline. (i) Define utility, TU and MU. (ii) State the law. (iii) Schedule (Table 4.1) and diagram (Figure 4.1). (iv) Assumptions. (v) Exceptions. (vi) Importance.

Q2. Explain indifference curve analysis and the budget line, and show consumer's equilibrium.

Outline. (i) Ordinal approach and definition of indifference curve. (ii) Properties with reasons. (iii) Budget line, equation, slope, shifts. (iv) Equilibrium at tangency (Figure 4.2). (v) Limitations. (vi) Applications.

LECTURE 5

Consumer's Equilibrium, Demand and Consumer Surplus

5.1 Consumer's Equilibrium

DEFINITION — CONSUMER'S EQUILIBRIUM

Consumer's equilibrium is that situation in which a consumer, with a given income and given prices, buys such a combination of goods that gives maximum satisfaction and from which there is no tendency to change.

There are two ways to explain consumer's equilibrium, matching the two utility approaches.

Equilibrium under the cardinal (marginal utility) approach

The consumer reaches equilibrium when the marginal utility per rupee spent is equal for all goods, and total spending equals income:

$$MU_x / P_x = MU_y / P_y = MU \text{ of money}$$

If MU_x/P_x is greater than MU_y/P_y, the consumer buys more of X and less of Y. As more X is bought its marginal utility falls, and the two ratios become equal. At that point equilibrium is reached.

Equilibrium under the ordinal (indifference curve) approach

The consumer is in equilibrium at the point where the budget line is **tangent** to the highest attainable indifference curve. At this tangency point, the marginal rate of substitution equals the price ratio:

$$MRS_{xy} = P_x / P_y$$

At any other point on the budget line, the consumer is on a lower indifference curve, or the slope of the indifference curve differs from the price ratio, giving an incentive to move. The tangency point is therefore the point of maximum satisfaction (Figure 4.2 of Lecture 4).

5.2 Derivation of the Demand Curve

The demand curve can be derived from the indifference curve diagram by changing the price of one good and noting the resulting equilibrium quantities.

Steps in deriving the demand curve

1. Start with an initial budget line for good X and find the tangency point with an indifference curve. This gives the quantity of X demanded at the initial price.
2. Suppose the price of X falls. The budget line rotates outward along the X-axis (the consumer can now afford more X).
3. A new tangency point with a higher indifference curve gives a larger quantity of X.
4. Repeat for another price fall to get a third quantity.

5. Plot each price against the corresponding quantity demanded. The resulting curve slopes downward — this is the demand curve.

The movement from a higher price to a lower price causes two effects. The **substitution effect** makes X cheaper relative to Y, so the consumer substitutes X for Y. The **income effect** means the consumer's real purchasing power rises, so more of both goods is bought. Both effects normally raise the quantity of X demanded as its price falls, giving the downward-sloping demand curve.

5.3 Demand — Meaning

> **DEFINITION — DEMAND**
>
> Demand means the quantity of a commodity that a consumer is willing and able to buy at a given price during a given period of time. Demand is a want backed by ability and willingness to pay.

Three things must be present for demand: **desire**, **ability to pay**, and **willingness to pay**, along with a given price and time. Demand is always expressed with reference to a price and a period.

Kinds of demand

Table 5.1 — Kinds of demand

Kind of demand	Meaning	Example
Price demand	Quantity demanded at a given price, other things unchanged	Rice demanded at ₹30/kg
Income demand	Quantity demanded at a given income	More milk demanded as income rises
Cross demand	Quantity of one good demanded as the price of a related good changes	Tea demanded when coffee price rises
Direct (autonomous) demand	Demand for goods that satisfy wants directly	Demand for food
Derived demand	Demand for goods used to produce other goods	Demand for fertiliser and tractors
Joint (complementary) demand	Goods demanded together	Car and fuel; seed and fertiliser
Composite demand	One good demanded for several uses	Milk for drinking, curd, butter, ghee
Competitive demand	Demand for substitutes	Tea and coffee; rice and wheat

5.4 Law of Demand

STATEMENT — LAW OF DEMAND

Other things being equal, when the price of a commodity rises its quantity demanded falls, and when the price falls its quantity demanded rises. There is an inverse relationship between price and quantity demanded.

Assumptions of the law of demand

- Income of the consumer does not change.
- Tastes and preferences do not change.
- Prices of related goods do not change.
- No expectation of future price change.
- The commodity is a normal good.

Reasons for the downward-sloping demand curve

- **Income effect.** A fall in price raises real income, so more is bought.
- **Substitution effect.** A cheaper good is substituted for costlier substitutes.
- **New buyers.** A lower price brings in buyers who could not afford the good before.
- **Diminishing marginal utility.** To induce the consumer to buy more, the price must fall as marginal utility falls.

Exceptions to the law of demand

- **Giffen goods.** For very inferior staple foods, a fall in price may reduce quantity demanded because real income rises and people switch to better food.
- **Conspicuous (Veblen) goods.** Prestige goods such as diamonds may be bought more when they become more expensive, because high price signals status.
- **Expectation of further price change.** If people expect prices to rise, they may buy more now even at a higher price.
- **Ignorance and habit.** Consumers may buy the same quantity regardless of price.

Demand schedule and demand curve

Table 5.2 — Individual demand schedule

Price of rice (₹/kg)	Quantity demanded (kg/month)
50	10
40	20
30	30
20	40
10	50



Figure 5.1 — The demand curve plots the price on the vertical axis against quantity demanded on the horizontal axis. It slopes downward from left to right, showing the inverse price–quantity relationship.

Determinants of demand

- **Price of the commodity** — the main determinant; higher price, lower demand.
- **Income of the consumer** — for normal goods, higher income raises demand.
- **Prices of related goods** — substitutes (higher price of a substitute raises demand) and complements (higher price of a complement lowers demand).
- **Tastes and preferences** — advertising and fashion change demand.
- **Expectations of future prices** — expected price rise raises present demand.
- **Population and its composition** — more people mean more demand.
- **Distribution of income** — a more equal distribution can raise demand for essentials.
- **Government policy** — taxes raise the effective price; subsidies lower it.

> **Movement along versus shift of the demand curve.** A change in the commodity's own price causes a **movement along** the demand curve (a change in quantity demanded). A change in any other determinant (income, tastes, related prices) causes a **shift** of the whole demand curve (a change in demand).

5.5 Consumer Surplus

DEFINITION — CONSUMER SURPLUS

Consumer surplus is the difference between what a consumer is willing to pay for a good and what they actually pay. It is the excess of total utility over total expenditure.

$$\text{Consumer surplus} = \text{Willingness to pay} - \text{Actual payment} = \text{Total utility} - (\text{Price} \times \text{Quantity})$$

On a demand curve, consumer surplus is the area under the demand curve and above the price line. If the market price is uniform, consumers who would have paid more enjoy a surplus.

Importance of consumer surplus

- It measures the **welfare gain** consumers obtain from a market.
- It helps evaluate the effect of **price changes and government policies** — for example, a subsidy raises consumer surplus.
- It is used in **cost–benefit analysis** of public projects.
- It explains why public services (like a free irrigation canal or a public park) provide large value even when priced low or free.
- It helps in **price discrimination** and in setting utility tariffs.

Worked Example 5.1 — Consumer surplus from a demand schedule

Problem. A consumer's demand schedule for a good is given. The market price is ₹30. Find the consumer surplus.

Quantity	1	2	3	4	5
Price willing to pay (₹)	50	45	35	30	20

Solution. At a market price of ₹30, the consumer buys 4 units (the fifth unit is worth only ₹20, which is below ₹30, so it is not bought).

- Surplus on 1st unit = 50 − 30 = ₹20
- Surplus on 2nd unit = 45 − 30 = ₹15
- Surplus on 3rd unit = 35 − 30 = ₹5
- Surplus on 4th unit = 30 − 30 = ₹0

Total consumer surplus = 20 + 15 + 5 + 0 = **₹40**.

Check by the formula. Total willingness to pay = 50 + 45 + 35 + 30 = ₹160. Actual payment = 4 × 30 = ₹120. Consumer surplus = 160 − 120 = **₹40**. Both methods agree.

5.6 Demand, Elasticity and Total Revenue

The relationship between price elasticity and total revenue is an important practical tool.

Table 5.3 — Price change, elasticity and total revenue

Elasticity	If price rises	If price falls
Elastic ($E_d > 1$)	TR falls	TR rises
Unit elastic ($E_d = 1$)	TR unchanged	TR unchanged
Inelastic ($E_d < 1$)	TR rises	TR falls

This explains a central fact of agriculture: because the demand for food is inelastic, a bumper harvest that lowers the price can actually reduce farmers' total income. This is the "paradox of plenty", and it is the main economic reason for MSP and price-support policies.

Exam Focus — Lecture 5

Key points to remember

- Consumer's equilibrium: MU per rupee equal across goods (cardinal), or MRS = price ratio (ordinal).
- The demand curve is derived by changing price and noting successive tangency points.
- Demand = desire + ability + willingness to pay, at a given price and time.
- Kinds of demand: price, income, cross, direct, derived, joint, composite, competitive.
- Law of demand: inverse price–quantity relation, other things equal; reasons are income effect, substitution effect, new buyers, diminishing marginal utility.
- Exceptions: Giffen goods, Veblen goods, expectations, habit.
- Consumer surplus = willingness to pay − actual payment; it measures consumer welfare.

Short-answer questions with model answers

Q1. State the law of demand and give two exceptions.

Model answer. Other things being equal, quantity demanded rises as price falls and falls as price rises. Exceptions: (i) Giffen goods, where a price fall may reduce demand for an inferior staple; (ii) Veblen (conspicuous) goods, where a higher price increases demand because of prestige.

Q2. What is consumer surplus?

Model answer. Consumer surplus is the difference between what a consumer is willing to pay and what they actually pay. On a demand diagram it is the area under the demand curve and above the price line. It measures the welfare consumers gain from a purchase.

Q3. Distinguish between a movement along and a shift of the demand curve.

Model answer. A change in the commodity's own price causes a movement along the same demand curve (change in quantity demanded). A change in any other determinant such as income, tastes or related prices causes the entire demand curve to shift (change in demand).

Long-answer questions with answer outlines

Q1. Explain consumer's equilibrium under both the cardinal and ordinal approaches, and derive the demand curve.

Outline. (i) Meaning of equilibrium. (ii) Cardinal condition $MU_x/P_x = MU_y/P_y$. (iii) Ordinal condition MRS = price ratio, tangency. (iv) Derivation of the demand curve by changing price. (v) Conclusion.

Q2. Discuss the law of demand, its assumptions, reasons and exceptions, with a schedule and diagram.

Outline. (i) Statement. (ii) Schedule (Table 5.2) and diagram (Figure 5.1). (iii) Assumptions. (iv) Reasons for downward slope. (v) Exceptions. (vi) Determinants of demand. (vii) Conclusion.

LECTURE 6

Elasticity of Demand, Standard of Living and Engel's Law

6.1 Concept of Elasticity of Demand

The law of demand tells us the direction of the price–quantity relationship, but not its strength. Elasticity measures that strength — how responsive quantity demanded is to a change in price, income or the price of a related good.

DEFINITION — ELASTICITY OF DEMAND

Elasticity of demand is the degree of responsiveness of the quantity demanded of a commodity to a change in its price, the consumer's income, or the price of a related good.

6.2 Price Elasticity of Demand

DEFINITION — PRICE ELASTICITY OF DEMAND (E_D)

Price elasticity of demand is the ratio of the percentage change in quantity demanded to the percentage change in price. It measures how much quantity demanded responds to a price change.

$$E_d = (\%\ \text{change in quantity demanded}) \div (\%\ \text{change in price}) = (\Delta Q / Q) \div (\Delta P / P)$$

Types of price elasticity

Table 6.1 — Types of price elasticity of demand

Type	Value of E_d	Meaning	Example
Perfectly elastic	$E_d = \infty$	Any rise in price reduces demand to zero	A firm in perfect competition
Relatively elastic	$E_d > 1$	Quantity changes more than proportionately	Luxuries; fruits in a city market
Unit elastic	$E_d = 1$	Quantity changes in the same proportion	Textbook case
Relatively inelastic	$E_d < 1$	Quantity changes less than proportionately	Food grains, salt
Perfectly inelastic	$E_d = 0$	Quantity does not change at all	Life-saving medicines

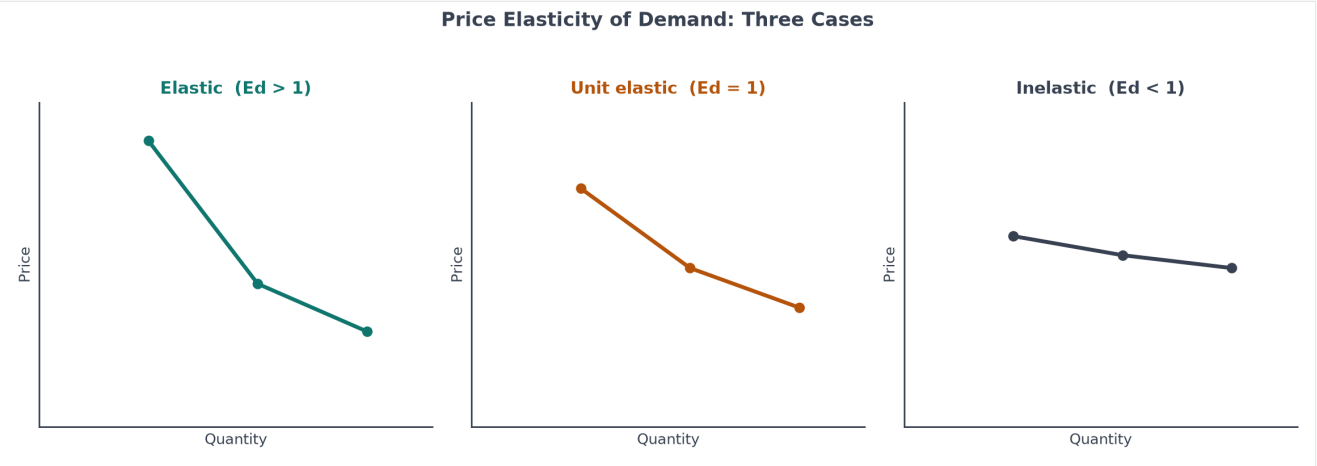


Figure 6.1 — Three cases of price elasticity of demand. Left: elastic ($E_d > 1$), a small price fall brings a large rise in quantity. Middle: unit elastic ($E_d = 1$). Right: inelastic ($E_d < 1$), quantity responds only slightly.

Measurement of price elasticity

(a) Percentage (or proportionate) method

This method divides the percentage change in quantity by the percentage change in price.

$$E_d = (\Delta Q / Q) \div (\Delta P / P) = (\Delta Q / \Delta P) \times (P / Q)$$

(b) Arc method (mid-point method)

When the price change is large, the percentage method gives different answers depending on whether we start from the old or the new price. The arc method avoids this by using the average of the initial and final values.

$$E_d = [\Delta Q / ((Q_1 + Q_2)/2)] \div [\Delta P / ((P_1 + P_2)/2)]$$

(c) Total outlay (expenditure) method

This method looks at what happens to total spending (price × quantity) when price changes. If total outlay rises when price falls, demand is elastic; if it stays the same, demand is unit elastic; if it falls, demand is inelastic.

(d) Point (geometric) method

On a straight-line demand curve, elasticity at a point equals the ratio of the lower segment to the upper segment of the curve measured from that point.

6.3 Income Elasticity of Demand

DEFINITION — INCOME ELASTICITY (E_Y)

Income elasticity of demand is the ratio of the percentage change in quantity demanded to the percentage change in consumer income.

$$E_y = (\%\text{ change in quantity demanded}) \div (\%\text{ change in income})$$

- $E_y > 0$ — normal good (demand rises with income).
- $E_y > 1$ — luxury good (demand rises faster than income).
- $0 < E_y < 1$ — necessity (demand rises more slowly than income).
- $E_y < 0$ — inferior good (demand falls as income rises).

6.4 Cross Elasticity of Demand

DEFINITION — CROSS ELASTICITY (E_{XY})

Cross elasticity of demand is the ratio of the percentage change in the quantity demanded of good X to the percentage change in the price of good Y.

$$E_{xy} = (\%\text{ change in quantity of X}) \div (\%\text{ change in price of Y})$$

- $E_{xy} > 0$ — substitutes (tea and coffee): a rise in the price of Y raises demand for X.
- $E_{xy} < 0$ — complements (car and fuel): a rise in the price of Y lowers demand for X.
- $E_{xy} = 0$ — unrelated goods (rice and cement).

6.5 Factors Influencing Elasticity of Demand

- **Nature of the commodity.** Necessities have inelastic demand; luxuries have elastic demand.
- **Availability of substitutes.** More substitutes mean more elastic demand.
- **Share of income spent.** Goods that take a large share of the budget (like food) tend to have inelastic demand; small items may be elastic.
- **Number of uses.** A good with many uses (milk, electricity) has more elastic demand.
- **Time period.** Demand is more elastic in the long run, when consumers can adjust.
- **Habit and addiction.** Habitual goods (tobacco) have inelastic demand.
- **Possibility of postponement.** If purchase can be postponed, demand is elastic.
- **Level of price.** At very high or very low prices, demand may become less elastic.

6.6 Importance of Elasticity of Demand

- **Price policy for farmers.** Because food demand is inelastic, a bumper crop can actually lower farmers' total income — a key reason for MSP and procurement.
- **Taxation.** Governments tax goods with inelastic demand (fuel, tobacco) to raise revenue without much fall in demand.
- **Pricing by firms.** A monopolist sets a higher price where demand is inelastic.
- **Devaluation and exports.** Exports respond to price, so elasticity matters for trade policy.
- **Wage bargaining.** Elasticity of demand for labour affects employment and wages.

- **Marketing and storage.** Knowledge of elasticity guides when to store and when to sell produce.

6.7 Standard of Living

DEFINITION — STANDARD OF LIVING

Standard of living refers to the level of comfort, goods and services available to a person, family or community, and the degree to which their material and social needs are satisfied.

The standard of living depends on income, the distribution of income, prices, the availability of goods and services, and public facilities such as education, health and housing. It is often studied through the **pattern of family expenditure** — how a family divides its spending among food, clothing, housing, education and other items.

6.8 Engel's Law of Family Expenditure

STATEMENT — ENGEL'S LAW

Engel's law states that as family income rises, the proportion (share) of income spent on food falls, even though the absolute amount spent on food rises. The share spent on other items, such as education, health and recreation, rises.

The law was stated by the German statistician Ernst Engel after studying family budgets. It follows from the fact that food is a necessity with income-inelastic demand (E_y less than one): as income rises, food spending rises but more slowly than income, so its share falls.

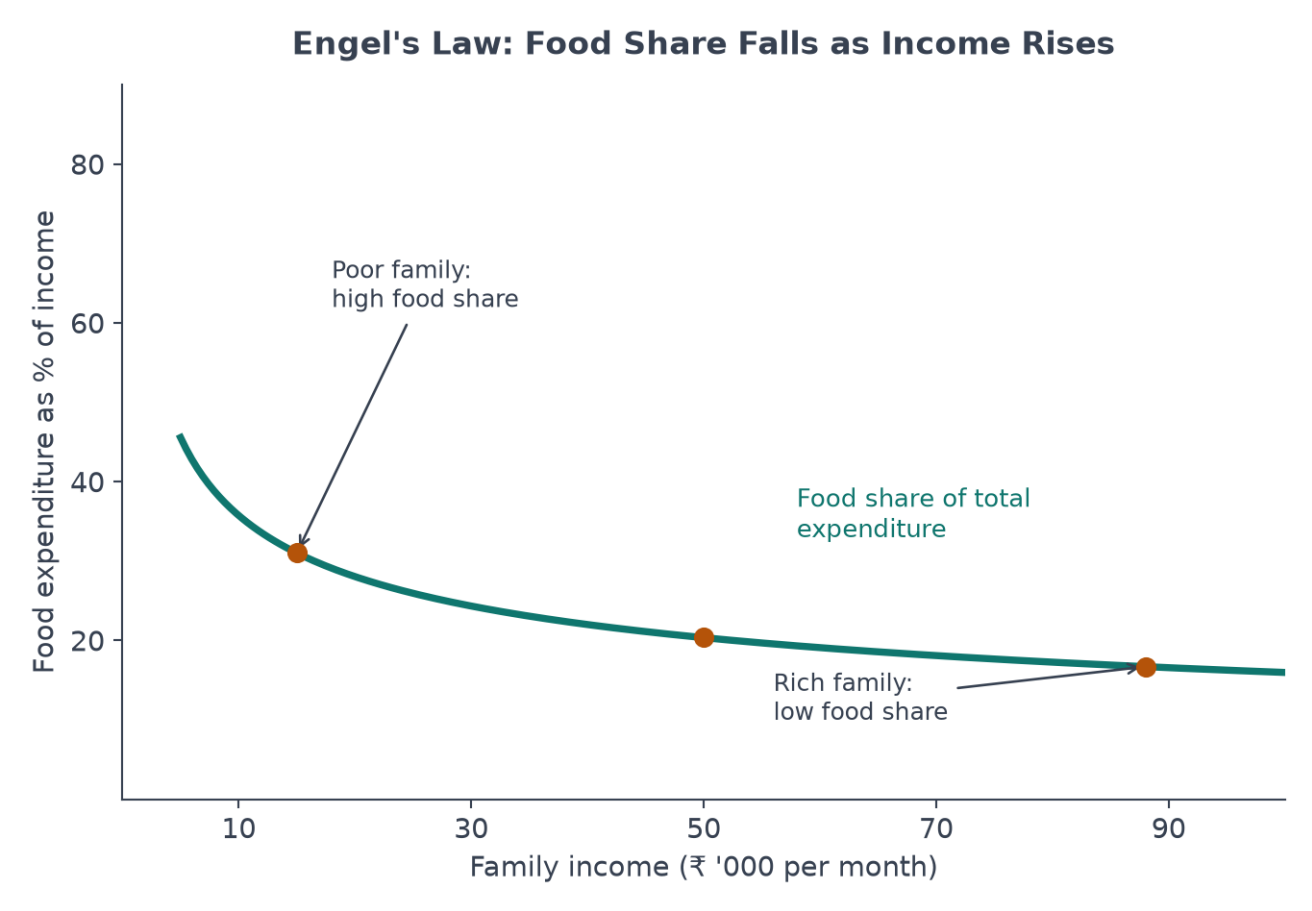


Figure 6.2 — Engel's law. As family income rises, the percentage of income spent on food falls. Poor families spend a high share on food; rich families spend a low share.

Importance of Engel's law

- It measures the **standard of living**: a lower food share indicates a higher standard of living.
- It helps in **demand forecasting** for food and other goods as incomes rise.
- It guides **agricultural policy**: as incomes rise, the demand for food grows slowly, so farmers must diversify into higher-value products.
- It explains the **shift in consumption patterns** in developing economies.

Worked Example 6.1 — Price elasticity by the percentage method

Problem. The price of tomato falls from ₹40 to ₹32 per kg and quantity demanded rises from 100 kg to 130 kg. Find E_d and state the type.

Solution.

- % change in quantity = (130 − 100)/100 × 100 = 30%
- % change in price = (32 − 40)/40 × 100 = −20%
- E_d = 30 / (−20) = −1.5. Taking the absolute value, E_d = **1.5**.

Interpretation. Since $E_d > 1$, demand is **relatively elastic** — a 1% fall in price raises quantity demanded by 1.5%.

Worked Example 6.2 — Price elasticity by the arc method

Problem. Price falls from ₹10 to ₹6 and quantity demanded rises from 80 to 120 units. Find E_d by the arc method.

Solution.

- $\Delta Q = 120 - 80 = 40$; average $Q = (80 + 120)/2 = 100$.
- $\Delta P = 6 - 10 = -4$; average $P = (10 + 6)/2 = 8$.
- $E_d = (40/100) \div (-4/8) = 0.40 \div (-0.50) = -0.8$.
- Absolute value = **0.8**, so demand is **relatively inelastic**.

Worked Example 6.3 — Income and cross elasticity

Problem. (a) A family's income rises from ₹20,000 to ₹24,000 per month and its demand for milk rises from 30 to 36 litres. Find income elasticity. (b) The price of coffee rises from ₹200 to ₹250 per kg and the demand for tea rises from 500 to 600 kg. Find cross elasticity.

Solution.

- (a) % change in quantity = $(36 - 30)/30 \times 100 = 20\%$; % change in income = $(24{,}000 - 20{,}000)/20{,}000 \times 100 = 20\%$. $E_y = 20/20 =$ **1.0**. Milk is a normal good.
- (b) % change in quantity of tea = $(600 - 500)/500 \times 100 = 20\%$; % change in price of coffee = $(250 - 200)/200 \times 100 = 25\%$. $E_{xy} = 20/25 =$ **+0.8**. Since E_{xy} is positive, tea and coffee are **substitutes**.

6.9 Indicators of the Standard of Living

The standard of living is judged not by income alone but by a range of indicators:

- **Per capita income** and its distribution.
- **Pattern of expenditure** — a lower food share (Engel's law) indicates a higher standard.
- **Nutrition and health** — calorie intake, life expectancy, infant mortality.
- **Education** — literacy and years of schooling.
- **Housing, sanitation and clean drinking water.**
- **Access to public services** and leisure.

These indicators together form the basis of composite measures such as the Human Development Index, which shows that income is only one part of well-being.

Exam Focus — Lecture 6

Key points to remember

- Elasticity measures responsiveness; price elasticity E_d = %ΔQ ÷ %ΔP.
- Five types: perfectly elastic, elastic, unit elastic, inelastic, perfectly inelastic.
- Methods: percentage, arc (mid-point), total outlay and point methods.
- Income elasticity: positive for normal, >1 for luxury, negative for inferior goods.
- Cross elasticity: positive for substitutes, negative for complements.
- Food demand is inelastic — hence MSP and procurement to protect farmers' income.
- Engel's law: the share of income spent on food falls as income rises — a measure of the standard of living.

Short-answer questions with model answers

Q1. Define price elasticity of demand and give the formula.

Model answer. Price elasticity of demand is the ratio of the percentage change in quantity demanded to the percentage change in price: E_d = (%ΔQ) ÷ (%ΔP). It measures how responsive quantity demanded is to price.

Q2. State Engel's law.

Model answer. Engel's law states that as family income rises, the proportion of income spent on food falls, though the absolute amount spent on food rises. It indicates a rising standard of living.

Q3. What does a positive cross elasticity between two goods indicate?

Model answer. A positive cross elasticity indicates that the two goods are substitutes: when the price of one rises, the demand for the other rises (for example, tea and coffee).

Long-answer questions with answer outlines

Q1. Explain the concept and measurement of price elasticity of demand, and the factors influencing it.

Outline. (i) Concept and definition. (ii) Types with the value of E_d (Table 6.1, Figure 6.1). (iii) Methods — percentage, arc, total outlay, point. (iv) Factors influencing elasticity. (v) Importance. (vi) Conclusion.

Q2. Discuss income and cross elasticity of demand, and explain Engel's law of family expenditure.

Outline. (i) Income elasticity — formula and types of goods. (ii) Cross elasticity — formula, substitutes and complements. (iii) Engel's law — statement, reason, Figure 6.2. (iv) Importance of Engel's law for policy. (v) Conclusion.

LECTURE 7

Production: Process, Factors and Input–Output Relations

7.1 Meaning of Production

DEFINITION — PRODUCTION

Production is the process of creating utility, that is, of transforming inputs (factors of production) into outputs (goods and services) that satisfy human wants.

In economics, production does not mean only "making" a physical thing. It means **creating utility** — adding value by changing the form, place or time of goods. A farmer who grows paddy creates form utility; a transporter who moves vegetables to the city creates place utility; a cold-storage owner who keeps potatoes until prices rise creates time utility.

Creation of utility — the forms

- **Form utility** — changing the form of a good (paddy into rice, milk into curd).
- **Place utility** — moving goods to where they are wanted (mangoes from Theni to Chennai).
- **Time utility** — storing goods until they are wanted (storing grain until the lean season).
- **Possession (ownership) utility** — transferring ownership through trade.
- **Service utility** — personal services such as the advice of an extension officer.

7.2 Factors of Production

DEFINITION — FACTORS OF PRODUCTION

Factors of production are the resources used to produce goods and services. They are classified into four groups: land, labour, capital and organisation (enterprise).

Each factor earns a reward: land earns **rent**, labour earns **wages**, capital earns **interest**, and organisation earns **profit**. These rewards are the subject of distribution theory (Lecture 11).

Table 7.1 — Factors of production and their characteristics

Factor	Meaning	Chief characteristics	Reward
Land	All free gifts of nature — soil, water, minerals, climate	Fixed in supply; immobile; heterogeneous in quality; subject to diminishing returns; a passive factor	Rent
Labour	All human physical and mental effort used in production	Inseparable from the worker; perishable (cannot be stored); heterogeneous; can be mobile; the worker sells labour, not himself	Wages
Capital	Man-made, produced means of production	Produced by man; productive; depreciable; mobile; created by saving and investment; divisible	Interest

Factor	Meaning	Chief characteristics	Reward
Organisation (Enterprise)	The work of combining and managing the other three factors and bearing risk	Combines factors; takes decisions; bears risk and uncertainty; innovative; earns residual income	Profit

Land — characteristics in detail

- **Fixed supply.** The total area of land is fixed; its supply cannot be increased (though its productive power can be raised).
- **Immobile.** Land cannot be shifted from one place to another.
- **Heterogeneous.** Different plots differ in fertility, location and climate.
- **Passive factor.** Land does not act on its own; it must be combined with labour and capital.
- **Diminishing returns.** Applying more inputs to the same land eventually yields less extra output.

Labour — characteristics in detail

- **Inseparable from the labourer.** Labour cannot be separated from the person who provides it.
- **Perishable.** A day's labour lost cannot be stored and used later.
- **Heterogeneous.** Labour differs in skill, efficiency and training.
- **Human element.** Labour is influenced by feelings, working conditions and fair treatment.
- **Bargaining power.** Labour is less mobile and often weaker in bargaining than capital.

Capital — characteristics in detail

- **Man-made.** Capital is created by human effort, not a free gift of nature.
- **Productive.** Capital produces a flow of income and a return (interest).
- **Depreciable.** Capital wears out and must be replaced; this is why depreciation is a cost.
- **Mobile.** Unlike land, capital can be moved from one place or use to another.
- **Requires saving.** Capital formation requires saving, which means sacrificing present consumption.

> **Fixed versus variable factors.** A **fixed factor** cannot be changed in the short run (land, a tractor, a tubewell). A **variable factor** can be changed quickly (seed, fertiliser, casual labour, water). The distinction between fixed and variable factors is the basis of the short-run and long-run cost concepts in Lecture 8.

7.3 Production Function

DEFINITION — PRODUCTION FUNCTION

A production function is the technological relationship between inputs (factors of production) and the maximum output that can be produced from them, given the existing technology.

$$Q = f(L, La, K, M, T \ldots) \quad \text{where } Q = \text{output}, L = \text{labour}, La = \text{land}, K = \text{capital}, M = \text{management}, T = \text{technology}$$

A production function shows the maximum output obtainable from each combination of inputs. It is a purely technological relation; it does not tell us the price or the profit, only what is physically possible.

Short run and long run production function

- **Short-run production function.** At least one factor is fixed; output is changed by varying the variable factors. This gives the law of variable proportions (Lecture 8).
- **Long-run production function.** All factors are variable; output is changed by changing the scale of all inputs together. This gives returns to scale (Lecture 8).

Important product concepts

- **Total product (TP).** The total output produced with a given quantity of inputs.
- **Average product (AP).** Output per unit of the variable input; AP = TP / number of units of input.
- **Marginal product (MP).** The addition to total output from using one more unit of the variable input; MP = ΔTP / Δinput.

$$AP = TP / Q_{input} \quad | \quad MP = \Delta TP / \Delta Q_{input} \quad | \quad TP = \Sigma MP$$

The relationship between AP and MP is important: when MP is greater than AP, AP rises; when MP is less than AP, AP falls; and when MP equals AP, AP is at its maximum.

7.4 Input–Output Relationship

In farm economics, the relationship between inputs and output is studied in three forms, depending on what is allowed to change.

(a) Factor–product relationship

Here one input is varied while others are held constant, and we study how output responds. This gives the law of diminishing returns. The key decision rule is that a farmer should use the variable input up to the point where the value of the marginal product equals the price of the input.

$$\text{Use input until} \quad VMP = P_{input} \quad (\text{or} \quad MP \times P_{output} = \text{price of input})$$

(b) Factor–factor relationship

Here two inputs are varied and output is held constant, to find the cheapest combination of inputs. The principle is substitution: a farmer can use more labour and less machinery (or vice versa) to produce the same output, and will choose the combination that costs least.

$$\text{Least-cost combination:} \quad MP_L / P_L = MP_K / P_K$$

(c) Product–product relationship

Here the same resources can produce more than one product, and the farmer decides how to allocate them among competing enterprises (for example, how much land to paddy and how much to sugarcane). The best combination is where the marginal rate of substitution between products equals the inverse ratio of their prices.

Optimum product combination: $MRPS_{12} = P_1 / P_2$

Types of input–output response

Table 7.2 — Types of returns in input–output relationships

Type	Meaning	Example
Constant returns	Each additional unit of input adds the same amount of output	Rare in agriculture; roughly in some assembly work
Increasing returns	Each additional unit of input adds more output than before	Early stage of fertiliser use on poor soil
Decreasing (diminishing) returns	Each additional unit of input adds less output than before	Applying more and more fertiliser to the same plot
Negative returns	Additional units reduce total output	Too much fertiliser burns the crop

Worked Example 7.1 — Computing TP, AP and MP

Problem. A farmer applies increasing bags of fertiliser to one hectare of paddy. Compute the total, average and marginal product and identify the stage of diminishing returns.

Bags of fertiliser	1	2	3	4	5	6
Total product (quintals)	10	24	39	50	56	58

Solution. MP = change in TP; AP = TP ÷ bags.

Bags	TP	MP	AP
1	10	10	10.0
2	24	14	12.0
3	39	15	13.0
4	50	11	12.5
5	56	6	11.2
6	58	2	9.7

Interpretation. MP rises up to the 3rd bag (10, 14, 15) — the stage of increasing returns. From the 4th bag onwards MP falls (11, 6, 2) — the stage of **diminishing returns**. AP is maximum (13.0) when MP equals AP at the 3rd bag, which is the boundary between the two stages.

7.5 The Law of Diminishing Returns in Agriculture

The law of diminishing returns operates with special force in agriculture because land is fixed. This has three important consequences for farm management.

- **Intensive cultivation has limits.** Applying more and more inputs to the same plot raises output only up to a point.

- **Optimum input use.** A rational farmer stops applying an input where the value of its marginal product equals its price — not at the point of maximum output.
- **Extensive cultivation.** Beyond the limit of intensive farming, output is raised by bringing more land under cultivation.

Farmers who ignore the law and apply "more and more" fertiliser or water actually reduce profit and may harm the crop (negative returns). This is why soil testing and balanced fertilisation are promoted.

Exam Focus — Lecture 7

Key points to remember

- Production is the creation of utility — form, place, time, possession and service utility.
- Four factors of production: land (rent), labour (wages), capital (interest), organisation (profit).
- Land is fixed, immobile and heterogeneous; labour is inseparable and perishable; capital is man-made and depreciable.
- A production function shows the maximum output from given inputs, at a given technology.
- TP, AP and MP: AP = TP/input, MP = ΔTP/Δinput; AP is maximum where MP = AP.
- Three input–output relations: factor–product, factor–factor and product–product.
- Returns may be constant, increasing, diminishing or negative.

Short-answer questions with model answers

Q1. Define production and explain "creation of utility".

Model answer. Production is the process of creating utility by transforming inputs into outputs. Utility is created by changing the form (paddy to rice), place (moving vegetables to the city), time (cold storage), ownership (trade) or through services.

Q2. What are the four factors of production and their rewards?

Model answer. Land (reward: rent), labour (wages), capital (interest) and organisation or enterprise (profit).

Q3. Distinguish between average product and marginal product.

Model answer. Average product is output per unit of the variable input (TP ÷ input). Marginal product is the addition to total output from one more unit of input (ΔTP ÷ Δinput). AP is maximum when MP equals AP.

Long-answer questions with answer outlines

Q1. Explain the factors of production and their characteristics.

Outline. (i) Meaning. (ii) Land — characteristics. (iii) Labour — characteristics. (iv) Capital — characteristics. (v) Organisation — functions and risk-bearing. (vi) Rewards table (Table 7.1). (vii) Conclusion.

Q2. Discuss the concept of the production function and the input–output relationships in agriculture.

Outline. (i) Production function — definition and equation. (ii) Short-run and long-run. (iii) TP, AP, MP. (iv) Factor–product, factor–factor and product–product relations with decision rules. (v) Types of returns (Table 7.2). (vi) Conclusion.

LECTURE 8

Laws of Returns, Cost and Revenue

8.1 Laws of Returns

The laws of returns describe how output responds when inputs are increased. There are two laws, matching the short run and the long run.

- **Law of variable proportions** (short run) — some factors are fixed, one factor is varied.
- **Law of returns to scale** (long run) — all factors are varied together.

8.2 Law of Variable Proportions

> **STATEMENT**
>
> The law of variable proportions states that, as more and more units of a variable factor are applied to fixed factors, total product first increases at an increasing rate, then increases at a diminishing rate, and finally declines.

This law is also called the law of diminishing returns in its second stage. It operates because, with a fixed factor, the variable factor becomes relatively more abundant and the proportion between them becomes less and less ideal.

Assumptions

- The state of technology is constant.
- Some factors are fixed in supply.
- The variable factor units are homogeneous.
- Input prices are constant.

The three stages

- **Stage I — Increasing returns.** TP rises at an increasing rate, MP rises, and AP rises. The fixed factor is too large relative to the variable factor, so adding variable input raises efficiency. MP is at its maximum at the end of this stage.
- **Stage II — Diminishing returns.** TP rises at a decreasing rate, MP falls (but is positive), and AP falls after reaching its maximum. The ideal proportion has been passed. This stage ends where MP becomes zero and TP is maximum.
- **Stage III — Negative returns.** TP falls, MP is negative. The variable factor is so abundant that it interferes with production. No rational producer operates here.

The rational producer operates in **Stage II**, because it is the stage where the fixed factor is fully utilised and every extra unit of input still adds some output. Within Stage II, the exact point is where the value of the marginal product equals the price of the input.

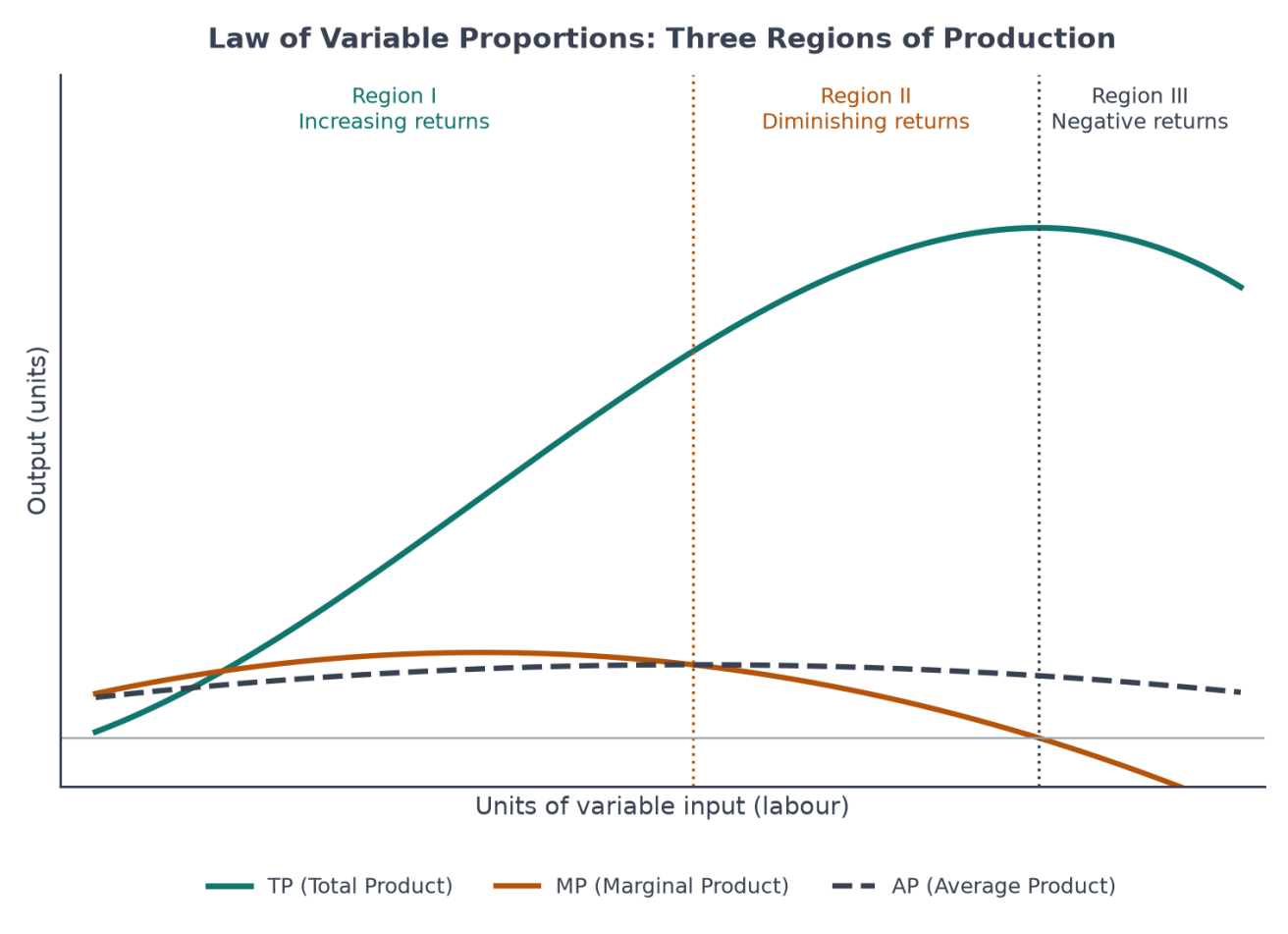


Figure 8.1 — The three regions of production. Region I ends where the average product is maximum (MP = AP). Region II, the rational stage, ends where marginal product is zero and total product is maximum. Region III has negative marginal product.

Why does the law operate?

- **Scarcity of the fixed factor.** With fixed land, extra labour has less and less land to work on.
- **Imperfect substitutability.** Labour and land cannot be substituted perfectly, so the ideal proportion is soon passed.
- **Limits to specialisation.** Beyond a point, extra workers cannot be given more specialised tasks.

8.3 Law of Returns to Scale

STATEMENT

The law of returns to scale states that, when all factors of production are increased in the same proportion, output may increase in a greater proportion (increasing returns to scale), the same proportion (constant returns to scale), or a smaller proportion (decreasing returns to scale).

- **Increasing returns to scale.** Output rises more than proportionately (for example, all inputs doubled, output more than doubled). Causes: internal and external economies of scale, greater specialisation, indivisibilities.
- **Constant returns to scale.** Output rises in the same proportion as inputs. The most efficient scale of operation, where economies are exhausted.

- **Decreasing returns to scale.** Output rises less than proportionately. Causes: diseconomies of scale, management difficulties, rising costs of coordination.

If all inputs are multiplied by k, output is multiplied by:
> k → increasing returns to scale; = k → constant; < k → decreasing

8.4 Cost Concepts

DEFINITION — COST OF PRODUCTION

Cost of production is the total money value of all resources used in producing a good, that is, the sum of payments made to the factors of production.

Table 8.1 — Cost concepts and their definitions

Concept	Definition	Formula
Total fixed cost (TFC)	Cost that does not change with output; incurred even at zero output	TFC = constant
Total variable cost (TVC)	Cost that changes directly with output	TVC = f(output)
Total cost (TC)	Sum of total fixed and total variable cost	TC = TFC + TVC
Average fixed cost (AFC)	Fixed cost per unit of output; falls continuously	AFC = TFC / Q
Average variable cost (AVC)	Variable cost per unit of output	AVC = TVC / Q
Average total cost (AC)	Total cost per unit of output	AC = TC / Q = AFC + AVC
Marginal cost (MC)	Addition to total cost from producing one more unit	MC = ΔTC / ΔQ = ΔTVC / ΔQ
Opportunity cost	Value of the next best alternative given up	Not a money figure; a concept
Explicit cost	Actual payments made to outsiders	Wages, rent, seed, fertiliser paid
Implicit cost	Value of self-owned resources used	Imputed rent of own land, family labour
Money (accounting) cost	Cost as recorded in the books (explicit only)	Explicit costs
Real (economic) cost	Explicit plus implicit cost	Accounting cost + implicit cost

Table 8.2 — Short-run versus long-run costs

Basis	Short run	Long run
Factors	Some fixed, some variable	All factors variable
Fixed cost	Present (TFC > 0)	Absent (all costs variable)
Curves	TFC, TVC, TC, AFC, AVC, AC, MC	LAC (long-run average cost), LMC
Law operating	Law of variable proportions	Law of returns to scale
Scale of plant	Plant size fixed	Plant size can be changed

Basis	Short run	Long run
Shape of AC	U-shaped due to fixed factor	U-shaped (envelope) due to economies and diseconomies of scale

Behaviour of the short-run cost curves

- **AFC** falls continuously as output rises, because a fixed cost is spread over more units. It approaches but never touches the horizontal axis (a rectangular hyperbola).
- **AVC, AC and MC** are U-shaped. They first fall (as fixed factors are better utilised) and then rise (as diminishing returns set in).
- **MC** cuts both AVC and AC at their minimum points. When MC is below AC, AC falls; when MC is above AC, AC rises.
- The gap between AC and AVC is AFC, which shrinks as output rises, so AC and AVC come closer together at high output.



Figure 8.2 — Short-run cost curves. Left: total fixed, total variable and total cost. Right: average cost, average variable cost and marginal cost, with marginal cost cutting average cost at its minimum.

8.5 Revenue Concepts

DEFINITIONS

Total revenue (TR) is the total money received from selling a given output. **Average revenue (AR)** is revenue per unit sold, which equals price. **Marginal revenue (MR)** is the addition to total revenue from selling one more unit.

$$TR = P \times Q \quad | \quad AR = TR / Q = Price \quad | \quad MR = \Delta TR / \Delta Q$$

Revenue under perfect competition and monopoly

- **Perfect competition.** The firm is a price taker, so price is constant. Hence AR = MR = price, and TR rises as a straight line from the origin. AR and MR are the same horizontal line.

- **Monopoly.** To sell more, the monopolist must lower the price. Hence AR slopes downward and MR falls faster than AR (MR is below AR). TR rises, reaches a maximum where MR = 0, then falls.

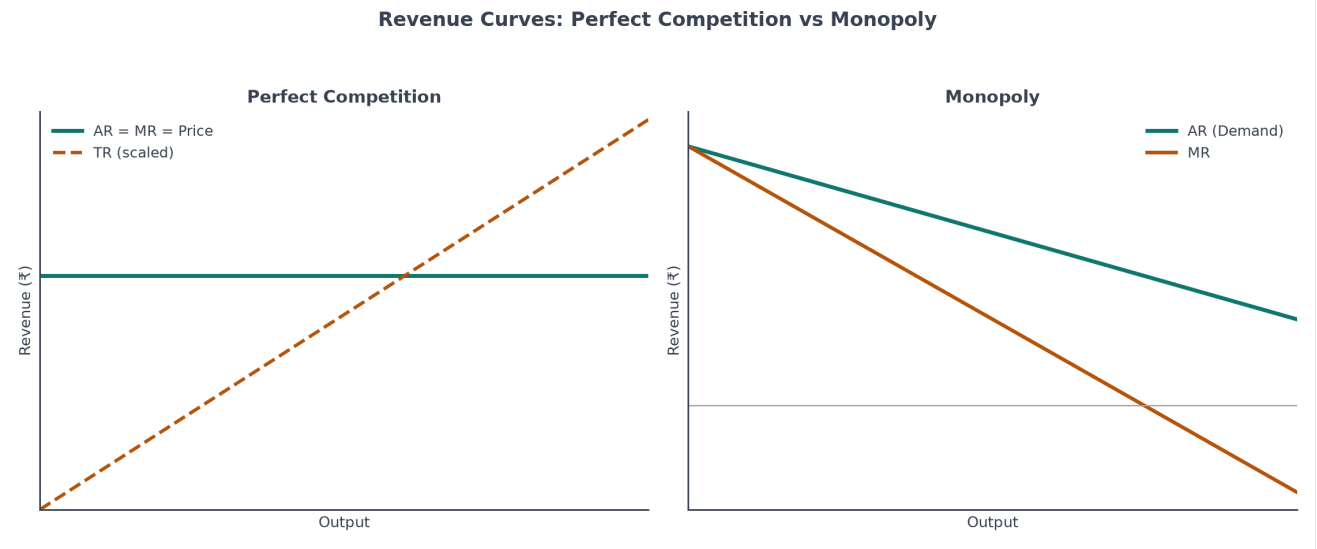


Figure 8.3 — Revenue curves. Left (perfect competition): average revenue equals marginal revenue equals price, a horizontal line, with total revenue rising steadily. Right (monopoly): average revenue slopes down and marginal revenue falls more steeply.

Profit maximisation

A firm maximises profit at the output where the difference between total revenue and total cost is greatest. This occurs where marginal cost equals marginal revenue, with MC rising through MR.

$$\text{Profit} = TR - TC \quad | \quad \text{Profit is maximum where } MR = MC \text{ (and MC is rising)}$$

Break-even analysis

The break-even point is the output at which total revenue equals total cost, so profit is zero. Below it the firm makes a loss; above it a profit.

$$\text{Break-even output } Q^* = TFC / (\text{Price} - AVC \text{ per unit}) = TFC / \text{Contribution per unit}$$

Worked Example 8.1 — Cost and revenue table with profit

Problem. A firm has TFC = ₹100. Its variable costs and the price (₹20 per unit) are given. Complete the table and find the profit-maximising output.

Output	0	1	2	3	4	5	6
TVC (₹)	0	18	30	40	52	68	90

Solution. TC = TFC + TVC = 100 + TVC. TR = 20 × Q. Profit = TR − TC.

Q	TVC	TC	TR	Profit	MC	MR
0	0	100	0	−100	—	—
1	18	118	20	−98	18	20
2	30	130	40	−90	12	20
3	40	140	60	−80	10	20
4	52	152	80	−72	12	20
5	68	168	100	−68	16	20
6	90	190	120	−70	22	20

Interpretation. The loss is smallest (−₹68) at output = 5, where MC (₹16) is below MR (₹20); at output 6, MC (₹22) exceeds MR, so profit falls. The firm maximises profit (minimises loss) at **output 5**. This is where MR = MC as closely as the data allow.

Worked Example 8.2 — Break-even output

Problem. A farmer's fixed cost is ₹60,000 per season. The selling price of the produce is ₹30 per kg and the average variable cost is ₹18 per kg. Find the break-even output.

Solution.

- Contribution per kg = Price − AVC = 30 − 18 = ₹12.
- Break-even output = TFC ÷ contribution = 60,000 ÷ 12 = **5,000 kg**.

Check. At 5,000 kg: TR = 30 × 5,000 = ₹1,50,000. TC = 60,000 + 18 × 5,000 = ₹1,50,000. TR = TC, so profit is zero — the break-even point. Above 5,000 kg the farmer earns a profit.

8.6 Long-Run Average Cost Curve

In the long run all factors are variable, so the firm can choose the size of its plant. The **long-run average cost (LAC)** curve shows the lowest average cost of producing each output when the firm can build the most efficient plant for that output.

- The LAC curve is the **envelope** of a family of short-run average cost curves — each short-run curve is tangent to the LAC curve at one output.

- It is U-shaped because of **economies of scale** (falling LAC) followed by **diseconomies of scale** (rising LAC).
- The point at which the LAC is minimum is the **optimum scale of the firm** — the plant size that gives the lowest cost per unit.
- Economies of scale arise from specialisation, bulk buying, indivisible machinery and better management. Diseconomies arise from the difficulties of managing a very large organisation.

> LAC = minimum possible average cost for each output, when plant size is variable

For Indian agriculture, the LAC concept explains why very small holdings have a high cost per unit (they cannot use machinery fully) while very large farms may face management diseconomies. It supports the case for cooperative use of machinery and for custom-hiring services.

Exam Focus — Lecture 8

Key points to remember

- Law of variable proportions: three stages — increasing, diminishing, negative returns; rational stage is Stage II.
- Law of returns to scale: increasing, constant and decreasing returns when all inputs change together.
- TC = TFC + TVC; AC = AFC + AVC; MC = ΔTC/ΔQ.
- AFC falls continuously; AC, AVC and MC are U-shaped; MC cuts AC at its minimum.
- TR = P×Q; AR = price; MR = ΔTR/ΔQ.
- Under perfect competition AR = MR = price (horizontal); under monopoly MR < AR (falling).
- Profit is maximum where MR = MC; break-even output = TFC ÷ (Price − AVC).

Short-answer questions with model answers

Q1. State the law of variable proportions.

Model answer. As more units of a variable factor are applied to fixed factors, total product first rises at an increasing rate, then at a diminishing rate, and finally falls. The rational producer operates in the second (diminishing returns) stage.

Q2. Why is the average fixed cost curve a rectangular hyperbola?

Model answer. Because AFC = TFC ÷ Q and TFC is constant, AFC falls continuously as output rises, approaching the horizontal axis but never touching it. This shape is a rectangular hyperbola.

Q3. Define marginal cost and marginal revenue.

Model answer. Marginal cost is the addition to total cost from producing one more unit (MC = ΔTC/ΔQ). Marginal revenue is the addition to total revenue from selling one more unit (MR = ΔTR/ΔQ). Profit is maximised where MR = MC.

Long-answer questions with answer outlines

Q1. Explain the law of variable proportions with its stages and diagram.

Outline. (i) Statement. (ii) Assumptions. (iii) Three stages with TP, AP, MP. (iv) Figure 8.1. (v) Why the law operates. (vi) Rational stage. (vii) Conclusion.

Q2. Explain the cost concepts and the behaviour of short-run cost curves, and the revenue concepts.

Outline. (i) Cost concepts table (Table 8.1). (ii) Short-run versus long-run costs (Table 8.2). (iii) Behaviour of TFC, TVC, TC, AFC, AVC, AC, MC (Figure 8.2). (iv) Revenue concepts — TR, AR, MR (Figure 8.3). (v) Profit maximisation MR = MC. (vi) Conclusion.

LECTURE 9

Supply: Law, Determinants and Elasticity

9.1 Meaning of Supply

DEFINITION — SUPPLY

Supply is the quantity of a commodity that producers are willing and able to offer for sale at a given price during a given period of time.

Like demand, supply is expressed with reference to a price and a period. A producer may be willing to sell 100 quintals of paddy at ₹2,300 per quintal this season; that is the supply at that price. Supply is a flow concept, measured per unit of time.

9.2 Stock versus Supply

DEFINITIONS

Stock is the total quantity of a commodity available with a producer at a point of time. **Supply** is that part of the stock which the producer is willing to offer for sale at a given price during a period.

Table 9.1 — Stock versus supply

Basis	Stock	Supply
Meaning	Total quantity available at a point of time	Quantity offered for sale at a given price
Nature	Stock concept	Flow concept (per period)
Relation to price	Not directly related to price	Directly related to price
Relation	Stock is the basis of supply	Supply is that part of stock brought to market
Example	A godown holding 5,000 bags of rice	2,000 bags offered for sale at the current price

Stock can be greater than, equal to or less than supply. A farmer may hold 5,000 bags of paddy in stock but sell only 2,000 bags now, expecting a higher price later. Stock is potential supply; supply is actual market offer.

9.3 Law of Supply

STATEMENT — LAW OF SUPPLY

Other things being equal, when the price of a commodity rises its quantity supplied rises, and when the price falls its quantity supplied falls. There is a direct (positive) relationship between price and quantity supplied.

Assumptions

- Cost of production does not change.
- The state of technology is constant.

- Prices of other goods do not change.
- Producers' goals and the government's policy (taxes, subsidies) remain unchanged.
- No change in the number of sellers.

Reasons for the upward-sloping supply curve

- **Profit motive.** A higher price raises the profit from each unit, so producers supply more.
- **Higher cost of additional output.** To produce more, less efficient resources must be used, so higher output requires a higher price to cover the rising cost.
- **New entrants.** A higher price attracts new producers into the market.
- **Shifting from other uses.** Higher price makes the good more attractive than alternative products.

Exceptions to the law of supply

- **Perishable goods.** Sellers must sell whatever they have, regardless of price.
- **Agricultural goods in the short run.** The crop is already produced, so supply is fixed at harvest.
- **Expectation of a further price rise.** Producers may hold back stock, reducing current supply.
- **Rare and artistic goods.** Supply cannot be increased even if price rises.

Supply schedule and supply curve

Table 9.2 — Individual supply schedule

Price of paddy (₹/quintal)	Quantity supplied (quintals)
1,500	10
1,800	20
2,100	30
2,400	40
2,700	50

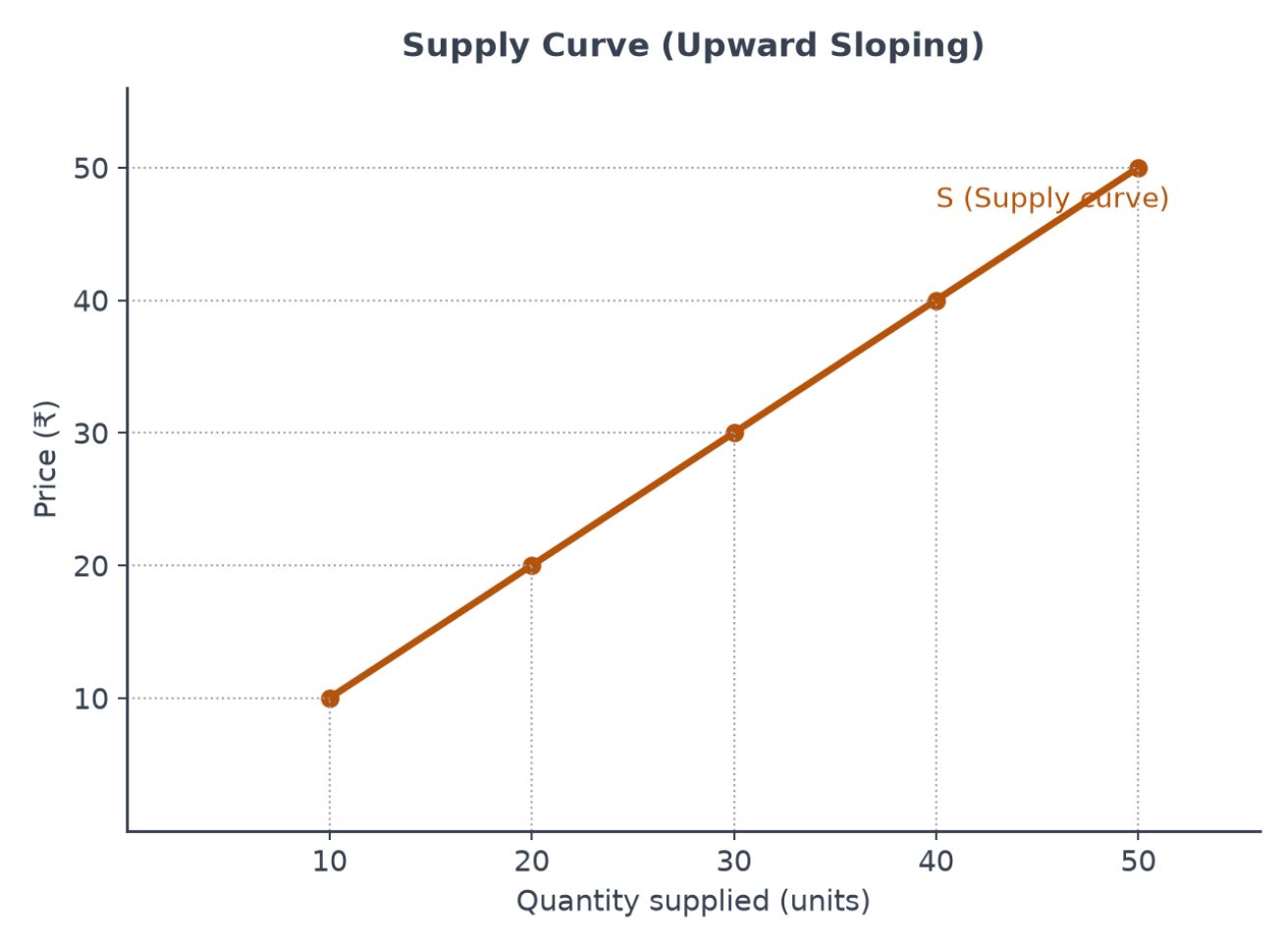


Figure 9.1 — The supply curve slopes upward from left to right, showing that a higher price calls forth a larger quantity supplied.

Determinants of supply

- **Price of the commodity** — the main determinant; higher price, higher supply.
- **Cost of production** — lower costs raise supply; higher input prices reduce it.
- **Technology** — improved technology lowers cost and raises supply.
- **Prices of related goods** — a rise in the price of an alternative crop may reduce the supply of the present crop.
- **Government policy** — taxes reduce supply; subsidies and MSP raise it.
- **Weather and natural factors** — a good monsoon raises farm supply.
- **Number of producers** — more sellers mean greater market supply.
- **Expectations of future prices** — expectation of a price rise may reduce current supply.
- **Infrastructure and transport** — better roads and storage improve market supply.

> **Movement along versus shift of the supply curve.** A change in the commodity's own price causes a **movement along** the supply curve (change in quantity supplied). A change in cost, technology, taxes or weather causes a **shift** of the whole supply curve (change in supply).

9.4 Elasticity of Supply

DEFINITION — ELASTICITY OF SUPPLY (E_S)

Elasticity of supply is the ratio of the percentage change in quantity supplied to the percentage change in price. It measures how responsive supply is to a price change.

$$E_S = (\%\ \text{change in quantity supplied}) \div (\%\ \text{change in price}) = (\Delta Q / Q) \div (\Delta P / P)$$

Types of elasticity of supply

Table 9.3 — Types of elasticity of supply

Type	Value of E_S	Meaning
Perfectly elastic	$E_S = \infty$	Any quantity supplied at one price (horizontal curve)
Relatively elastic	$E_S > 1$	Supply changes more than proportionately to price
Unit elastic	$E_S = 1$	Supply changes in the same proportion (passes through origin)
Relatively inelastic	$E_S < 1$	Supply changes less than proportionately
Perfectly inelastic	$E_S = 0$	Supply does not change with price (vertical curve)

Factors influencing elasticity of supply

- **Time period.** Supply is more elastic in the long run, when producers can adjust fully.
- **Nature of the commodity.** Manufactured goods are more elastic in supply than farm products.
- **Gestation period.** Long-gestation crops (rubber, coconut, mango) have inelastic supply.
- **Storage facilities.** Good storage makes supply more elastic by allowing producers to hold stock.
- **Cost structure.** If extra output can be produced at similar cost, supply is elastic.
- **Availability of inputs and mobility of factors.** Easy availability raises elasticity of supply.

Worked Example 9.1 — Elasticity of supply

Problem. The price of groundnut rises from ₹5,000 to ₹6,000 per quintal and the quantity supplied rises from 200 to 260 quintals. Find the elasticity of supply.

Solution.

- % change in quantity supplied = (260 − 200)/200 × 100 = 30%
- % change in price = (6,000 − 5,000)/5,000 × 100 = 20%
- E_S = 30 / 20 = **1.5**.

Interpretation. Since $E_S > 1$, supply is **relatively elastic**: a 1% rise in price brings a 1.5% rise in quantity supplied.

Worked Example 9.2 — Market equilibrium price and quantity

Problem. In a local market, demand is $Q_d = 100 - 2P$ and supply is $Q_s = 20 + 3P$. Find the equilibrium price and quantity.

Solution. At equilibrium $Q_d = Q_s$:

- $100 - 2P = 20 + 3P$
- $100 - 20 = 3P + 2P$
- $80 = 5P \rightarrow P =$ **₹16**.
- $Q = 100 - 2(16) = 100 - 32 =$ **68 units**.

Check. $Q_s = 20 + 3(16) = 20 + 48 = 68$. Both equal 68, so equilibrium price is ₹16 and quantity 68 units.

9.5 Shifts in Demand and Supply and Market Equilibrium

When demand or supply shifts, the equilibrium price and quantity change. The direction of change depends on which curve shifts.

Table 9.4 — Effect of a shift on equilibrium

Change	Equilibrium price	Equilibrium quantity
Demand rises (shift right)	Rises	Rises
Demand falls (shift left)	Falls	Falls
Supply rises (shift right)	Falls	Rises
Supply falls (shift left)	Rises	Falls

Application. A good monsoon shifts the supply of food grains to the right, lowering price and raising quantity. A rise in consumer income shifts demand to the right, raising both price and quantity. Understanding these shifts helps explain price movements in mandis and the need for price stabilisation.

Exam Focus — Lecture 9

Key points to remember

- Supply is the quantity offered for sale at a given price and time; it is a flow.
- Stock is total availability at a point of time; supply is the part offered for sale.
- Law of supply: direct relationship between price and quantity supplied.
- Reasons for upward slope: profit motive, rising cost of extra output, new entrants, shifting from other uses.
- Exceptions: perishable goods, fixed harvest, expectation of price rise, rare goods.
- Elasticity of supply $E_s = \%\Delta Q_s \div \%\Delta P$; five types.
- Market equilibrium is where demand equals supply.

Short-answer questions with model answers

Q1. Distinguish between stock and supply.

Model answer. Stock is the total quantity available with a producer at a point of time, while supply is that part of the stock which the producer is willing to sell at a given price during a period. Stock is a stock concept and supply a flow concept.

Q2. State the law of supply.

Model answer. Other things being equal, quantity supplied rises as price rises and falls as price falls; there is a direct relationship between price and quantity supplied.

Q3. Why is the supply of farm products relatively inelastic in the short run?

Model answer. Because the crop is already planted and its output cannot be changed quickly; the biological time lag and the fixed area under the crop make supply unresponsive to price changes within a season.

Long-answer questions with answer outlines

Q1. Explain the law of supply with a schedule and diagram, and state its assumptions and exceptions.

Outline. (i) Meaning of supply. (ii) Statement and schedule (Table 9.2). (iii) Diagram (Figure 9.1). (iv) Assumptions. (v) Reasons for upward slope. (vi) Exceptions. (vii) Determinants. (viii) Conclusion.

Q2. Discuss the elasticity of supply, its types and the factors influencing it.

Outline. (i) Definition and formula. (ii) Types with the value of E_s (Table 9.3). (iii) Factors influencing elasticity of supply. (iv) Importance. (v) Conclusion.

LECTURE 10

Market Structure and Digital Marketing

10.1 Meaning of Market

DEFINITION — MARKET

In economics, a market is not a place but an arrangement through which buyers and sellers of a commodity come into contact with one another and settle the price and quantity of the commodity bought and sold.

Thus a market may be a physical place (a mandi), or it may be spread over a whole region or the whole world. What makes a market is contact between buyers and sellers, not a building. The essential features of a market are: a commodity, buyers and sellers, and a price that emerges from their interaction.

10.2 Types of Market

Markets may be classified in several ways.

- **By area:** local, regional, national and international markets.
- **By time:** very short period (market period), short period and long period markets.
- **By competition:** perfectly competitive and imperfectly competitive markets (monopoly, monopolistic competition, oligopoly).
- **By regulation:** regulated markets (APMC mandis) and unregulated markets.
- **By nature of the commodity:** produce markets, factor markets and financial markets.

The classification by competition is the most important for price theory, and is discussed below.

10.3 Perfect Competition

DEFINITION — PERFECT COMPETITION

Perfect competition is a market structure in which there are a very large number of buyers and sellers of a homogeneous product, so that no single buyer or seller can influence the price, and the firm is a price taker.

Features of perfect competition

- **Very large number of buyers and sellers.** Each is too small to influence price.
- **Homogeneous product.** The product of one seller is identical to that of another.
- **Free entry and exit.** Firms can enter or leave the industry freely.
- **Perfect knowledge.** Buyers and sellers know the market price.
- **Perfect mobility of factors.** Factors can move freely between uses.
- **No transport cost and no government interference.**

Because the firm is a price taker, its average revenue equals marginal revenue equals price, and the firm's demand curve is a horizontal line (Figure 8.3, left panel). The firm's equilibrium output is where MC equals the given price.

> **Agricultural link.** The market for a single farmer's paddy or wheat closely resembles perfect competition: the farmer's output is homogeneous with others', the number of farmers is huge, and no single farmer can influence the market price. This is why farmers are "price takers" and why MSP is needed to support their income.

10.4 Imperfect Competition

Imperfect competition covers all market structures in which firms have some control over price.

Monopoly

> **DEFINITION — MONOPOLY**
>
> Monopoly is a market in which there is a single seller of a product that has no close substitutes, and entry of new firms is blocked.

- **Features:** one seller, no close substitute, strong barriers to entry (patents, licences, control of raw material, huge capital).
- **Price control:** the monopolist is a price maker, but is limited by the demand curve.
- **Revenue:** AR slopes downward; MR falls faster than AR (Figure 8.3, right panel).
- **Example:** Indian Railways (a near-monopoly in rail transport).

Monopolistic competition

> **DEFINITION — MONOPOLISTIC COMPETITION**
>
> Monopolistic competition is a market with many sellers of a differentiated product, where each seller has some control over price because of product differentiation, but faces competition from close substitutes.

- **Features:** many sellers, product differentiation (branding, quality, packaging), free entry and exit, some price control, heavy advertising.
- **Example:** soaps, toothpaste, branded food products.

Oligopoly

> **DEFINITION — OLIGOPOLY**
>
> Oligopoly is a market in which a few large sellers dominate, and each seller's price and output decisions are interdependent — each must consider the likely reaction of rivals.

- **Features:** few sellers, interdependence, price rigidity, non-price competition, possibility of collusion.
- **Example:** cement, telecom, automobiles, petroleum.

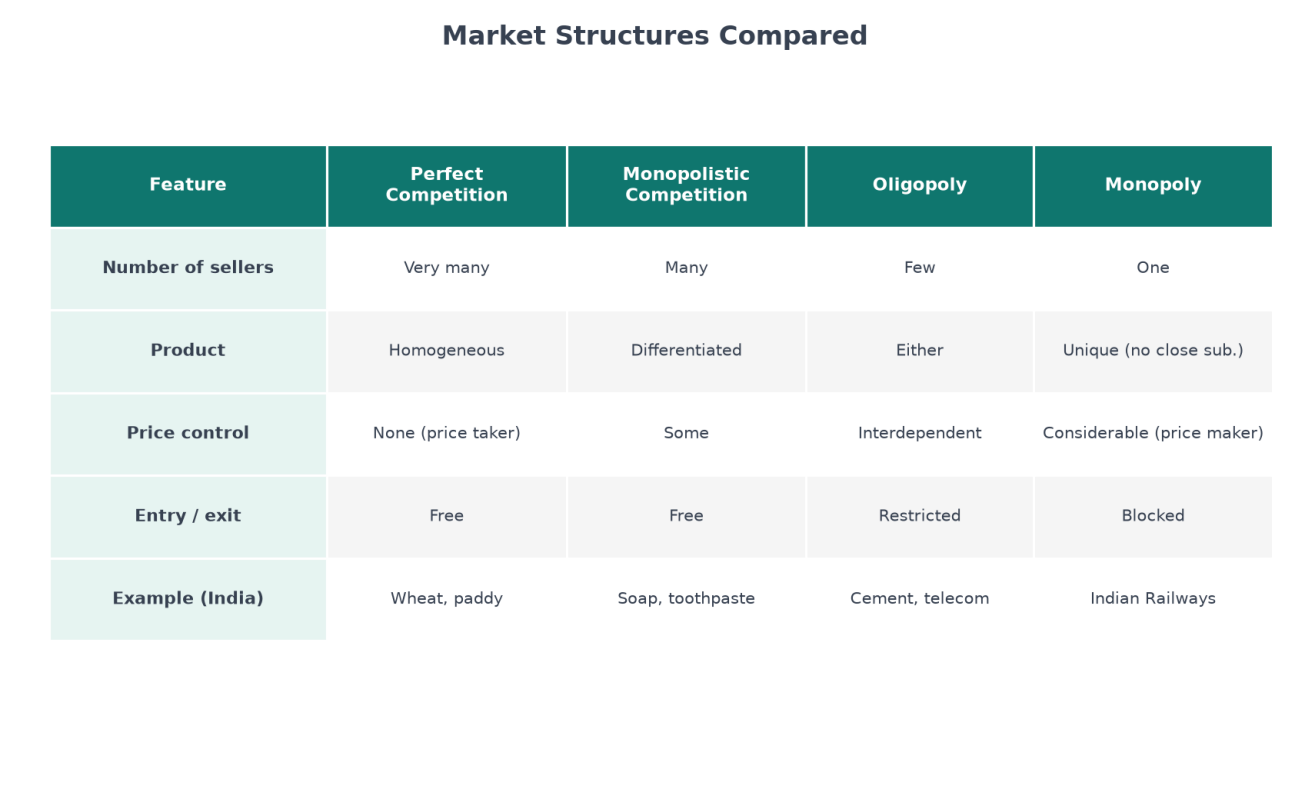


Figure 10.1 — A visual comparison of perfect competition, monopolistic competition, oligopoly and monopoly across number of sellers, product nature, price control, entry conditions and Indian examples.

Table 10.1 — Features of the four market structures

Feature	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Sellers	Very many	Many	Few	One
Product	Homogeneous	Differentiated	Either	Unique
Price control	None	Some	Interdependent	Considerable
Entry	Free	Free	Restricted	Blocked
Demand curve	Horizontal	Downward, elastic	Kinked / uncertain	Downward, inelastic
Indian example	Wheat, paddy (farmer level)	Soap, toothpaste	Cement, telecom	Indian Railways

10.5 Price and Output Determination — A Summary

- **Perfect competition.** The firm takes the market price and produces where MC = price. In the long run, entry and exit drive profit to normal, so price equals the minimum average cost.
- **Monopoly.** The firm produces where MR = MC and charges the price shown by the demand curve at that output. Because MR is below price, the monopolist produces less and charges more than under competition.
- **Monopolistic competition.** In the short run, firms may earn super-normal profit; in the long run, entry of rivals removes it, leaving only normal profit and excess capacity.

- **Oligopoly.** Because of interdependence, price tends to be rigid, and firms often compete through advertising, quality and service rather than price.

10.6 Digital Marketing — Meaning and Scope

> **DEFINITION — DIGITAL MARKETING**
>
> Digital marketing is the use of electronic channels — the internet, mobile phones, electronic trading platforms and social media — to connect buyers and sellers, discover prices, and buy and sell goods and services.

Scope in agriculture

- **Electronic trading platforms** that connect farmers to distant buyers and improve price discovery.
- **Mobile apps and SMS services** that give farmers real-time market prices (for example, Agmarknet data).
- **Direct-to-consumer sales** through websites, e-commerce and social media.
- **Digital payments** through UPI, bank transfers and mobile wallets.
- **Input and advisory services** delivered online — weather forecasts, soil-health cards, e-extension.

eNAM — the National Agriculture Market

eNAM is a pan-India electronic trading portal that networks the existing APMC mandis to create a unified national market for agricultural commodities. It was launched on 14 April 2016 and is implemented with the Small Farmers' Agribusiness Consortium (SFAC) as the lead agency. It allows transparent online auction and price discovery based on the quality of produce, and facilitates online payment to farmers.

Role of digital marketing in agricultural markets

- **Better price discovery.** Farmers can see prices across many markets and choose the best.
- **Reduction of middlemen.** Direct contact with buyers reduces the chain of intermediaries.
- **Wider market access.** A farmer in a remote village can reach buyers across the country.
- **Transparency.** Online auctions reduce the scope for manipulation and information asymmetry.
- **Faster payment.** Online settlement reduces delays in payment.
- **Reduced waste.** Better matching of supply and demand reduces losses of perishable produce.

Worked Example 10.1 — Comparing a competitive and a monopoly price

Problem. A market's demand is P = 100 − Q. Marginal cost is constant at ₹20. Find (a) the perfectly competitive price and output and (b) the monopoly price and output, and compare.

Solution.

- (a) **Perfect competition:** price = MC → 100 − Q = 20 → Q = 80, P = **₹20**.
- (b) **Monopoly:** TR = P×Q = (100 − Q)Q = 100Q − Q². MR = 100 − 2Q. Set MR = MC: 100 − 2Q = 20 → 2Q = 80 → Q = 40. Price = 100 − 40 = **₹60**.
- **Comparison:** the monopolist produces only 40 units and charges ₹60, whereas competition gives 80 units at ₹20. The monopolist restricts output and charges a higher price — the classic welfare loss of monopoly.

10.7 Marketing Channels and Price Spread

Between the farmer and the final consumer lies a chain of intermediaries. The **marketing channel** is the path along which produce moves from producer to consumer.

- **Farmer → Consumer** (direct sale, as in uzhavar sandhai) — the farmer gets the largest share.
- **Farmer → Village trader → Wholesaler → Retailer → Consumer** — the long chain, where the farmer's share is small.
- **Farmer → Commission agent → Wholesaler → Retailer → Consumer** (typical APMC mandi chain).

DEFINITION — PRICE SPREAD

Price spread is the difference between the price paid by the consumer and the price received by the producer. It covers the costs and margins of all the intermediaries in the marketing channel.

Reducing the price spread — through direct marketing, cooperatives, eNAM and farmer producer organisations — raises the farmer's share of the consumer's rupee.

Exam Focus — Lecture 10

Key points to remember

- A market is an arrangement for contact between buyers and sellers, not necessarily a place.
- Perfect competition: many buyers and sellers, homogeneous product, free entry, price taker.
- Monopoly: one seller, no close substitute, blocked entry, price maker.
- Monopolistic competition: many sellers of differentiated products, some price control.
- Oligopoly: few interdependent sellers, price rigidity.
- Under perfect competition AR = MR = price; under monopoly MR < AR.
- Digital marketing, especially eNAM, improves price discovery, transparency and market access for farmers.

Short-answer questions with model answers

Q1. Define a market in economics.

Model answer. A market is an arrangement through which buyers and sellers of a commodity come into contact to settle the price and quantity of the commodity bought and sold. It need not be a physical place.

Q2. What is the difference between monopoly and monopolistic competition?

Model answer. In monopoly there is a single seller with no close substitutes and blocked entry, so the firm has considerable price control. In monopolistic competition there are many sellers of differentiated products with free entry, so each firm has only some price control.

Q3. What is eNAM?

Model answer. eNAM (National Agriculture Market) is a pan-India electronic trading portal that networks APMC mandis into a unified national market for agricultural commodities, launched in 2016, to improve price discovery, transparency and farmers' access to buyers.

Long-answer questions with answer outlines

Q1. Explain the meaning and types of market, and the basic features of perfect and imperfect competition.

Outline. (i) Meaning of market. (ii) Types by area, time, competition and regulation. (iii) Features of perfect competition. (iv) Monopoly, monopolistic competition and oligopoly. (v) Table 10.1 and Figure 10.1. (vi) Price and output determination summary. (vii) Conclusion.

Q2. Discuss digital marketing — its meaning, scope and role in agricultural markets, with reference to eNAM.

Outline. (i) Meaning of digital marketing. (ii) Scope in agriculture. (iii) eNAM — features and objectives. (iv) Role: price discovery, reduced middlemen, transparency, faster payment. (v) Challenges. (vi) Conclusion.

LECTURE 11

Distribution Theory: Rent, Wages and Population

11.1 Meaning of Distribution

DEFINITION — DISTRIBUTION

Distribution theory deals with how the total product (or national income) is shared among the factors of production — land, labour, capital and organisation — in the form of rent, wages, interest and profit.

Production creates income; distribution shares it. The rewards are: **rent** for land, **wages** for labour, **interest** for capital and **profit** for enterprise. Distribution theory explains how each reward is determined.

11.2 Factor Market and Pricing of Factors

DEFINITION — FACTOR MARKET

A factor market is the market in which the services of the factors of production are bought and sold. Here households supply factors and firms demand them.

The price of a factor (rent, wage, interest) is determined by the demand for and supply of that factor, just as the price of a good is determined by the demand for and supply of the good. But there is an important difference: the demand for a factor is a **derived demand** — it is derived from the demand for the goods the factor helps to produce.

Demand for a factor

A firm demands a factor up to the point where the value of the marginal product of the factor equals the factor's price.

$$\text{Firm hires a factor until} \quad \text{VMP} = \text{factor price} \quad \text{where VMP} = \text{MP} \times \text{price of output}$$

Supply of a factor

- **Land** — fixed in supply, so its supply curve is vertical.
- **Labour** — supply depends on population, wages and working conditions.
- **Capital** — supply depends on saving and the rate of interest.

11.3 Rent

DEFINITION — RENT (ECONOMIC RENT)

Economic rent is the payment made for the use of land and other factors whose supply is fixed. In the Ricardian sense, rent is that part of the produce which is paid to the landlord for the use of the original and indestructible powers of the soil.

Ricardian theory of rent

- Rent arises because land differs in fertility and location.
- On the least fertile (marginal) land, the produce just covers the cost of cultivation, so there is no rent. This is the **no-rent land**.
- On better land, the same effort yields more produce. The surplus over the marginal land's produce is rent.
- Hence rent is a **differential surplus** — it is high for good land and zero for marginal land.
- Rent is a surplus, not a cost of production, so it does not enter the price of the product. Rather, rent is determined by the price.

Quasi-rent

> **DEFINITION — QUASI-RENT**
>
> Quasi-rent is the earnings of a factor whose supply is fixed only in the short run but variable in the long run. It is a temporary surplus earned by machines and other man-made capital.

In the short run, a machine or a building cannot be changed, so its earnings may exceed its variable cost; this surplus is quasi-rent. In the long run, when the machine wears out and is replaced, the supply becomes variable and quasi-rent disappears. Thus quasi-rent is a short-run phenomenon, whereas true rent is a long-run phenomenon on land.

Table 11.1 — Rent versus quasi-rent

Basis	Rent	Quasi-rent
Factor	Land (free gift of nature)	Man-made capital (machines, buildings)
Supply	Fixed in both short and long run	Fixed only in the short run
Duration	Permanent (long run)	Temporary (short run)
Nature	Differential surplus	Short-run surplus over variable cost
Example	Rent of fertile delta land	Earnings of a tractor in the short run

11.4 Wages

> **DEFINITION — WAGES**
>
> Wages are the reward paid for the services of labour. **Money (nominal) wage** is the amount of money paid to a worker. **Real wage** is the quantity of goods and services that money wage can buy.

$$\text{Real wage} = \text{Money wage} \div \text{Price level (cost of living)}$$

Real wage is the true measure of a worker's well-being. If money wages rise by 10% but prices rise by 12%, the worker is worse off, because real wage has fallen.

Theories of wages

- **Subsistence theory.** Wages tend to settle at the minimum needed for the worker's survival and reproduction.
- **Wage-fund theory.** Wages depend on the fund set aside for paying labour divided by the number of workers.
- **Marginal productivity theory.** Under competition, wages equal the value of the marginal product of labour. This is the modern explanation.
- **Bargaining theory.** Wages are settled by bargaining between workers and employers, influenced by the strength of trade unions.

Factors determining real wages

- Level of money wages.
- Price level and the cost of living.
- Availability of wage goods and public services.
- Efficiency, skill and productivity of labour.
- Bargaining power of workers and government labour policy.

11.5 Population

Importance of the study of population

- Population is both the source of labour supply and the market for goods.
- Its size, growth and composition affect production, consumption and the standard of living.
- Rapid growth strains food, land, water, jobs, housing and public services.
- Population studies guide policies on family welfare, education, health and employment.

Malthusian theory of population

> **STATEMENT**
>
> Malthus argued that population tends to grow in geometric progression (2, 4, 8, 16 ...) while the means of subsistence grow only in arithmetic progression (2, 4, 6, 8 ...), so population tends to outrun food supply, leading to poverty, disease and famine unless checked.

- **Positive checks** (which raise the death rate): famine, disease, war and misery.
- **Preventive checks** (which lower the birth rate): moral restraint, late marriage and celibacy.

Criticism. Malthus underestimated the growth of food production made possible by technology (the Green Revolution is a good example) and the role of education, urbanisation and contraception in lowering birth rates. His forecast did not materialise in the developed world. But his warning remains relevant for regions where population grows faster than food and jobs.

Optimum theory of population

> **STATEMENT**
>
> Optimum population is that size of population which, combined with the available resources and capital, gives the maximum per capita income (or maximum output per head).

If population is below the optimum, there is under-population and adding workers raises output per head. If population is above the optimum, there is over-population and adding workers lowers output per head. The optimum is not fixed: it rises with capital, technology and resources.

Table 11.2 — Malthusian versus optimum theory of population

Basis	Malthusian theory	Optimum theory
Focus	Population versus food supply	Population versus resources and capital
Key idea	Population grows faster than food	Best size gives maximum per capita income
Checks	Positive and preventive checks	No automatic checks; policy to reach optimum
Nature	Pessimistic	Balanced and dynamic

Natural and socio-economic determinants of population

- **Natural (biological):** birth rate and death rate.
- **Social:** marriage age, customs, caste and religious attitudes, status of women, education (especially female education).
- **Economic:** income level, employment, cost of raising children, urbanisation.
- **Health and medical facilities:** lower death rates and longer life expectancy.
- **Migration:** movement of people into and out of a region.

Population policy and programmes in India

India was among the first countries in the world to launch a national family planning programme, in 1952. The present framework is the **National Population Policy, 2000**, which works on a voluntary and target-free approach. Its objectives are: to meet the unmet need for contraception and reproductive health care (immediate), to bring the total fertility rate (TFR) down to replacement level of 2.1 (medium term), and to achieve a stable population by 2045, consistent with sustainable development (long term).

Related programmes include Mission Parivar Vikas, the National Health Mission's family-planning services, and incentives such as the promotion of the small-family norm and delayed marriage of girls. As a result of these efforts, India's TFR has fallen steadily over recent decades, and a majority of states have already reached replacement-level fertility.

Worked Example 11.1 — Population growth rate

Problem. A district's population was 8,00,000 in 2011 and 9,20,000 in 2021. Find (a) the decadal growth rate and (b) the simple annual growth rate.

Solution.

- Increase = 9,20,000 − 8,00,000 = 1,20,000.
- (a) Decadal growth rate = (1,20,000 / 8,00,000) × 100 = **15%**.
- (b) Simple annual growth rate = 15% ÷ 10 = **1.5% per year**.

Note. For greater accuracy, the compound annual growth rate (CAGR) is used: CAGR = [(9,20,000/8,00,000)^(1/10) − 1] × 100 ≈ **1.41% per year**.

11.6 Agricultural Wages in India

Wages in Indian agriculture have several special features:

- **Payment in kind and cash.** Farm workers are often paid partly in grain, meals or other goods.
- **Seasonal employment.** Work is available mainly during sowing and harvest, causing seasonal unemployment.
- **Wide variation** between regions, crops and sexes.
- **Weak bargaining power** because of the large supply of unskilled labour and the absence of strong unions.
- **Low real wages** relative to non-farm work, which drives migration to cities.

Minimum wages for agricultural labour are fixed by state governments. Programmes such as the Mahatma Gandhi National Rural Employment Guarantee Scheme provide a floor to rural wages by offering alternative employment in the lean season.

Exam Focus — Lecture 11

Key points to remember

- Distribution theory explains how income is shared as rent, wages, interest and profit.
- Factor demand is derived demand; a firm hires a factor until VMP = factor price.
- Ricardian rent is a differential surplus; marginal land earns no rent.
- Quasi-rent is the short-run surplus earned by man-made capital.
- Real wage = money wage ÷ price level; it measures true well-being.
- Malthus: population grows geometrically, food arithmetically; checks are positive and preventive.
- Optimum population gives the maximum per capita income.
- India's National Population Policy 2000 targets replacement-level TFR (2.1) and a stable population by 2045.

Short-answer questions with model answers

Q1. Define economic rent and quasi-rent.

Model answer. Economic rent is the payment for the use of land and other factors whose supply is fixed; in the Ricardian sense it is the differential surplus of better land over marginal land. Quasi-rent is the short-run surplus earned by man-made capital whose supply is fixed only in the short run.

Q2. Distinguish between money wage and real wage.

Model answer. Money wage is the amount of money paid to a worker; real wage is the quantity of goods and services that money wage can buy, i.e. money wage divided by the price level. Real wage is the true measure of a worker's standard of living.

Q3. State the optimum theory of population.

Model answer. Optimum population is the size of population which, with the given resources and capital, yields the maximum per capita income. Below it there is under-population; above it there is over-population.

Long-answer questions with answer outlines

Q1. Explain the Ricardian theory of rent and the concept of quasi-rent.

Outline. (i) Meaning of rent. (ii) Ricardian theory — differential surplus, marginal land, no-rent land. (iii) Rent as surplus, not cost. (iv) Quasi-rent — definition and example. (v) Rent versus quasi-rent table (Table 11.1). (vi) Conclusion.

Q2. Discuss the theories of population and the population policy of India.

Outline. (i) Importance of population study. (ii) Malthusian theory and criticism. (iii) Optimum theory. (iv) Comparison table (Table 11.2). (v) Natural and socio-economic determinants. (vi) India's National Population Policy 2000 and programmes. (vii) Conclusion.

LECTURE 12

Interest, Profit and Business Organisations

12.1 Interest

DEFINITION — INTEREST

Interest is the price paid for the use of capital. It is the reward to the owner of capital for parting with it for a period of time.

Capital is a scarce resource. A person who lends money or capital gives up the use of it for a period, and expects a return for that sacrifice. Interest is that return. It is usually expressed as a percentage per year, called the rate of interest.

Pure interest and gross interest

DEFINITIONS

Pure (net) interest is the payment made purely for the use of capital funds, with no allowance for risk, inconvenience or management. **Gross interest** is the total payment made by a borrower, which includes pure interest plus other charges.

Gross interest has four components:

- **Pure interest** — the reward for the use of capital.
- **Risk premium** — compensation for the risk that the borrower may not repay.
- **Reward for inconvenience** — for the trouble of lending and managing the loan.
- **Reward for management** — for the work of looking after the investment.

$$\text{Gross interest} = \text{Pure interest} + \text{Risk premium} + \text{Reward for inconvenience} + \text{Reward for management}$$

In agriculture, the difference between pure and gross interest explains why moneylenders charge very high rates: their risk premium and management charges are large, while the pure interest is small. Institutional credit through the Kisan Credit Card carries a low rate because risk is pooled and management costs are low.

Theories of interest

- **Loanable-funds theory.** The rate of interest is determined by the demand for and supply of loanable funds. Demand comes from investment; supply comes from savings and new money.
- **Liquidity-preference theory (Keynes).** Interest is the reward for parting with liquidity. People prefer to hold money for transactions, precaution and speculation; the rate of interest is the price that persuades them to give up cash.
- **Productivity theory.** Capital is productive, so it earns a return; interest is the reward for that productivity.

- **Abstinence theory.** Interest is the reward for the saving (abstinence from consumption) that creates capital.

Factors affecting the rate of interest

- Demand for and supply of capital funds.
- Rate of inflation (higher expected inflation raises the nominal rate).
- Central bank policy (the repo rate set by the Reserve Bank of India).
- Risk of the borrower and the purpose of the loan.
- Duration and liquidity of the loan.

12.2 Profit

DEFINITION — PROFIT

Profit is the reward of the entrepreneur for organising production, taking decisions and bearing risk. It is the residual income left after all other factors have been paid.

$$\text{Accounting profit} = \text{Total revenue} - \text{Explicit cost}$$
$$\text{Economic profit} = \text{Total revenue} - (\text{Explicit cost} + \text{Implicit cost})$$

Economic (pure) profit and normal profit

- **Normal profit** is the minimum return necessary to keep the entrepreneur in the business. It is a part of cost, because without it the entrepreneur would leave.
- **Economic (super-normal or pure) profit** is the profit over and above normal profit. It is the surplus left after all costs, including implicit costs and normal profit, are met.
- In the long run, under perfect competition, economic profit tends to disappear and only normal profit remains, because new firms enter the industry.

Sources of profit

- **Risk and uncertainty.** The entrepreneur bears the risk of price, weather and demand changes.
- **Innovation.** Introducing new products, methods or markets gives temporary profit.
- **Monopoly power.** Control over the market can yield sustained profit.
- **Efficiency.** Lower costs than rivals raise profit.

12.3 Forms of Business Organisations

Production is carried on through different forms of business organisation. Each has its own advantages and disadvantages, and the choice depends on the scale, capital, risk and control desired.

Table 12.1 — Forms of business organisations compared

Form	Meaning	Advantages	Disadvantages
Sole proprietorship	Owned and run by one person	Easy to start; full control; quick decisions; profit not shared	Unlimited liability; limited capital; limited life; heavy risk

Form	Meaning	Advantages	Disadvantages
Partnership	Two or more persons share ownership and profit	More capital; shared risk; combined skills	Unlimited liability; possible disputes; limited capital
Joint Hindu family firm	Business carried on by members of a Hindu undivided family	Simple; family loyalty; continuity	Limited to family; unlimited liability; slow decisions
Joint stock company	An artificial person with shareholders; limited liability	Large capital; limited liability; continuity; transferable shares	Complex to form; double taxation; separation of ownership and control
Cooperative society	Voluntary association for mutual benefit	Member service; one member one vote; government support	Limited capital; weak management; political interference
Public corporation	Owned by government for a public purpose	Serves public interest; large scale; security	Bureaucracy; inefficiency; political interference

Cooperatives in Indian agriculture

Cooperative societies are especially important in Indian agriculture. They provide:

- **Credit** — Primary Agricultural Cooperative Credit Societies (PACS) give short-term and medium-term loans.
- **Marketing** — marketing cooperatives help farmers sell produce collectively and get better prices.
- **Processing** — sugar factories, milk plants and cotton ginning run on a cooperative basis.
- **Input supply** — cooperatives supply seed, fertiliser and implements at fair prices.

The **cooperative dairy model** (as in Gujarat's Amul and similar unions in Tamil Nadu) shows how small producers can gain from collective marketing and processing, which they could not achieve individually.

Worked Example 12.1 — Simple and compound interest

Problem. A farmer borrows ₹50,000 at 9% per annum. Find (a) the simple interest for 2 years and (b) the compound interest for 2 years (compounded annually).

Solution.

- (a) Simple interest = P × R × T / 100 = 50,000 × 9 × 2 / 100 = **₹9,000**.
- (b) Amount under compound interest = $P\,(1 + R/100)^{T}$ = $50{,}000 \times (1.09)^{2}$ = 50,000 × 1.1881 = ₹59,405.
- Compound interest = 59,405 − 50,000 = **₹9,405**.

Comparison. Compound interest is ₹405 more than simple interest for the same period, because interest is charged on interest.

Worked Example 12.2 — Economic profit

Problem. A farmer's total revenue is ₹8,00,000. His explicit costs (seed, fertiliser, hired labour, diesel) are ₹5,00,000. He uses his own land, which could be rented for ₹80,000, and his own labour, worth ₹70,000. Find accounting profit and economic profit.

Solution.

- Accounting profit = TR − explicit cost = 8,00,000 − 5,00,000 = **₹3,00,000**.
- Implicit cost = own land rent (₹80,000) + own labour (₹70,000) = ₹1,50,000.
- Economic profit = TR − (explicit + implicit) = 8,00,000 − (5,00,000 + 1,50,000) = **₹1,50,000**.

Interpretation. The accounting profit looks large, but the economic profit is smaller because the farmer's own resources have an opportunity cost. If economic profit were negative, the farmer would be better off renting out the land and working elsewhere.

12.4 Risk and Uncertainty in Agriculture

Because profit is the reward for bearing risk, risk is central to agricultural business.

- **Production risk** — weather, pests and disease reduce output.
- **Price risk** — market prices may fall below the cost of production.
- **Personal and institutional risk** — illness, accidents and changes in policy.

Farmers manage risk through diversification (growing several crops), crop insurance (Pradhan Mantri Fasal Bima Yojana), irrigation to reduce weather dependence, and price support through MSP. Reducing risk raises the willingness of farmers to invest and adopt new technology.

Exam Focus — Lecture 12

Key points to remember

- Interest is the reward for the use of capital; it is a price for parting with liquidity.
- Pure interest is only the reward for capital; gross interest adds risk, inconvenience and management charges.
- Accounting profit = TR − explicit cost; economic profit = TR − (explicit + implicit cost).
- Normal profit is part of cost; super-normal (economic) profit is above normal profit.
- Sources of profit: risk, innovation, monopoly power, efficiency.
- Forms of business: sole proprietorship, partnership, joint Hindu family firm, joint stock company, cooperative, public corporation.
- Cooperatives are important in Indian agriculture for credit, marketing, processing and input supply.

Short-answer questions with model answers

Q1. Distinguish between pure interest and gross interest.

Model answer. Pure interest is the payment made only for the use of capital funds. Gross interest is the total payment made by a borrower, which includes pure interest plus a risk premium, reward for inconvenience and reward for management.

Q2. Define economic profit.

Model answer. Economic profit is total revenue minus total economic cost, where economic cost includes both explicit costs (actual payments) and implicit costs (the value of self-owned resources). It is the surplus over and above normal profit.

Q3. Name any four forms of business organisation.

Model answer. Sole proprietorship, partnership, joint stock company and cooperative society (also joint Hindu family firm and public corporation).

Long-answer questions with answer outlines

Q1. Explain the concepts of pure interest and gross interest, and the theories of interest.

Outline. (i) Meaning of interest. (ii) Pure versus gross interest with components. (iii) Theories — loanable funds, liquidity preference, productivity, abstinence. (iv) Factors affecting the rate of interest. (v) Conclusion.

Q2. Discuss profit and the forms of business organisations, with special reference to cooperatives in Indian agriculture.

Outline. (i) Meaning of profit; normal versus economic profit. (ii) Sources of profit. (iii) Forms of business (Table 12.1). (iv) Cooperatives — credit, marketing, processing, inputs. (v) Conclusion.

LECTURE 13

National Income: Concepts and Measurement

13.1 Meaning of National Income

DEFINITION — NATIONAL INCOME

National income is the total money value of all final goods and services produced by a country in a year, together with the net income earned from abroad. It is the aggregate of the incomes earned by the factors of production of a country during a period.

National income measures the economic performance of a country. It is the sum of the values added by all producing units, or equivalently, the sum of all factor incomes (rent, wages, interest and profit) generated in the economy during a year.

13.2 Importance of National Income

- **Measures economic performance** and the rate of growth of the economy.
- **Shows the standard of living** through per capita income.
- **Helps compare** the economy over time and with other countries.
- **Guides policy** on taxation, public expenditure, planning and investment.
- **Reveals the structure** of the economy — the shares of agriculture, industry and services.
- **Basis of planning** — targets for growth and the allocation of resources.

13.3 The Circular Flow of Income

National income moves in a circular flow among the sectors of the economy. In the simplest two-sector model, households and firms are linked by two sets of flows through the factor market and the product market.

- **Through the factor market:** households supply factor services (land, labour, capital) to firms; firms pay factor incomes (rent, wages, interest, profit) to households.
- **Through the product market:** households spend money on goods and services; firms supply goods and services to households.

The two flows are two sides of the same coin: the money paid out by firms as factor incomes comes back to them as consumer expenditure. In this simple model, income = output = expenditure. Adding government and the foreign sector gives the four-sector model.

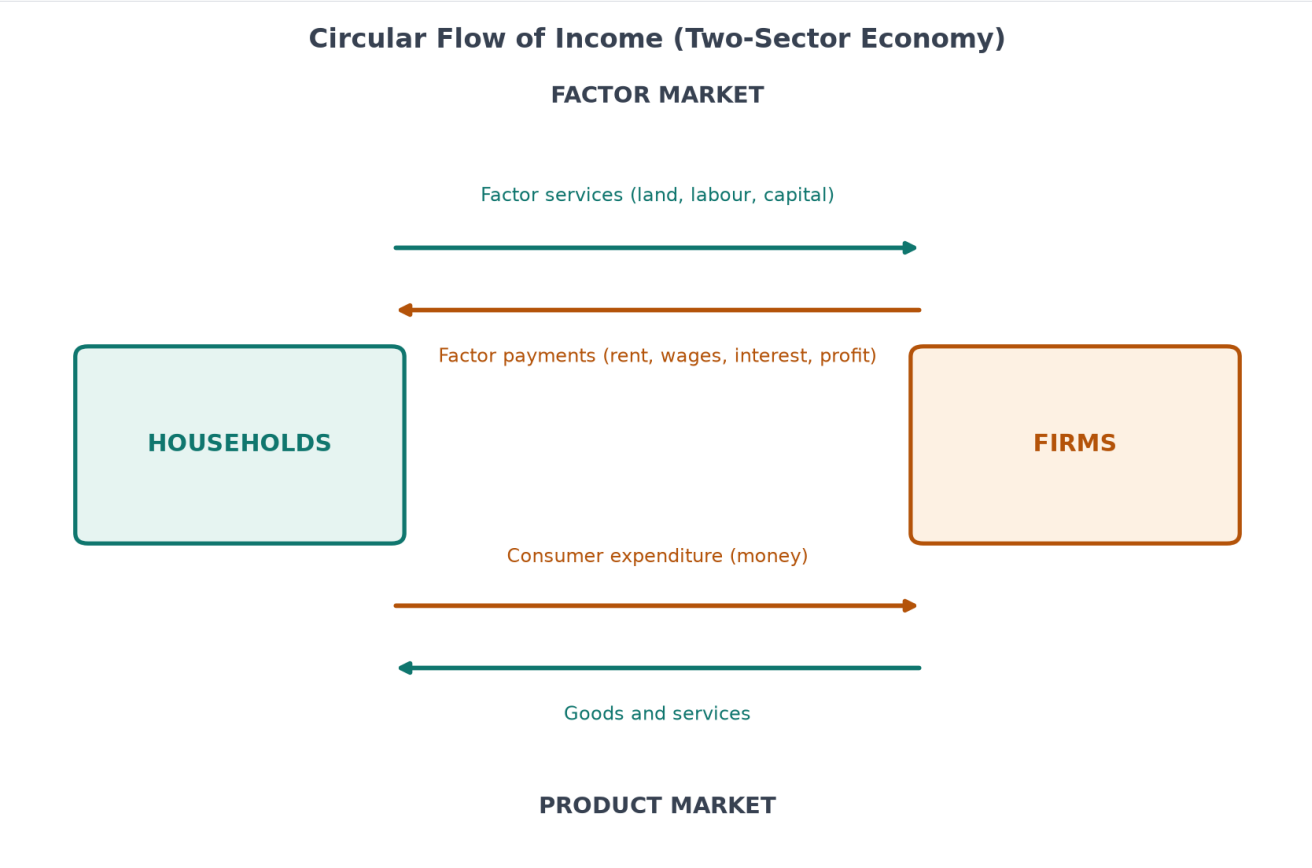


Figure 13.1 — The circular flow of income in a two-sector economy. Households supply factor services to firms through the factor market and receive factor payments; they spend on goods and services through the product market, which returns the money to firms.

13.4 Concepts of National Income

KEY CONCEPTS

Gross Domestic Product (GDP) — value of all final goods and services produced within the domestic territory of a country in a year. **Gross National Product (GNP)** — GDP plus net factor income from abroad. **Net National Product (NNP)** — GNP minus depreciation. **NNP at factor cost** is national income.

- **GDP at market price** = value of output at market prices.
- **GDP at factor cost** = GDP at market price − indirect taxes + subsidies.
- **GNP** = GDP + net factor income from abroad (NFIA).
- **NNP** = GNP − depreciation.
- **National income (NNP at factor cost)** = NNP at market price − indirect taxes + subsidies.
- **Per capita income** = National income ÷ population.
- **Personal income** = income actually received by households.
- **Disposable income** = personal income − direct taxes; it is the income available for spending or saving.

$$\text{GDP} \rightarrow +\text{NFIA} = \text{GNP} \rightarrow -\text{Depreciation} = \text{NNP} \rightarrow -\text{Indirect taxes} + \text{Subsidies} = \text{National Income}$$

Table 13.1 — Measures of national income and their relationships

Measure	Meaning	Relationship
GDP (market price)	Output within domestic territory at market prices	Base measure
GDP (factor cost)	Output within territory at factor cost	GDP_{mp} − indirect taxes + subsidies
GNP	Income of residents, wherever earned	GDP + NFIA
NNP	GNP after deducting depreciation	GNP − depreciation
National income	NNP at factor cost	NNP_{mp} − indirect taxes + subsidies
Per capita income	Average income per person	National income ÷ population
Disposable income	Income available for spending/saving	Personal income − direct taxes

13.5 National Income Accounting and Approaches to Measurement

There are three ways to measure national income, all of which must give the same total because income = output = expenditure.

(a) Product (value-added) method

Add the money value of all final goods and services produced in the economy. To avoid double counting, only the **value added** at each stage is counted, or only the value of **final** goods.

$$\text{National income} = \Sigma\,(\text{Value of output} - \text{Intermediate consumption}) + \text{Net factor income from abroad}$$

(b) Income method

Add all the factor incomes earned in producing the national product: compensation of employees + rent + interest + profit + mixed income of the self-employed, plus net factor income from abroad.

$$\text{National income} = \text{Compensation of employees} + \text{Rent} + \text{Interest} + \text{Profit} + \text{Mixed income} + \text{NFIA}$$

(c) Expenditure method

Add all final expenditures on the national product: private consumption + investment + government expenditure + net exports.

$$\text{GDP} = C + I + G + (X - M)$$

Difficulties in the measurement of national income

- **Double counting.** The same output may be counted more than once unless value added is used.
- **Non-monetised transactions.** A large part of farm output is consumed at home and is not sold in the market.
- **Unpaid services.** Services of housewives and family labour are not counted.
- **Illiteracy and ignorance.** Producers may not keep accounts or report accurately.
- **Problem of imputation.** It is hard to value self-consumed output and own-account services.
- **Underground economy.** Incomes from illegal or hidden activities are not reported.
- **Price changes.** Inflation distorts comparisons unless real (constant-price) figures are used.
- **Transfer payments.** Pensions and scholarships are not payments for production and must be excluded.

Worked Example 13.1 — National income by the product (value-added) method

Problem. A simple economy has three sectors. Using the value-added method, find GDP at factor cost.

Sector	Output (₹ crore)	Intermediate consumption (₹ crore)
Agriculture	500	100
Industry	800	350
Services	600	200

Solution. Value added = output − intermediate consumption.

- Agriculture: 500 − 100 = 400
- Industry: 800 − 350 = 450
- Services: 600 − 200 = 400
- GDP at factor cost = 400 + 450 + 400 = **₹1,250 crore**.

Worked Example 13.2 — National income by the income method

Problem. Given: compensation of employees ₹900 crore; rent ₹200 crore; interest ₹150 crore; profit ₹250 crore; mixed income of self-employed ₹300 crore; net factor income from abroad −₹50 crore. Find national income.

Solution.

- Domestic factor income = 900 + 200 + 150 + 250 + 300 = ₹1,800 crore.
- National income = domestic factor income + NFIA = 1,800 + (−50) = **₹1,750 crore**.

Worked Example 13.3 — GDP by the expenditure method

Problem. Private consumption = ₹2,000 crore; investment = ₹800 crore; government expenditure = ₹600 crore; exports = ₹300 crore; imports = ₹400 crore. Find GDP.

Solution.

- GDP = C + I + G + (X − M)
- = 2,000 + 800 + 600 + (300 − 400)
- = 3,400 − 100 = **₹3,300 crore**.

Note. All three methods must give the same GDP, because total output, total income and total expenditure are three views of the same flow.

13.6 Real versus Nominal GDP and the GDP Deflator

National income figures at current prices can rise simply because prices have risen. To see the true change in output, we use real GDP.

- **Nominal (money) GDP** is measured at current-year prices.
- **Real GDP** is measured at the prices of a fixed base year, so it shows the change in actual output.
- **GDP deflator** is an index of the price level, obtained by dividing nominal GDP by real GDP.

$$\text{Real GDP} = (\text{Nominal GDP} \div \text{GDP deflator}) \times 100 \quad | \quad \text{GDP deflator} = (\text{Nominal GDP} \div \text{Real GDP}) \times 100$$

For example, if nominal GDP rises by 12% in a year while the deflator rises by 8%, real GDP has grown by only about 4%. Real GDP is the correct measure for judging economic growth and the standard of living over time.

13.7 National Income as a Measure of Welfare — Limitations

National income is a useful but imperfect measure of welfare. Its limitations include:

- It ignores the **distribution of income** — a high average income may hide widespread poverty.
- It does not value **non-market activities** such as household work and self-consumed farm output.
- It ignores **leisure** and the quality of life.
- It does not deduct the **cost of environmental damage** or the depletion of natural resources.
- It counts "regrettable necessities" (such as defence and pollution clean-up) as if they added welfare.

Because of these gaps, economists also use measures such as the Human Development Index, which combines income with education and health, to judge the true standard of living.

Exam Focus — Lecture 13

Key points to remember

- National income is the money value of all final goods and services produced in a year, plus net income from abroad.
- GDP is domestic product; GNP = GDP + NFIA; NNP = GNP − depreciation.
- National income = NNP at factor cost; per capita income = national income ÷ population.
- The circular flow shows income = output = expenditure.
- Three measurement methods: product (value added), income and expenditure.
- Expenditure method: GDP = C + I + G + (X − M).
- Difficulties: double counting, non-monetised output, unpaid services, illiteracy, imputation, underground economy.

Short-answer questions with model answers

Q1. Define national income.

Model answer. National income is the total money value of all final goods and services produced by a country in a year, together with net income from abroad. It equals the sum of all factor incomes (rent, wages, interest and profit) earned during the year.

Q2. Distinguish between GDP and GNP.

Model answer. GDP is the value of all final goods and services produced within the domestic territory of a country. GNP is GDP plus net factor income from abroad, i.e. the income earned by the country's residents wherever they are.

Q3. State the three methods of measuring national income.

Model answer. (i) Product or value-added method, (ii) income method, and (iii) expenditure method. All three give the same total because output, income and expenditure are three views of the same flow.

Long-answer questions with answer outlines

Q1. Explain the concepts of national income and the circular flow of income.

Outline. (i) Meaning and importance. (ii) Circular flow — two-sector model (Figure 13.1). (iii) Concepts — GDP, GNP, NNP, national income, per capita income (Table 13.1). (iv) Conclusion.

Q2. Discuss the approaches to the measurement of national income and the difficulties involved.

Outline. (i) Product/value-added method with example. (ii) Income method with example. (iii) Expenditure method with example. (iv) Why all three agree. (v) Difficulties in measurement. (vi) Conclusion.

LECTURE 14

Money, Prices and Agricultural Finance

14.1 Barter System and Its Problems

DEFINITION — BARTER

Barter is the direct exchange of goods and services for other goods and services, without the use of money.

Before money was invented, people exchanged goods directly — say, a bag of rice for a goat. Barter works only in a small, simple economy. As trade grew, its problems became severe.

Problems of the barter system

- **Lack of double coincidence of wants.** Each trader must want exactly what the other has to offer. A farmer with rice must find someone who has cloth and wants rice at the same time.
- **Lack of a common measure of value.** There is no common unit to express the value of different goods.
- **Indivisibility of certain goods.** A large item such as a cow cannot be divided to buy a small item such as salt.
- **Difficulty in storing value.** Goods like grain and milk cannot be stored for long, so wealth cannot be saved easily.
- **Problem of deferred payments.** It is hard to make future payments (loans) when values keep changing.
- **Lack of standard for deferred payment and transfer of value.** There is no easy way to send value from one place to another.

These problems led to the invention of money.

14.2 Evolution of Money

Money evolved in stages, each solving a problem of the earlier stage:

1. **Commodity money.** Useful goods such as cattle, salt, grain, shells and beads served as money.
2. **Metallic money.** Gold, silver and copper coins, because they were durable, divisible and portable.
3. **Paper money.** Notes issued by governments and banks, backed by law, because carrying metal was inconvenient.
4. **Credit money.** Cheques, drafts and bills of exchange, which transfer bank deposits.
5. **Plastic money.** Credit and debit cards.
6. **Digital money.** Electronic transfers, mobile wallets, UPI and central bank digital currency — money that exists only as entries in a computer.

Indian example. In India, coins and currency notes are issued by the Reserve Bank of India, while coins are minted by the Government. The Unified Payments Interface (UPI) has made digital money a part of everyday transactions, including payments at agricultural markets and to farmers under schemes such as PM-KISAN.

14.3 Meaning and Functions of Money

DEFINITION — MONEY

Money is anything that is generally accepted as a medium of exchange and as a measure of value, and that is used to settle debts and store wealth.

Primary functions

- **Medium of exchange.** Money is accepted in payment for goods and services, removing the need for a double coincidence of wants.
- **Measure of value (unit of account).** Money provides a common unit in which the value of all goods is expressed.

Secondary functions

- **Store of value.** Money can be saved for future use because it is durable and stable in value.
- **Standard of deferred payment.** Money allows loans and future payments to be fixed and settled.
- **Transfer of value.** Money can be moved from one place or person to another easily.

Contingent functions

- Basis of credit and the banking system.
- Distribution of national income.
- Measurement and maximisation of utility.
- Basis of the price system.

14.4 Classification of Money

Table 14.1 — Classification of money

Type	Meaning	Example
Full-bodied money	Money whose commodity value equals its face value	Gold and silver coins
Token money	Money whose face value exceeds its commodity value	Modern coins of small denomination
Legal tender money	Money that must be accepted by law	Rupee notes and coins
Optional money	Money accepted by choice, not by law	Cheques, drafts
Representative money	Paper money representing a claim on metal	Convertible notes (historically)
Credit money	Claims against banks and institutions	Bank deposits, cheques
Near money	Assets easily convertible into money	Bills, bonds, savings deposits

14.5 Money Supply

DEFINITION — MONEY SUPPLY

Money supply is the total stock of money in circulation with the public at a point of time. It excludes the money held by banks and the government.

The Reserve Bank of India measures money supply in several forms:

- **M1 (narrow money)** = currency with the public + demand deposits with banks + other deposits with the RBI.
- **M2** = M1 + savings deposits with post office savings banks.
- **M3 (broad money)** = M1 + net time deposits with banks. M3 is the most commonly used measure.
- **M4** = M3 + total post office deposits.

Money supply is affected by the central bank's policy, the government's deficit, and the behaviour of banks and the public.

14.6 General Price Index

DEFINITION — PRICE INDEX

A price index measures the average change in the prices of a group of goods and services between two periods, relative to a base period whose value is taken as 100.

$$\text{Price index} = (\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times 100$$

Important price indices in India include the **Consumer Price Index (CPI)**, which measures retail prices and is used for inflation targeting and dearness allowance, and the **Wholesale Price Index (WPI)**, which measures wholesale prices. A price index is used to convert nominal (money) values into real values.

$$\text{Real value} = (\text{Nominal value} \div \text{Price index}) \times 100$$

14.7 Inflation and Deflation

DEFINITIONS

Inflation is a sustained rise in the general price level, which reduces the purchasing power of money. **Deflation** is a sustained fall in the general price level, which raises the purchasing power of money.

Types of inflation

- **By speed:** creeping (mild), walking, running and hyper-inflation.
- **By cause:** demand-pull (demand exceeds supply) and cost-push (rising costs of production, especially wages and inputs).

Causes of inflation

- Increase in money supply relative to output.

- Rise in demand (demand-pull).
- Rise in the cost of production (cost-push).
- Shortage of goods, including food and fuel.
- Hoarding, black marketing and speculation.

Effects of inflation

- Falls in the value of money and a rise in the cost of living.
- Redistribution of income — debtors gain, creditors lose; fixed-income earners suffer.
- Fall in savings and a rise in speculative activity.
- Adverse effect on the balance of payments, as exports become dearer.

Effects of deflation

- Falls in prices, profits and employment.
- Fall in demand, leading to recession.
- Burden on debtors, who must repay in more valuable money.

$$\text{Inflation rate} = [(\text{Price index in current year} - \text{Price index in previous year}) \div \text{Price index in previous year}] \times 100$$

Worked Example 14.1 — Inflation rate and real income

Problem. The CPI was 150 in 2023 and 162 in 2024. A worker's money wage rose from ₹300 to ₹320 per day. Find (a) the inflation rate and (b) whether the real wage rose or fell.

Solution.

- (a) Inflation rate = [(162 − 150)/150] × 100 = (12/150) × 100 = **8%**.
- (b)
  Real wage in 2024 at 2023 prices = money wage × (CPI 2023 / CPI 2024) = 320 × (150/162) = **₹296.30**.
- The real wage fell from ₹300 to ₹296.30, even though the money wage rose. Since the money wage rose by (20/300)×100 = 6.7%, which is less than the 8% inflation, the worker is worse off in real terms.

14.8 Agricultural and Public Finance

DEFINITION — FINANCE

Finance is the study of how money is raised, managed and spent by individuals, firms and governments.

Micro finance versus macro finance

Table 14.2 — Micro finance versus macro finance

Basis	Micro finance	Macro finance
Unit	The individual farm, firm or household	The economy as a whole
Concern	How an individual raises and uses funds	Money supply, credit, public revenue and expenditure
Example	A farmer's crop loan and repayment plan	The RBI's monetary policy; the Union Budget
Goal	Maximise the individual's return	Price stability and full employment

Agricultural finance is a part of micro finance. It deals with the money matters of farming — the supply of credit to farmers for production and investment, and the recovery of loans.

Need for agricultural finance

- **Seasonality.** Farm expenses are spread over the season but income comes mainly at harvest, creating a gap that must be bridged by credit.
- **Long gestation.** Perennial crops and livestock take years to yield returns, so long-term capital is needed.
- **Low savings.** Small and marginal farmers have little surplus to invest from their own savings.
- **High cost of inputs.** Seed, fertiliser, pesticides and machinery require substantial funds.
- **Risk and uncertainty.** Crop failure and price swings make insurance and credit essential.
- **Capital formation.** Credit is needed for irrigation, land development and mechanisation.

Sources of agricultural credit in India

- **Institutional:** cooperative societies (PACS), commercial banks, regional rural banks, land development banks and microfinance institutions. The **Kisan Credit Card (KCC)** provides flexible crop credit, and the **Modified Interest Subvention Scheme** gives short-term crop loans at concessional rates.
- **Non-institutional:** moneylenders, traders, commission agents and landlords — often quick but at very high interest rates.

14.9 Quantity Theory of Money

STATEMENT

The quantity theory of money states that the general price level is directly proportional to the quantity of money in circulation, other things being equal.

Fisher's equation of exchange: $M \times V = P \times T$

Here M is the quantity of money, V is the velocity of circulation (the number of times money changes hands), P is the general price level and T is the volume of transactions. If V and T are stable, a rise in M leads to a proportional rise in P — that is, inflation. This links money supply directly to the price level and underlies the central bank's policy of controlling inflation by managing money supply.

Exam Focus — Lecture 14

Key points to remember

- Barter has the problems of double coincidence of wants, no common measure, indivisibility and difficulty of storing value.
- Money evolved: commodity → metallic → paper → credit → plastic → digital.
- Primary functions of money: medium of exchange and measure of value; secondary: store of value and standard of deferred payment.
- Money supply: M1 narrow money; M3 broad money (most used).
- A price index measures the average change in prices; real value = nominal ÷ index × 100.
- Inflation is a sustained rise in the general price level; deflation is a sustained fall.
- Agricultural finance bridges the seasonal gap between farm expenses and income; KCC is a key institutional source.

Short-answer questions with model answers

Q1. State the problems of the barter system.

Model answer. Lack of double coincidence of wants, lack of a common measure of value, indivisibility of certain goods, difficulty in storing value, and problems of deferred payment and transfer of value.

Q2. Define inflation and state its main cause.

Model answer. Inflation is a sustained rise in the general price level that reduces the purchasing power of money. Its main cause is an increase in money supply relative to output, often through demand-pull or cost-push pressures.

Q3. Why is agricultural finance needed?

Model answer. Because of the seasonal gap between expenses and income, the long gestation of perennial crops, low savings of small farmers, high input costs, and the risks of crop failure and price swings.

Long-answer questions with answer outlines

Q1. Explain the evolution, meaning and functions of money, and the classification of money.

Outline. (i) Barter and its problems. (ii) Evolution stages. (iii) Meaning of money. (iv) Functions — primary, secondary, contingent. (v) Classification table (Table 14.1). (vi) Conclusion.

Q2. Discuss inflation and deflation, and the need for agricultural finance.

Outline. (i) Inflation — meaning, types, causes, effects. (ii) Deflation — meaning and effects. (iii) Inflation rate formula and real value. (iv) Agricultural finance — meaning, micro versus macro (Table 14.2). (v) Need for agricultural finance. (vi) Sources, including KCC. (vii) Conclusion.

LECTURE 15

Public Finance, Economic Systems and International Trade

15.1 Public Revenue and Public Expenditure

DEFINITION — PUBLIC FINANCE

Public finance is the branch of economics that studies the revenue and expenditure of the government, and how the government manages its income and spending to achieve its objectives.

Public finance has three main parts: **public revenue** (how the government raises money), **public expenditure** (how it spends money), and **public debt** (how it borrows when revenue falls short of expenditure).

Public revenue

Public revenue is the income of the government. Its main sources are:

- **Tax revenue** — direct taxes (income tax, corporation tax) and indirect taxes (GST, customs, excise).
- **Non-tax revenue** — fees, fines, licence fees, interest receipts, dividends from public enterprises and profits from the RBI.
- **Capital receipts** — borrowings, loans recovered and disinvestment.

Public expenditure

Public expenditure is the spending of the government. It is classified as:

- **Revenue expenditure** — spending on day-to-day running, salaries, subsidies and interest payments; it does not create assets.
- **Capital expenditure** — spending on assets such as roads, dams, irrigation and buildings; it creates assets.
- **Developmental and non-developmental** expenditure — the former promotes growth; the latter is for administration and defence.

Reasons for the growth of public expenditure

- Growth of population and the need for more public services.
- Defence and internal security needs.
- Welfare programmes, subsidies and poverty alleviation.
- Infrastructure and development investment.
- Servicing of public debt and interest payments.
- Urbanisation and rising expectations of citizens.

15.2 Tax — Meaning and Types

DEFINITION — TAX

A tax is a compulsory payment made by individuals and firms to the government, without any direct return of benefit to the payer. It is a legal obligation, not a voluntary contribution.

A good tax system should be equitable (fair), certain, convenient to pay, economical to collect, and should not harm production. Taxes are broadly of two kinds: direct and indirect.

Table 15.1 — Direct versus indirect taxes

Basis	Direct tax	Indirect tax
Incidence and impact	Fall on the same person (impact = incidence)	Can be shifted to others (impact ≠ incidence)
Levied on	Income and wealth of persons	Goods and services
Nature	Progressive (higher income pays more)	Often regressive in effect on the poor
Collection	Collected directly from the assessee	Collected through sellers and passed on
Examples	Income tax, corporation tax, capital gains tax	GST, customs duty, excise duty
Evasion	More scope for evasion	Less scope for evasion

15.3 Agricultural Taxation

Agricultural taxation is the taxation of agricultural income, land and produce. In India, the power to tax agricultural income rests mainly with the states, while the Centre taxes non-agricultural income.

Forms of agricultural taxation

- **Land revenue.** A tax on land, historically the main source of revenue from agriculture; now largely abolished or very low in most states.
- **Agricultural income tax.** A tax on income from agriculture, levied by some states.
- **Market fees and cesses.** Charged on the sale of produce in regulated markets.
- **Indirect taxes on inputs and produce.** Taxes on fertiliser, diesel, machinery and processed agricultural goods.

Issues in agricultural taxation

- Low tax base because agricultural income is exempt from central income tax and hard to assess.
- Political sensitivity — farmers form a large part of the electorate.
- Small and uneconomic holdings leave little taxable surplus.
- Poor record-keeping and the unorganised nature of the sector.

Balance of considerations. Heavy taxation of agriculture can hurt small farmers, so policy has generally moved from taxing agriculture towards supporting it — through MSP, input subsidies and direct income support such as PM-KISAN.

15.4 GST and Its Implications for the Indian Economy

DEFINITION — GOODS AND SERVICES TAX (GST)

GST is a comprehensive indirect tax on the supply of goods and services, levied as a single tax across the country, replacing many earlier indirect taxes such as excise duty, service tax and VAT.

GST was introduced in India on 1 July 2017. It is a **destination-based** tax (taxed where goods are consumed) and a **value-added** tax (tax paid on inputs is set off against tax on output). It has three components: **CGST** (central), **SGST** (state) and **IGST** (for inter-state trade).

Implications for the Indian economy

- **One nation, one tax.** A single tax replaced a maze of taxes, reducing cascading and simplifying compliance.
- **Wider market.** A common market removed inter-state barriers, helping trade and the movement of goods.
- **Transparency and formalisation.** The input-credit mechanism encourages formal record-keeping.
- **Boost to logistics.** Fewer check posts and faster movement of goods.
- **Cooperative federalism.** Centre and states jointly decide rates through the GST Council.

GST and agriculture

- Most **unprocessed agricultural produce** is exempt from GST — fresh fruits, vegetables, grains, milk and unprocessed food grains carry nil GST, so the burden on the farmer and consumer is minimal.
- Services such as loading, unloading, packing and warehousing of agricultural produce are also nil-rated.
- Some **processed products** (such as tea, jaggery, de-husked pulses and processed spices) fall outside the definition of "agricultural produce" and attract GST.
- Certain **pre-packaged and labelled** cereals and flours attract GST at a low rate, which affects the packaged-food segment.
- **Inputs** such as fertiliser, machinery and pesticides attract GST at various rates, which becomes part of the cost of cultivation.

15.5 Economic Systems

DEFINITION — ECONOMY

An economy is the organised system by which a society produces, distributes and consumes goods and services. Its main functions are to decide what, how and for whom to produce, and to allocate resources.

Functions of an economy

- Deciding what to produce and in what quantities.
- Deciding how to produce (choice of technique).
- Deciding for whom to produce (distribution of output).

- Ensuring efficient use of resources and economic growth.
- Providing for the future through saving and investment.

Table 15.2 — Capitalistic, socialistic and mixed economies

Basis	Capitalistic	Socialistic	Mixed
Ownership	Private	Public (state)	Both private and public
Price mechanism	Market decides (price mechanism)	Central planning decides	Both market and planning
Profit motive	Central	Limited; social welfare is the aim	Both profit and welfare
Role of government	Minimal (laissez-faire)	Dominant	Regulator and participant
Consumer sovereignty	High	Low	Moderate
Example	USA (historically)	Former USSR, Cuba	India

India is a mixed economy. Private enterprise operates in agriculture, industry and services, while the state plays a major role in infrastructure, banking, defence and welfare. Agriculture itself is largely private, but the government influences it heavily through MSP, procurement, credit, subsidies and regulation.

Elements of economic planning

- **Objectives** — growth, employment, equity and self-reliance.
- **Targets** — quantitative goals for output, investment and income.
- **Priorities** — the order in which sectors are developed.
- **Resource mobilisation** — taxes, savings and foreign aid to fund the plan.
- **Instruments** — fiscal, monetary and physical controls.
- **Evaluation** — review of achievements against targets.

India adopted five-year plans from 1951, with agriculture, industry and infrastructure as priorities. In recent years, planning has shifted to a market-oriented framework with the NITI Aayog, but the elements of goal-setting and resource allocation remain.

15.6 International Trade and Balance of Payments

> **DEFINITION — INTERNATIONAL TRADE**
>
> International trade is the exchange of goods and services between countries. It includes exports (goods sold abroad) and imports (goods bought from abroad).

Why countries trade

- **Differences in resources.** Countries differ in climate, land and minerals; India can grow spices and tea that colder countries cannot.
- **Comparative advantage.** A country gains by specialising in goods it can produce at a lower opportunity cost and trading for others.

- **Economies of scale.** Larger markets allow lower-cost production.
- **Variety and technology.** Trade gives access to goods and technology not available at home.

Balance of payments

> **DEFINITION — BALANCE OF PAYMENTS (BOP)**
>
> The balance of payments is a systematic record of all economic transactions between the residents of a country and the rest of the world during a year.

The BoP has two main accounts:

- **Current account** — trade in goods (visible) and services (invisible), plus unilateral transfers. The **balance of trade** is the difference between exports and imports of goods.
- **Capital account** — transactions in assets and liabilities, such as foreign investment, loans and changes in reserves.

$$\text{Balance of trade} = \text{Exports of goods} - \text{Imports of goods}$$
$$\text{Current account balance} = \text{Trade balance} + \text{Net services} + \text{Net transfers}$$
$$\text{BoP balance} = \text{Current account} + \text{Capital account} = 0 \text{ (in principle)}$$

A **trade deficit** occurs when imports exceed exports; a **trade surplus** when exports exceed imports. A persistent deficit may weaken the currency and require borrowing, while a surplus strengthens the currency. India generally runs a trade deficit that is partly offset by services exports (software) and remittances.

Worked Example 15.1 — GST and the tax on a sale

Problem. A farmer sells unprocessed paddy to a miller for ₹2,00,000. The miller processes it and sells rice (packaged and labelled) for ₹2,60,000, on which GST at 5% applies. The miller also paid ₹2,000 GST on packaging material. Compute the miller's output GST, input credit and net GST payable.

Solution.

- The farmer's sale of unprocessed paddy is exempt, so no GST arises at the farm level.
- Output GST on the miller's sale = 5% of ₹2,60,000 = **₹13,000**.
- Input tax credit available = ₹2,000 (GST paid on packaging).
- Net GST payable = 13,000 − 2,000 = **₹11,000**.

Interpretation. GST is a value-added tax: only the value added by the miller is taxed, because input credit is deducted. The unprocessed produce at the farm level bears no GST, which protects the farmer's price.

Worked Example 15.2 — Balance of trade and current account

Problem. A country's exports of goods = ₹500 crore; imports of goods = ₹620 crore; services exports = ₹150 crore; services imports = ₹60 crore; remittances received = ₹40 crore. Find the balance of trade and the current account balance.

Solution.

- Balance of trade = 500 − 620 = **−₹120 crore** (a trade deficit).
- Net services = 150 − 60 = ₹90 crore.
- Current account balance = (−120) + 90 + 40 = **+₹10 crore** (a small current account surplus).

Interpretation. Although the country has a trade deficit of ₹120 crore, strong services exports and remittances turn the current account into a small surplus. This is close to India's actual position, where services and remittances offset much of the goods deficit.

15.7 Canons of Taxation and Public Debt

Adam Smith laid down four canons (principles) of a good tax system:

- **Equality (equity).** Tax should be proportional to ability to pay; equals should be treated equally.
- **Certainty.** The time, manner and amount of tax should be clear to the payer, not arbitrary.
- **Convenience.** Tax should be collected at a time and in a manner convenient to the payer.
- **Economy.** The cost of collecting tax should be small relative to the revenue raised.

Modern canons add flexibility (tax should respond to changing needs) and productivity (tax should yield adequate revenue).

Public debt

When public expenditure exceeds public revenue, the government borrows, creating **public debt**. Debt may be internal (borrowed within the country) or external (borrowed from abroad). Public debt is used to finance development and to cover deficits, but heavy borrowing creates an interest burden and may crowd out private investment. The government must balance the need for spending against the cost of debt.

Exam Focus — Lecture 15

Key points to remember

- Public finance covers public revenue, public expenditure and public debt.
- A tax is a compulsory payment with no direct benefit; direct taxes cannot be shifted, indirect taxes can.
- Agricultural taxation is largely a state subject; land revenue and market fees are the main forms.
- GST is a destination-based, value-added indirect tax (CGST, SGST, IGST); unprocessed farm produce is exempt.
- Economic systems: capitalistic, socialistic and mixed; India is a mixed economy.
- Balance of payments records all external transactions; the current account includes trade in goods and services.
- Balance of trade = exports of goods − imports of goods.

Short-answer questions with model answers

Q1. Distinguish between direct and indirect taxes.

Model answer. A direct tax is levied on income and wealth and its impact and incidence fall on the same person (e.g. income tax). An indirect tax is levied on goods and services and can be shifted to the consumer (e.g. GST).

Q2. What is GST?

Model answer. GST (Goods and Services Tax) is a comprehensive, destination-based, value-added indirect tax on the supply of goods and services, introduced in India on 1 July 2017, with CGST, SGST and IGST components. It replaced many earlier indirect taxes.

Q3. Define balance of payments and balance of trade.

Model answer. The balance of payments is a record of all economic transactions between a country's residents and the rest of the world in a year. The balance of trade is the difference between exports and imports of goods alone.

Long-answer questions with answer outlines

Q1. Explain public revenue and public expenditure, and the meaning and types of taxes.

Outline. (i) Meaning of public finance. (ii) Public revenue sources. (iii) Public expenditure types and growth. (iv) Tax — meaning and canons. (v) Direct versus indirect taxes (Table 15.1). (vi) Agricultural taxation. (vii) Conclusion.

Q2. Discuss the economic systems and the elements of economic planning, and explain international trade and the balance of payments with GST and its implications.

Outline. (i) Economic systems — capitalistic, socialistic, mixed (Table 15.2). (ii) India as a mixed economy. (iii) Elements of economic planning. (iv) International trade — reasons. (v) Balance of payments and balance of trade. (vi) GST and its implications for India, including agriculture. (vii) Conclusion.

Appendix A — Model Question Papers

Two model question papers are given below, following the usual pattern of a theory paper: short-answer questions carrying fewer marks and long-answer questions carrying more marks. The suggested answer points are given after each paper.

Model Question Paper 1

Time: 3 hours Maximum marks: 100

Part A — Answer any eight of the following (8 × 5 = 40 marks)

1. Distinguish between microeconomics and macroeconomics.
2. State the law of diminishing marginal utility.
3. What are the characteristics of agriculture that make it different from industry?
4. Define consumer surplus. How is it measured?
5. Explain the factors influencing the elasticity of demand.
6. Distinguish between stock and supply.
7. What is a production function? Give the equation.
8. State the law of variable proportions.
9. Define direct and indirect taxes with examples.
10. What is the balance of payments?

Part B — Answer any four of the following (4 × 15 = 60 marks)

1. Explain the meaning, scope and subject matter of economics, and the central problems of an economy.
2. Explain the law of demand with a schedule and diagram, its assumptions, reasons and exceptions.
3. Discuss the law of variable proportions and the law of returns to scale.
4. Explain the meaning and types of market, and the features of perfect and imperfect competition.
5. Explain the concepts of national income and the three approaches to its measurement, with difficulties.
6. Discuss money — its evolution, functions and the problems of barter.

Suggested answer points — Model Question Paper 1

Part A. (1) Micro = individual units, price theory; macro = aggregates, income theory. (2) As more units are consumed, marginal utility falls; TU is maximum where MU = 0. (3) Dependence on nature, time lag, seasonality, perishability, small holdings, risk, diminishing returns. (4) Excess of willingness to pay over actual payment; area under the demand curve above price. (5) Substitutes, share of income, nature of good, number of uses, time, habit. (6) Stock = total available at a point of time; supply = part offered for sale at a price. (7) Technological relation between inputs and maximum output; Q = f(L, La, K, M). (8) Three stages — increasing, diminishing, negative returns; rational stage is the second. (9) Direct tax on income/wealth, not shiftable (income tax); indirect tax on goods, shiftable (GST). (10) A record of all economic transactions between a country and the rest of the world.

Part B. (1) Three definitions (wealth, welfare, scarcity); micro and macro; central problems of what, how and for whom. (2) Schedule, downward-sloping curve, assumptions, income and substitution effects, exceptions (Giffen, Veblen). (3) Short-run law of variable proportions with three stages; long-run returns to scale — increasing, constant, decreasing. (4) Meaning of market; types by competition; features of perfect competition, monopoly, monopolistic competition, oligopoly. (5) GDP, GNP, NNP, national income, per capita income; product, income and expenditure methods; difficulties. (6) Barter and its problems; evolution stages; functions of money — primary, secondary, contingent.

Model Question Paper 2

Time: 3 hours Maximum marks: 100

Part A — Answer any eight of the following (8 × 5 = 40 marks)

1. What is meant by ceteris paribus?
2. Define wealth, capital and income.
3. Explain the law of equi-marginal utility.
4. Define price elasticity of demand and state its types.
5. State Engel's law of family expenditure.
6. What are the four factors of production and their rewards?
7. Distinguish between fixed cost and variable cost.
8. What is the elasticity of supply?
9. Define economic rent and quasi-rent.
10. What is inflation? State two of its effects.

Part B — Answer any four of the following (4 × 15 = 60 marks)

1. Explain indifference curve analysis and the budget line, and show consumer's equilibrium.
2. Explain the concept and measurement of price, income and cross elasticity of demand.
3. Explain the cost concepts and the behaviour of the short-run cost curves.
4. Discuss digital marketing and its role in agricultural markets, with reference to eNAM.
5. Explain the theories of population and India's population policy.
6. Discuss GST and its implications for the Indian economy, especially agriculture.

Suggested answer points — Model Question Paper 2

Part A. (1) "Other things being equal"; holds other factors constant. (2) Wealth = stock of valued assets; capital = wealth used to produce more wealth; income = flow of receipts over a period. (3) Consumer maximises satisfaction when MU per rupee is equal across goods. (4) $\%\Delta Q \div \%\Delta P$; types — elastic, inelastic, unit, perfectly elastic, perfectly inelastic. (5) The share of income spent on food falls as income rises. (6) Land (rent), labour (wages), capital (interest), organisation (profit). (7) Fixed cost does not change with output; variable cost does. (8) $\%\Delta Q_s \div \%\Delta P$. (9) Rent = payment for land, a differential surplus; quasi-rent = short-run surplus of man-made capital. (10) Sustained rise in the general price level; effects — fall in real income of fixed-income earners, rise in cost of living.

Part B. (1) Indifference curve, properties, budget line, tangency point, MRS = price ratio. (2) Price elasticity with formula and methods; income elasticity types of goods; cross elasticity substitutes and complements. (3) TFC, TVC, TC, AFC, AVC, AC, MC; U-shaped curves; MC cuts AC at minimum. (4) Digital marketing meaning and scope; eNAM — features, price discovery, transparency, market access. (5) Malthusian theory and criticism; optimum theory; determinants; National Population Policy 2000. (6) GST meaning, components; implications — one nation one tax, wider market; agriculture — unprocessed produce exempt, processed items taxed.

Appendix B — Practice Numericals with Answers

These problems cover the numerical types most likely to appear in the examination. Attempt each one with the answer covered.

Elasticity of demand

Problem 1

Price falls from ₹50 to ₹40 and quantity demanded rises from 200 to 250 units. Find E_d.

Answer. $\%\Delta Q = (50/200)\times 100 = 25\%$; $\%\Delta P = (-10/50)\times 100 = -20\%$. $E_d = 25/20 = \mathbf{1.25}$ (elastic).

Problem 2

Price rises from ₹8 to ₹12 and quantity demanded falls from 100 to 70 units. Find E_d by the arc method.

Answer. $\Delta Q = -30$, average $Q = 85$; $\Delta P = +4$, average $P = 10$. $E_d = (-30/85) \div (4/10) = (-0.3529) \div 0.4 = -0.88$. Absolute value = **0.88** (inelastic).

Problem 3

Income rises from ₹15,000 to ₹18,000 and the demand for a good rises from 40 to 52 units. Find income elasticity and state the type of good.

Answer. $\%\Delta Q = (12/40)\times 100 = 30\%$; $\%\Delta Y = (3{,}000/15{,}000)\times 100 = 20\%$. $E_y = 30/20 = \mathbf{1.5}$; since $E_y > 1$, it is a luxury good.

Problem 4

The price of good Y rises from ₹100 to ₹120 and the demand for good X falls from 500 to 450 units. Find cross elasticity and state the relationship.

Answer. $\%\Delta Q_x = (-50/500)\times 100 = -10\%$; $\%\Delta P_y = (20/100)\times 100 = 20\%$. $E_{xy} = -10/20 = \mathbf{-0.5}$; negative, so the goods are **complements**.

Utility and consumer surplus

Problem 5

A consumer's demand schedule is: quantity 1, 2, 3, 4 with willingness to pay ₹60, ₹50, ₹40, ₹35. The market price is ₹40. Find consumer surplus.

Answer. At ₹40 the consumer buys 3 units (the 4th is worth ₹35 < ₹40). Surplus = (60−40) + (50−40) + (40−40) = 20 + 10 + 0 = **₹30**.

Cost, revenue and break-even

Problem 6

TFC = ₹200; TVC for outputs 1 to 5 = ₹100, ₹180, ₹240, ₹320, ₹450. Find TC, AC and MC for each output.

Answer.

Q	TVC	TC	AC	MC
1	100	300	300	100
2	180	380	190	80
3	240	440	146.7	60
4	320	520	130	80
5	450	650	130	130

Problem 7

Price = ₹25 per unit; AVC = ₹15 per unit; TFC = ₹80,000. Find the break-even output.

Answer. Contribution = 25 − 15 = ₹10. Break-even = 80,000 ÷ 10 = **8,000 units**.

National income

Problem 8

Find national income: compensation of employees ₹1,200 crore; rent ₹300 crore; interest ₹250 crore; profit ₹400 crore; mixed income ₹350 crore; NFIA −₹60 crore.

Answer. Domestic factor income = 1,200 + 300 + 250 + 400 + 350 = ₹2,500 crore. National income = 2,500 − 60 = **₹2,440 crore**.

Problem 9

Find GDP by the expenditure method: C = ₹5,000 crore; I = ₹1,800 crore; G = ₹1,500 crore; X = ₹700 crore; M = ₹900 crore.

Answer. GDP = 5,000 + 1,800 + 1,500 + (700 − 900) = 8,300 − 200 = **₹8,100 crore**.

Money, prices and growth

Problem 10

The CPI rises from 180 to 198. Find the inflation rate.

Answer. Inflation = [(198 − 180)/180] × 100 = (18/180)×100 = **10%**.

Problem 11

A population grows from 5,00,000 to 6,00,000 in 10 years. Find the decadal and the compound annual growth rate.

Answer. Decadal growth = (1,00,000/5,00,000)×100 = **20%**. CAGR = $[(6,00,000/5,00,000)^{1/10} - 1]\times100 = [(1.2)^{0.1} - 1]\times100 \approx$ **1.84%** per year.

Problem 12

Simple and compound interest on ₹40,000 at 10% for 2 years.

Answer. SI = 40,000×10×2/100 = **₹8,000**. CI amount = $40,000\times(1.1)^2$ = ₹48,400, so CI = **₹8,400**.

Appendix C — One-Line Revision Points

Quick one-line statements for last-minute revision.

- Economics studies the allocation of scarce resources among unlimited wants.
- Robbins: economics is the science of choice between scarce means with alternative uses.
- Microeconomics is price theory; macroeconomics is income theory.
- Positive economics is "what is"; normative is "what ought to be".
- Ceteris paribus means "other things being equal".
- Economic laws are statements of tendency, not certainties.
- Wealth and capital are stocks; income is a flow.
- Desire + ability to pay = want; want + willingness to pay = demand.
- Utility is the power to satisfy a want; it is subjective.
- Marginal utility falls as consumption rises; TU is maximum where MU = 0.
- Equi-marginal utility: $MU_x/P_x = MU_y/P_y$.
- Indifference curves are downward sloping, convex, and never intersect.
- Consumer equilibrium (ordinal): MRS = price ratio.
- Consumer surplus = willingness to pay − actual payment.
- Price elasticity of demand = %ΔQ ÷ %ΔP.
- Food demand is inelastic; hence MSP protects farmers' income.
- Engel's law: food share of income falls as income rises.
- Four factors of production: land, labour, capital, organisation.
- AP is maximum where MP = AP; TP is maximum where MP = 0.
- Law of variable proportions has three stages; the second is the rational stage.
- TC = TFC + TVC; MC cuts AC at its minimum.
- Under perfect competition, AR = MR = price; under monopoly, MR < AR.
- Profit is maximised where MR = MC.
- Break-even output = TFC ÷ (price − AVC).
- Supply is a flow; stock is the basis of supply.
- Perfect competition: many sellers, homogeneous product, price takers.
- Monopoly: one seller, no close substitute, price maker.
- eNAM is the electronic National Agriculture Market linking APMC mandis.
- Ricardian rent is a differential surplus; marginal land earns no rent.
- Quasi-rent is the short-run surplus of man-made capital.
- Real wage = money wage ÷ price level.
- Malthus: population grows geometrically, food arithmetically.
- Optimum population gives maximum per capita income.
- Pure interest is only for capital; gross interest adds risk and management charges.
- Economic profit = TR − (explicit + implicit cost).
- National income = NNP at factor cost; GNP = GDP + NFIA.
- GDP = C + I + G + (X − M).
- Barter fails for lack of a double coincidence of wants.
- Primary functions of money: medium of exchange and measure of value.

- Inflation is a sustained rise in the general price level.
- Direct taxes cannot be shifted; indirect taxes can.
- GST is a destination-based, value-added indirect tax; unprocessed farm produce is exempt.
- India is a mixed economy.
- Balance of trade = exports of goods − imports of goods.

Glossary of Key Terms

The glossary lists the important terms used in this book, with short definitions in simple language.

Table G.1 — Glossary

Term	Meaning
Economics	The social science that studies the production, distribution and consumption of goods and services, and the allocation of scarce resources among unlimited wants.
Microeconomics	The study of individual economic units — a consumer, a firm, a single market.
Macroeconomics	The study of the economy as a whole — national income, employment, the general price level.
Positive economics	Analysis of "what is"; factual and testable.
Normative economics	Analysis of "what ought to be"; based on value judgements.
Ceteris paribus	"Other things being equal"; an assumption holding other factors constant.
Rationality	Choosing the option that maximises satisfaction (consumer) or profit (firm).
Equilibrium	A state of rest where no agent has an incentive to change behaviour.
Economic law	A statement of a tendency about economic behaviour, conditional on other things being equal.
Utility	The power of a good or service to satisfy a human want.
Total utility	The sum of satisfaction from all units of a good consumed.
Marginal utility	The additional satisfaction from consuming one more unit of a good.
Law of diminishing marginal utility	As more units are consumed, marginal utility falls.
Equi-marginal utility	A consumer maximises satisfaction when the marginal utility per rupee is equal across goods.
Indifference curve	A curve showing combinations of two goods that give equal satisfaction.
Marginal rate of substitution	The rate at which a consumer will give up one good for another while remaining equally satisfied.
Budget line	The combinations of two goods a consumer can buy with a given income at given prices.
Consumer's equilibrium	The combination of goods that gives maximum satisfaction for a given income.
Demand	The quantity of a good that consumers are willing and able to buy at a given price and time.
Law of demand	Other things being equal, quantity demanded falls as price rises.
Giffen good	An inferior good for which a price fall may reduce demand.
Consumer surplus	The excess of what a consumer is willing to pay over what they actually pay.
Elasticity of demand	The responsiveness of quantity demanded to a change in price, income or related price.

Term	Meaning
Price elasticity of demand	% change in quantity demanded divided by % change in price.
Income elasticity	% change in quantity demanded divided by % change in income.
Cross elasticity	% change in quantity of one good divided by % change in price of another.
Engel's law	As income rises, the proportion of income spent on food falls.
Production	The process of creating utility by transforming inputs into outputs.
Factors of production	Land, labour, capital and organisation — the resources used to produce goods.
Production function	The technological relation between inputs and the maximum output obtainable.
Total product	The total output produced with given inputs.
Average product	Output per unit of the variable input (TP ÷ input).
Marginal product	The addition to total output from one more unit of input.
Law of variable proportions	As a variable factor increases on fixed factors, returns first rise, then diminish, then turn negative.
Returns to scale	How output changes when all inputs change in the same proportion.
Fixed cost	Cost that does not change with output in the short run.
Variable cost	Cost that changes with output.
Total cost	Sum of total fixed cost and total variable cost.
Average cost	Total cost per unit of output (TC ÷ Q).
Marginal cost	The addition to total cost from producing one more unit.
Opportunity cost	The value of the next best alternative given up.
Total revenue	Price times quantity sold.
Average revenue	Revenue per unit sold; equal to price.
Marginal revenue	The addition to total revenue from selling one more unit.
Break-even point	The output at which total revenue equals total cost.
Supply	The quantity of a good producers are willing to sell at a given price and time.
Stock	The total quantity of a good available at a point of time.
Law of supply	Other things being equal, quantity supplied rises as price rises.
Elasticity of supply	% change in quantity supplied divided by % change in price.
Market	An arrangement through which buyers and sellers settle price and quantity.
Perfect competition	A market with many buyers and sellers of a homogeneous product; firms are price takers.
Monopoly	A market with a single seller of a product with no close substitutes.
Monopolistic competition	A market with many sellers of differentiated products.

Term	Meaning
Oligopoly	A market with a few interdependent sellers.
eNAM	National Agriculture Market — an electronic trading portal linking APMC mandis.
Distribution theory	The theory of how income is shared among the factors of production.
Rent	The payment for the use of land; in the Ricardian sense a differential surplus.
Quasi-rent	The short-run surplus earned by man-made capital whose supply is fixed only in the short run.
Money wage	The amount of money paid to a worker.
Real wage	The goods and services that money wage can buy; money wage divided by the price level.
Interest	The price paid for the use of capital.
Pure interest	The reward purely for the use of capital funds.
Gross interest	Pure interest plus risk, inconvenience and management charges.
Profit	The residual reward of the entrepreneur after all other factors are paid.
Economic profit	Total revenue minus explicit and implicit costs.
Normal profit	The minimum return needed to keep the entrepreneur in business.
National income	The money value of all final goods and services produced in a year, plus net income from abroad.
GDP	Gross Domestic Product — output produced within the domestic territory.
GNP	Gross National Product — GDP plus net factor income from abroad.
NNP	Net National Product — GNP minus depreciation.
Per capita income	National income divided by population.
Barter	Direct exchange of goods for goods, without money.
Money	Anything generally accepted as a medium of exchange and measure of value.
Money supply	The total stock of money held by the public.
Price index	A measure of the average change in prices between two periods.
Inflation	A sustained rise in the general price level.
Deflation	A sustained fall in the general price level.
Public finance	The study of government revenue, expenditure and debt.
Tax	A compulsory payment to the government without direct benefit to the payer.
Direct tax	A tax on income or wealth that cannot be shifted.
Indirect tax	A tax on goods and services that can be shifted to the consumer.
GST	Goods and Services Tax — a destination-based value-added indirect tax.

Term	Meaning
Mixed economy	An economy with both private and public ownership (India).
Balance of payments	A record of all economic transactions between a country and the rest of the world.
Balance of trade	Exports of goods minus imports of goods.
MSP	Minimum Support Price — a guaranteed floor price for specified crops.
KCC	Kisan Credit Card — a scheme giving farmers timely and flexible credit.

Formula Sheet

All the important formulae of the course, collected for quick revision.

Utility and Consumer Behaviour

$$MU_n = TU_n - TU_{n-1} \;\mid\; TU = \Sigma\, MU \;\mid\; AU = TU / Q$$

Equi-marginal utility: $MU_x / P_x = MU_y / P_y = MU \text{ of money}$

Consumer's equilibrium (ordinal): $MRS_{xy} = P_x / P_y$

Budget line: $P_xQ_x + P_yQ_y = M \;\mid\; \text{slope} = -P_x/P_y$

Consumer surplus = Willingness to pay − Actual payment

Elasticity

$$E_d = (\%\Delta Q) \div (\%\Delta P) = (\Delta Q / \Delta P) \times (P / Q)$$

Arc elasticity: $E_d = [\Delta Q / ((Q_1+Q_2)/2)] \div [\Delta P / ((P_1+P_2)/2)]$

Income elasticity: $E_y = (\%\Delta Q) \div (\%\Delta Y)$ | Cross elasticity: $E_{xy} = (\%\Delta Q_x) \div (\%\Delta P_y)$

Elasticity of supply: $E_s = (\%\Delta Q_s) \div (\%\Delta P)$

Production

Production function: Q = f(L, La, K, M)

AP = TP / input | MP = ΔTP / Δinput

Least-cost combination: $MP_L/P_L = MP_K/P_K$

Profit-maximising input use: VMP = factor price

Cost and Revenue

TC = TFC + TVC | AC = TC / Q = AFC + AVC

$$AFC = TFC / Q \;|\; AVC = TVC / Q \;|\; MC = \Delta TC / \Delta Q$$

$$TR = P \times Q \;|\; AR = TR / Q = P \;|\; MR = \Delta TR / \Delta Q$$

$$Profit = TR - TC \;|\; Profit \text{ is maximum where } MR = MC$$

$$\text{Break-even output} = TFC \div (Price - AVC \text{ per unit})$$

Market Equilibrium

$$\text{Equilibrium where } Q_d = Q_s \text{ (demand = supply)}$$

Interest, Profit and National Income

$$\text{Simple interest} = P \times R \times T / 100 \;|\; \text{Compound amount} = P(1 + R/100)^T$$

$$\text{Accounting profit} = TR - \text{explicit cost} \;|\; \text{Economic profit} = TR - (\text{explicit} + \text{implicit cost})$$

$$GNP = GDP + NFIA \;|\; NNP = GNP - \text{depreciation}$$

$$\text{National income} = \text{NNP at factor cost} = NNP_{mp} - \text{indirect taxes} + \text{subsidies}$$

$$GDP = C + I + G + (X - M) \;|\; \text{Per capita income} = \text{National income} \div \text{population}$$

Money and Prices

$$\text{Price index} = (\Sigma P_1Q_0 / \Sigma P_0Q_0) \times 100$$

$$\text{Real value} = (\text{Nominal value} \div \text{Price index}) \times 100$$

$$\text{Inflation rate} = [(\text{Current index} - \text{Previous index}) \div \text{Previous index}] \times 100$$

$$\text{Real wage} = \text{Money wage} \div \text{Price level}$$

$$\text{Money supply: M1} = \text{currency} + \text{demand deposits} + \text{other deposits with RBI; M3} = \text{M1} + \text{net time deposits}$$

Population, Public Finance and Trade

$$\text{Population growth rate} = [(P_t - P_0) \div P_0] \times 100$$

$$\text{Compound annual growth rate} = [(P_t/P_0)^{1/n} - 1] \times 100$$

$$\text{Balance of trade} = \text{Exports of goods} - \text{Imports of goods}$$

$$\text{Current account balance} = \text{Trade balance} + \text{Net services} + \text{Net transfers}$$

References and Suggested Reading

The following standard texts and course materials are recommended for further reading. The chapters in this book are based on the standard treatment of these topics found in the works listed below.

Core textbooks

- Dewett, K. K. and Verma, J. D. — *Modern Economic Theory*. S. Chand & Company, New Delhi.
- Samuelson, P. A. and Nordhaus, W. D. — *Economics*. McGraw-Hill Education.
- Mankiw, N. G. — *Principles of Economics*. Cengage Learning.
- Ahuja, H. L. — *Principles of Microeconomics* and *Macroeconomics*. S. Chand & Company.

Agricultural economics texts

- Subba Reddy, S., Raghu Ram, P., Neelakanta Sastry, T. V. and Bhavani Devi, I. — *Agricultural Economics*. Oxford & IBH Publishing Co., New Delhi.
- Sadhu, A. N. and Singh, A. — *Fundamentals of Agricultural Economics*. Himalaya Publishing House.
- Reddy, S. S. et al. — *Agricultural Economics* (TNAU edition), Tamil Nadu Agricultural University.
- Kahlon, A. S. and Singh, K. — *Principles of Agricultural Economics*.

Course and institutional material

- Tamil Nadu Agricultural University — e-course material and lecture notes for AEC 101, *Principles of Agricultural Economics*.
- ICAR — e-course content and study material for undergraduate agricultural economics.
- Ministry of Agriculture and Farmers' Welfare, Government of India — annual MSP announcements and agricultural statistics.
- Reserve Bank of India — publications on money supply, credit and the balance of payments.
- NITI Aayog and the erstwhile Planning Commission — Five Year Plan documents and policy papers.
- National Population Policy, 2000 — Ministry of Health and Family Welfare, Government of India.

> **Note on examples.** The Indian and Tamil Nadu examples used in this book — MSP, agricultural markets, uzhavar sandhai, eNAM, farm finance, KCC, PM-KISAN and GST on farm produce — are drawn from the standard institutional material listed above and from the official publications of the Government of India. Students should refer to the latest official sources for current figures, as prices and schemes are revised from time to time.